



Electronics qn bank Sol

Electronic Device and Circuits (Pokhara University)



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1.10 SOLVED BOARD EXAMINATION PROBLEMS

1. What is a photo-diode?

[2013 F]

Solution: See the definition part 1.9.3

2. List at least 4 areas of application of a diode.

[2013 F]

Solution:

The diode finds use in many applications as enumerated below:

- As rectifier in dc power supplies
- In demodulation or detector circuits
- As switches in digital logic circuits
- In clipping circuits used as wave shaping circuits in computers, radars, radio and TV receivers
- In clamping networks employed as dc restorers in TV receivers and voltage multipliers

3. Write short notes on filters.

[2013 F]

Solution: See the definition part 1.8.3

4. With the help of the neat and clean curve explain I-V characteristics of the P-N junction diode and also discuss the forward and reverse bias of the P-N diode.

[2013 F, 2013 S, 2014 F, 2017 S]

Solution: See the definition part 1.4.1, 1.4.2, 1.5.1 and 1.5.2

5. Explain the zener diode as a voltage regulator with its V-I characteristics.

[2013 S, 2017 F, 2018 F]

Solution:

P-N junction

See the definition part 1.9.1

For the second part

The forward characteristics of a zener diode are similar to those of an ordinary P-N junction diode.

When reverse biased voltage is applied to a zener diode, it allows only a small amount of leakage current until the voltage is less than zener voltage. When reverse biased voltage applied to the zener diode reaches zener voltage, it starts allowing large amount of electric current. At

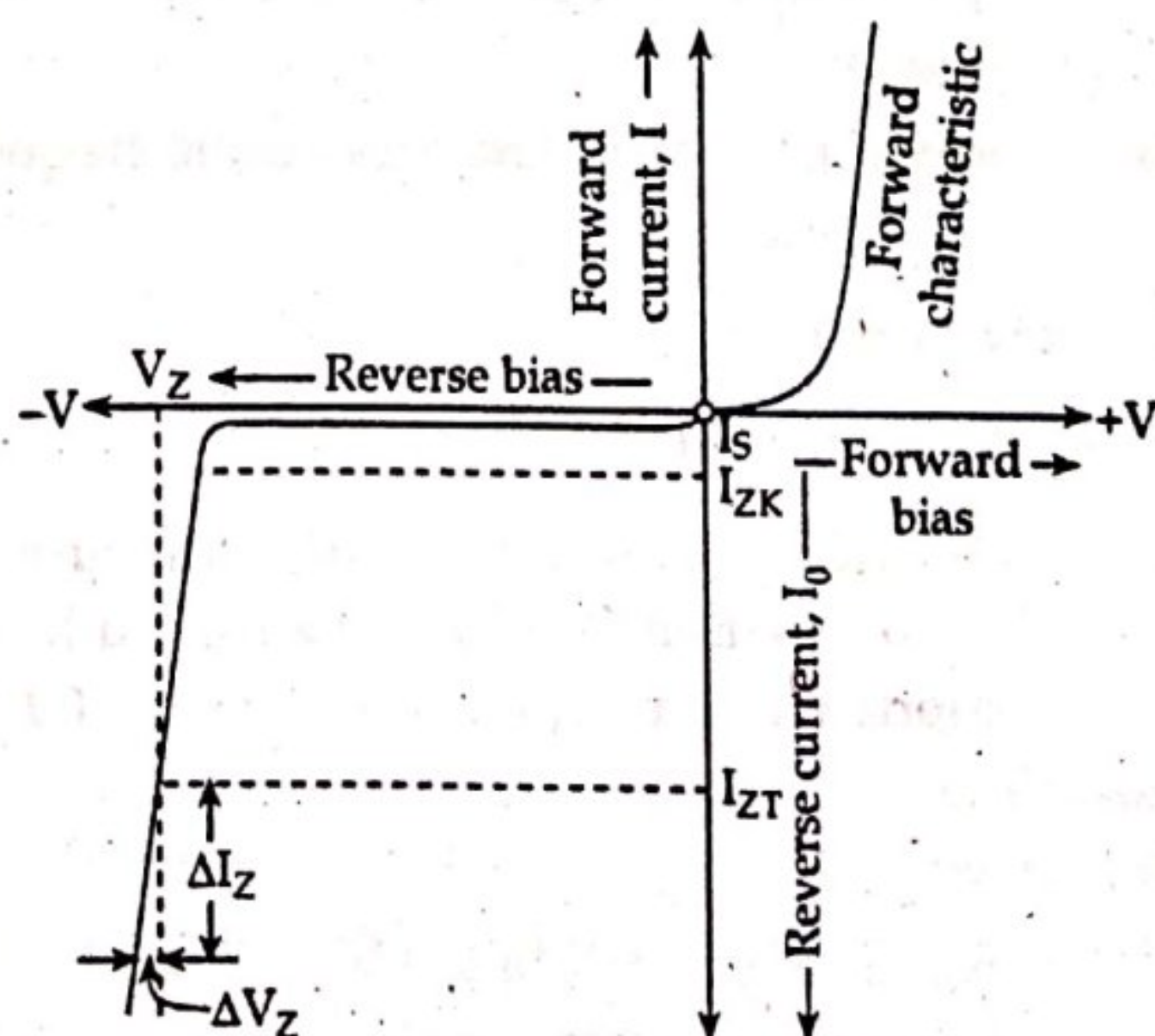


Figure: Zener diode characteristics with enlarged view of breakdown region

this point, a small increase in reverse voltage will rapidly increase the electric current. Because of this sudden rise in electric current, breakdown occurs called zener breakdown. However, zener diode exhibits a controlled breakdown that does damage the device. The zener breakdown voltage of the zener diode depends on the amount of doping applied.

6. Why a rectifier is used in electronic devices? How can we deploy 4 diodes to convert AC supply voltage to DC? Explain with circuit diagrams and waveforms. [2013 S, 2017 F]

Solution:

Rectifiers are used in the electronic devices to:

- derive dc from ac supply.
- demodulate the electric signals.
- create power through power rectifier.
- protect dc appliances from ac supply.
- detect amplified radio signals.

For the second part

See the definition part 1.8.2.2

7. Why full wave rectifier is more preferred over half wave rectifier? Support your answer with necessary mathematical derivations. [2013 F, 2014 F, 2016 S]

Solution:

Full wave rectifier is more preferred over half wave rectifier because of the following reasons:

- The rectification efficiency is double than half wave rectifier.
- Better transformer utilization factor.
- DC output voltage and current is higher hence, output power is higher.
- Ripple factor is less and ripple frequency is double hence easy to filter out.

For the second part

See the definition part 1.8.1 and 1.8.2

8. A silicon diode has a saturation current of 0.1 pA at 20°C. Find its current when it is forward biased by 0.55 V. Find current in same diode when temperature rises to 100°C. [2013 S, 2017 F, 2018 F]

Solution:

We have,

Applied forward bias, (V) = 0.55 V

Temperature, (T) = 20°C = 20 + 273 K = 293 K

Saturation current, (I_0) = 0.1 pA = 0.1×10^{-12} A

Volt equivalent at 293 K = $\frac{293}{11600}$

$$\therefore V_T = 0.02526 \text{ V}$$

We know that;

$$I = I_0 \left(e^{\frac{V}{V_T}} - 1 \right)$$

$$\text{or, } I = 0.1 \times 10^{-12} \left(e^{\frac{0.55}{0.02526}} - 1 \right)$$

$$\text{or, } I = 0.1 \times 10^{-12} \times 53463.73$$

$$\text{or, } I = 0.00000000534 \text{ A}$$

$$\therefore I = 5.346 \times 10^{-9} \text{ A} = 5.346 \text{ nA}$$

From experimental data, it is found that reverse saturation current I_0 increases 7% per °C for both silicon and germanium diode.

so, saturation current at 100°C is;

$$I'_0 = I_0 \times (1.07)^{100-20} = 0.1 \times 10^{-12} \times 224.234$$

$$\therefore I'_0 = 2.242 \times 10^{-11} \text{ A}$$

Hence, current when temperature rises to 100°C on same diode is;

$$I' = I'_0 \times 53463.73$$

$$\text{or, } I' = 2.242 \times 10^{-11} \times 53463.73$$

$$\therefore I' = 1.1988 \times 10^{-6} \text{ A} = 1.198 \text{ }\mu\text{A}$$

Here, $\eta = 2$; for silicon diode

9. Write short notes on π -filter. [2014 F]

Solution: See the definition part 1.8.3.1

10. Draw a circuit diagram and waveform of full-wave bridge rectifier. Determine the average, r.m.s. and form-factor of full-wave rectifier. [2014 S, 2015 S, 2016 F]

Solution: See the definition part 1.8.2.2

11. What is PN junction? Explain the action of PN junction diode under forward and reverse bias with necessary diagrams. [2014 S, 2015 F]

Solution:

P-N junction

A P-N junction is a boundary or interface between two types of semiconductor materials, P-type and N-type inside a single crystal of semiconductor.

For the second part

See the definition part 1.4.1 and 1.4.2

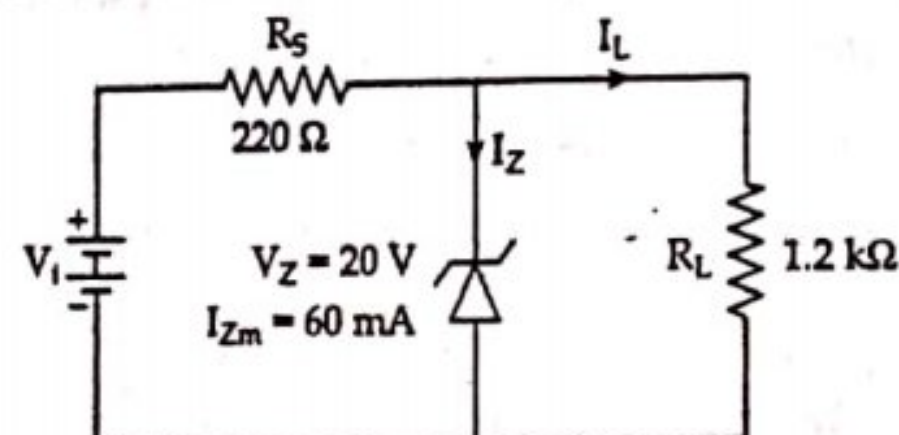
12. Why is an intrinsic semiconductor not adequate for making semiconductor device? [2014 S]

Solution:

Intrinsic semiconductors are not adequate for making semiconductor devices because of the following reasons:

- Intrinsic semiconductor can conduct only a small amount of current at normal temperature, to increase the current the semiconductor should be constantly heated which is not suitable. Hence, for devices that require large amount of current, intrinsic semiconductors cannot be used.
- In intrinsic semiconductor, when electrons get thermal energy, electrons move from valence band to conduction band. More will be the thermal energy, more will be the number of electrons in the conduction band and more will be the current. Hence, current is controlled by temperature.
- Electrical conductivity is very low.

13. Determine the range of values of V_{in} that will maintain the zener diode in the ON state. Also, find the maximum power that can be dissipated to the diode. [2014 S]



Solution:

Given that;

$$V_{i_{min}} = V_Z \times \left(\frac{R_L + R_S}{R_L} \right) = 20 \times \left(\frac{1200 + 220}{1200} \right) = 23.67 \text{ V}$$

$$I_L = \frac{V_L}{R_L} = \frac{V_Z}{R_L} = \frac{20}{1200} = 0.0166 \text{ A}$$

Also,

$$I_{R_{max}} = I_{ZM} + I_L = 60 + 16.66 = 76.66 \text{ mA} = 0.0766 \text{ A}$$

$$\therefore V_{i_{max}} = I_{R_{max}} \times R_S + V_Z = 0.0766 \times 220 + 20 = 36.87 \text{ V}$$

Hence, V_{in} should be within 23.67 V to 36.87 V for the zener diode to remain in ON state.

Maximum power dissipated in zener diode ($P_{Z_{max}}$) is given by;

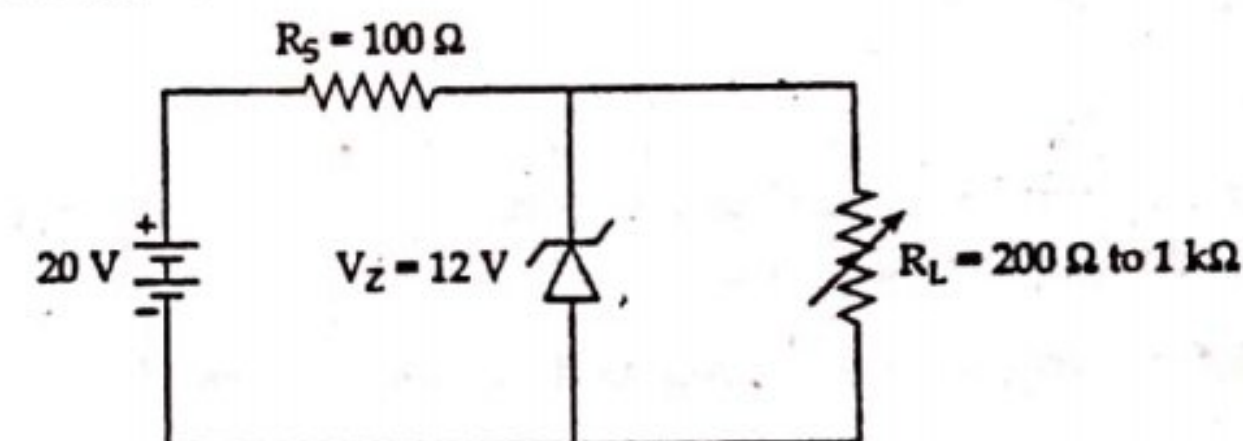
$$\begin{aligned} P_{Z_{max}} &= V_Z \times I_{Z_{max}} \\ &= V_Z \times (I_{S_{max}} - I_L) \end{aligned}$$

$$\begin{aligned} &= V_Z \times \left(\frac{V_{i_{max}} - V_Z}{R_S} - \frac{V_Z}{R_L} \right) \\ &= 20 \times \left(\frac{36.87 - 20}{220} - \frac{20}{1200} \right) \\ &= 20 \times (0.07668 - 0.0166) \end{aligned}$$

$$\therefore P_{Z_{max}} = 1.2016 \text{ Watt}$$

14. A zener regulator has 12 V zener voltage with variable load resistance. Calculate,

- maximum and minimum load current.
- voltage drop across series resistance.
- current through zener when the load is 200 ohm.
- minimum value of load resistance to ensure that zener diode is ON. [2015 F]



Solution:

Given that;

$$V_{in} = 20 \text{ V}$$

$$R_{L_{min}} = 200 \Omega$$

$$R_{L_{max}} = 1000 \Omega$$

$$V_Z = 12 \text{ V}$$

$$R_S = 100 \Omega$$

- i) Current through series resistance, (I_S) = $\frac{V_{in} - V_Z}{R_S} = \frac{20 - 12}{100} = 0.0800 \text{ A}$

Thus,

$$\begin{aligned} \text{Voltage drop across series resistance, } (V_{RS}) &= I_S \times R_S \\ &= 0.0800 \times 100 \\ &= 8 \text{ V} \end{aligned}$$

- ii) Current through zener diode when $R_L = 200 \Omega$

$$I_L = \frac{V_{out}}{R_L} = \frac{V_Z}{R_L} = \frac{12}{200} = 0.060 \text{ A}$$

Hence,

$$\begin{aligned} \text{Zener diode current, } (I_Z) &= I_S - I_L \\ &= 0.0800 - 0.060 = 0.020 \text{ A} \end{aligned}$$

$$\text{iii) Maximum load current, } (I_{L_{\max}}) = \frac{V_Z}{R_{L_{\min}}} \\ = \frac{12}{200} = 0.060 \text{ A}$$

$$\text{and, Minimum load current, } (I_{L_{\min}}) = \frac{V_Z}{R_{L_{\max}}} \\ = \frac{12}{1000} = 0.0120 \text{ A}$$

iv) Minimum value of load resistance to ensure that zener diode is ON is;

$$V_0 = V_Z = \frac{R_L \times V_{in}}{R_S + R_L}$$

$$\text{or, } R_{L_{\min}} = \frac{R_S \times V_Z}{V_{in} \times V_Z}$$

$$\text{or, } R_{L_{\min}} = \frac{R_S}{V_{in}} = \frac{100}{20}$$

$$\therefore R_{L_{\min}} = 5 \Omega$$

Hence, minimum resistance required is 5Ω to ensure that the zener diode is in the ON state.

15. Difference between zener and avalanche breakdown. [2015 F]

Solution:

	Zener breakdown	Avalanche breakdown
i)	This occurs at junction which being heavily doped having narrow depletion layers.	This occurs at junctions which being lightly doped having wide depletion layers.
ii)	This breakdown occurs within the depletion region.	This breakdown occurs outside the depletion region.
iii)	Low reverse potential is required for breakdown.	High reverse potential is required for breakdown.
iv)	Zener breakdown voltage is inversely proportional to the temperature.	Avalanche breakdown voltage is directly proportional to the temperature.
v)	Temperature coefficient of zener breakdown is negative.	Temperature coefficient of avalanche breakdown is positive.
vi)	Zener breakdown produces the electrons.	The avalanche breakdown produces the pairs of holes and electrons.
vii)	The zener breakdown V-I characteristics have a sharp curve.	The avalanche breakdown V-I characteristics do not have a sharp curve.

16. Difference between LED and photodiode. [2015 F]

Solution:

	LED	Photodiode
i)	LED is a two terminal device, which converts electrical energy into light energy.	Photodiode is two terminal device, which converts light energy into electrical energy.
ii)	It works on the principle of electro-luminance.	It works on the principle of photo conduction.
iii)	Biasing mode is forward biased only.	Biasing mode is reversed biased only.
iv)	Semiconductor used are gallium arsenide phosphide or gallium phosphide.	Semiconductor used are silicon and germanium.

17. What do you mean by ideal diode? Draw its V-I characteristics. Explain each point. [2015 S]

Solution:

Ideal diode

See the definition part 1.6.1

For the second part

i) An ideal diode does not offer any resistance to the flow of the current through it when it is in forward biased mode. This means that ideal diode will be a perfect conductor when forward biased.

ii) We know that;

$$I = \frac{V}{R}$$

$$\text{since, } R = 0, I = \infty$$

This indicates that there is no higher limit for the current, which can flow through the forward biased ideal diode.

iii) An ideal diode is expected to fully inhibit the flow of current through it under reverse biased condition. In other words, it is expected to mimic the behaviour of a perfect insulator when reverse biased.

iv) The ideal diode possesses infinite resistance when operating in reverse biased mode.

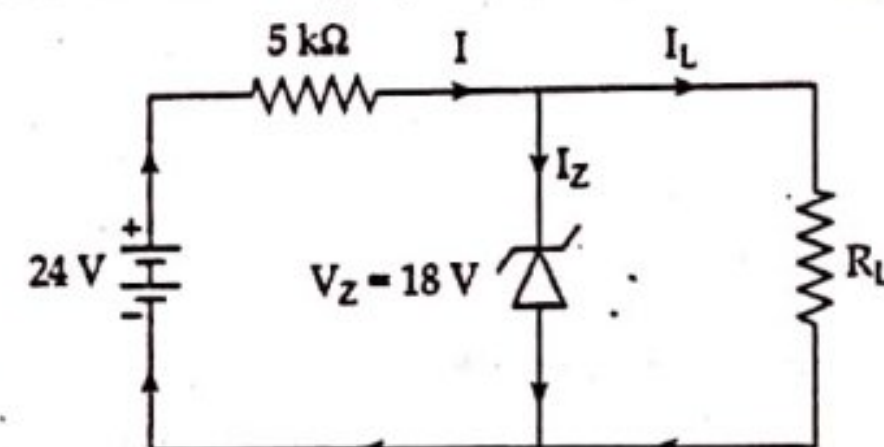
$$\text{i.e., } I = \frac{V}{\infty} = 0$$

Thus, it means that there will be no current flowing through the ideal diode when it is reversed biased, no matter how high the reverse voltage applied be.

18. Difference between majority and minority carrier. [2015 F]
Solution:

	Majority carrier	Minority carrier
i)	Majority carriers are those, which are present in a material in high concentration and take part in conduction of electric current.	Minority carriers are those, which are present in a material in low concentration and the current due to these carriers is always negligible.
ii)	The majority carriers in N-type materials are electrons.	The minority carriers in N-type materials are holes.
iii)	The majority carriers in P-type materials are holes.	The minority carriers in P-type materials are electrons.

19. For the zener diode regulator circuit shown below find;
i) calculate the current through 5 K-ohm resistance, zener diode and load resistance of 1 K-ohm.
ii) calculate load voltage (across R_L) [2015 S, 2016 S]



Solution:

Given that;

$$\begin{aligned} V_Z &= 18 \text{ V} \\ R_L &= 1 \text{ k}\Omega = 1000 \Omega \\ R_S &= 5 \text{ k}\Omega = 5000 \Omega \\ V_S &= 24 \text{ V} \end{aligned}$$

- i) Output voltage, $(V_L) = V_Z = 18 \text{ V}$
ii) Load current, $(I_L) = \frac{\text{Output voltage } (V_L)}{\text{Load resistance } (R_L)} = \frac{18}{1000} = 0.0180 \text{ A}$
Current through 5 K ohm resistance, $(I) = \frac{\text{Voltage drop across Series resistance}}{\text{Series resistance } (R_S)} = \frac{I \times R_S}{R_S} = \frac{V_S - V_L}{R_S} = \frac{24 - 18}{5000} = 0.00120 \text{ A}$
Current in zener diode, $(I_Z) = I - I_L = 0.00120 - 0.0180 = -0.01680 \text{ A} = -1.680 \times 10^{-2} \text{ A}$

20. Difference between intrinsic and extrinsic semiconductor. [2016 S, 2018 S]
Solution:

	Intrinsic semiconductor	Extrinsic semiconductor
i)	It is a pure semiconductor without any impurity in it.	It is prepared by doping a small quantity of impurity atoms to a pure semiconductor.
ii)	Its electrical conductivity is very low.	Its electrical conductivity is high.
iii)	In these, the energy gap is very small.	In these, the energy gap is more than that in pure semiconductors.
iv)	Pure group IV elements.	Group III or group V elements are introduced in group IV elements.
v)	Holes and electrons are equal.	Holes are more in P-type and electrons are more in N-type extrinsic semiconductor.
vi)	Fermi level lies between valence band conduction band.	Fermi level lies near valence band in P-type and near conduction band in N-type.
vii)	Its electrical conductivity is a function of temperature alone.	Its electrical conductivity depends upon the temperature as well as on the quantity of impurity atoms doped.

21. Calculate the load voltage and all the currents across the voltage regulator circuit for the given data $V_{in} = (60 - 100) \text{ V}$, $R_S = 1 \text{ k}\Omega$, $V_Z = 40 \text{ V}$ and $R_L = 1 \text{ k}\Omega$ [2016 S]

Solution:

Given that;

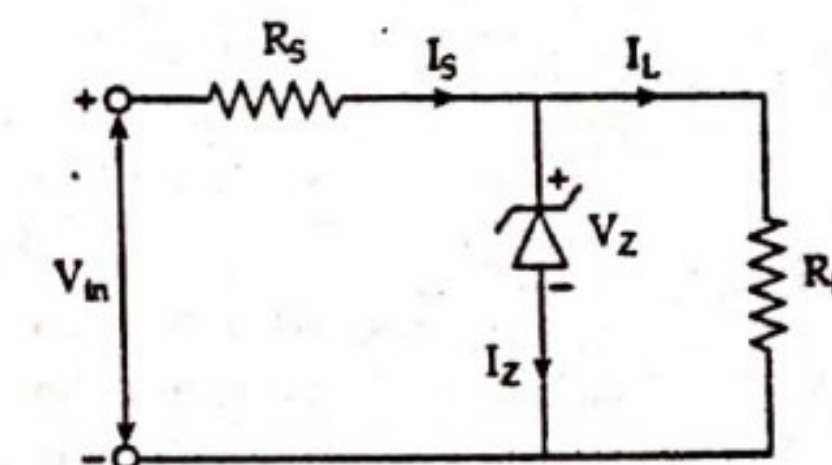
$$\begin{aligned} V_{in} &= (60 - 100) \text{ V} \\ R_S &= 1 \text{ k}\Omega \\ V_Z &= 40 \text{ V} \\ R_L &= 1 \text{ k}\Omega \end{aligned}$$

Now,

$$\text{Load current, } (I_L) = \frac{V_{out}}{R_L} = \frac{V_Z}{R_L} = \frac{40}{1000} = 0.04 \text{ A}$$

$$\begin{aligned} \text{Maximum supply current, } (I_{S_{max}}) &= \frac{V_{in_{max}} - V_Z}{R_S} = \frac{100 - 40}{1000} \\ &= 0.06 \text{ A} \end{aligned}$$

$$\text{Corresponding maximum zener current, } (I_{Z_{max}}) = I_{S_{max}} - I_L$$



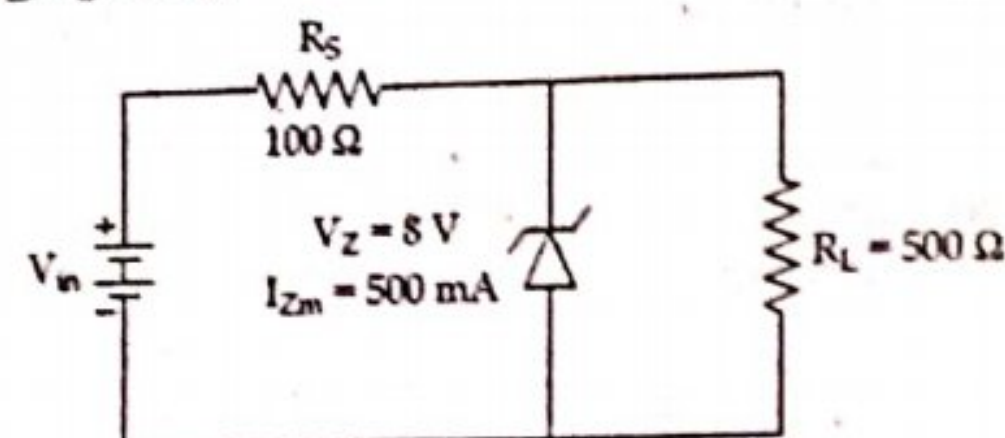
$$\begin{aligned} &= 0.06 - 0.04 \\ &= 0.02 \\ \text{and, Minimum supply current, } (I_{S_{\min}}) &= \frac{V_{in_{\min}} - V_Z}{R_S} = \frac{60 - 40}{1000} \\ &= 0.02 \text{ A} \end{aligned}$$

$$\begin{aligned} \text{Corresponding minimum zener current, } (I_{Z_{\min}}) &= I_{S_{\min}} - I_L \\ &= 0.02 - 0.04 \\ &= -0.02 \text{ A} \end{aligned}$$

Then,

$$\text{Load voltage, } (V_L) = I_L \times R_L = 0.04 \times 1000 = 40 \text{ V}$$

22. For the circuit shown below, assume $R_S = 100 \Omega$; determine,
 i) the value of V_{in} that ensures the zener diode is in ON state.
 ii) I_S, I_Z, I_L if $V_{in} = 30 \text{ V}$ [2016 F]



Solution:

Given that;

$$R_S = 1000 \Omega$$

$$V_Z = 8 \text{ V}$$

$$R_L = 500 \Omega$$

$$I_{ZM} = 500 \text{ mA} = 0.50 \text{ A}$$

- i) If $V_{in} = 30 \text{ V}$;

$$\text{Supply current, } (I_S) = \frac{V_{in} - V_Z}{R_S} = \frac{30 - 8}{100} = 0.22 \text{ A}$$

Now,

$$\text{Load current, } (I_L) = \frac{V_{out}}{R_L} = \frac{V_Z}{R_L} = \frac{8}{500} = 0.016 \text{ A}$$

$$\text{and, Zener diode current, } (I_Z) = I_S - I_L = 0.22 - 0.016 = 0.2040 \text{ A}$$

- ii) Value of V_{in} that ensures the zener diode is in the ON state,

$$V_{in_{\min}} = V_Z \times \left(\frac{R_L + R_S}{R_L} \right) = 8 \times \left(\frac{500 + 100}{100} \right) = 48 \text{ V}$$

Now,

$$\text{Load current, } (I_L) = \frac{V_L}{R_L} = \frac{V_Z}{R_L} = \frac{8}{500} = 0.0160 \text{ A}$$

Thus,

$$I_{R_{\max}} = I_{ZM} + I_L = 0.50 + 0.0160 = 0.5160 \text{ A}$$

Then,

$$V_{in_{\max}} = I_{R_{\max}} \times R_S + V_Z = 0.5160 \times 100 + 8 = 59.60 \text{ V}$$

Hence, the value of V_{in} should be $48 \leq V_{in} \leq 59.60 \text{ V}$.

23. Explain the barrier potential and formation of depletion region in the P-N junction. [2016 F]

Solution: See the definition part 1.3.2, 1.3.3 and 1.3.1

24. Write short notes on piecewise linear model of diode.

[2016 F, 2018 S]

Solution: See the definition part 1.6.3

25. Define fermi level. What is the difference between the band structure of a semiconductor, metal and insulators? Energy band gap of some unknown material given as;

- i) 15 eV ii) 0.7 eV iii) Overlapping

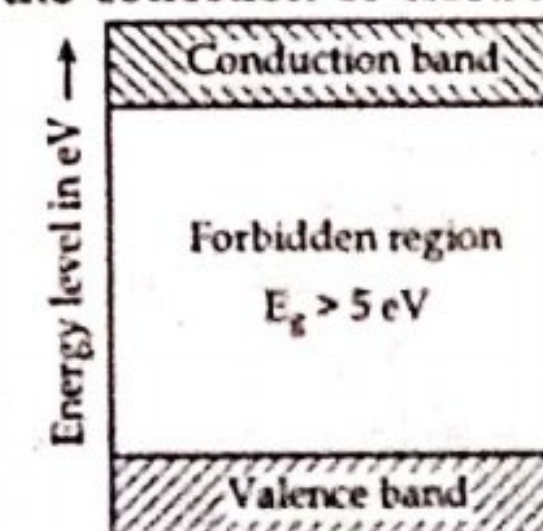
Distinguish their respective class as a semiconductor, metal and insulator. [2017 F]

Solution:

Fermi level

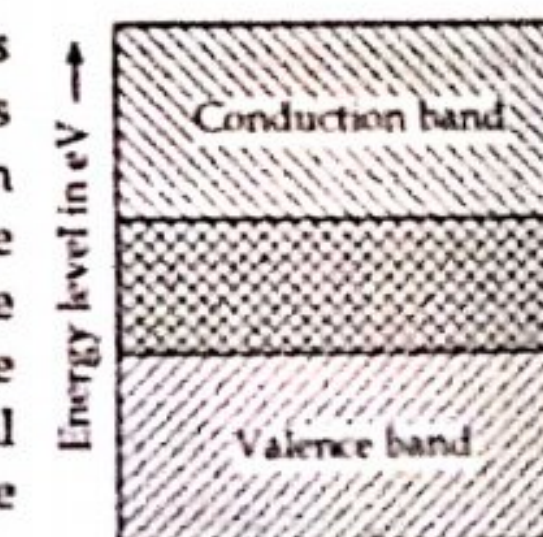
Fermi level is the term used to describe top of the collection of electron energy levels at absolute zero temperature.

In case of insulators, there is practically no electron in the conduction band and the valence band is filled. For an insulator, the valence and conduction bands are so apart (more than 5 eV) the energy which can be supplied to an electron from an applied field is too small to carry a particle from the filled band into vacant band.

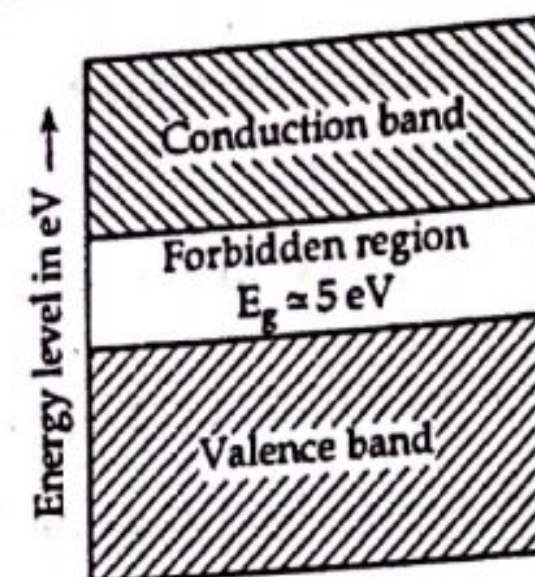


Since, the electron cannot acquire externally applied energy, conduction is impossible.

In case of conducting materials (metals), there is no forbidden gap and the conduction bands overlap each other. The orbits in the conduction band are very large. An electron in the conduction band experiences almost negligible nuclear attraction. Due to overlapping of valence and conduction bands, a slight potential difference across the conductor causes the free electrons to constitute electric current.



For semiconductors at a temperature of absolute zero (-273.15°C) the valence band is usually full and there may be no electron in the conduction band. In semiconductors material, both bands are so close (about 1 eV apart) than an electron can be lifted from the valence band to conduction band by imparting some energy to it.



For the second part

Given that;

i) 15 eV

It is an insulator material since, $E_g > 5 \text{ eV}$.

ii) 0.7 eV

It is semiconductor material since, $E_g \approx 1 \text{ eV}$.

iii) Overlapping

It is conducting material i.e., metal since there is no forbidden gap.

26. A germanium diode carries a current of 1 mA of room temperature when a forward bias of 0.15 V is applied, estimate the reverse saturation current at room temperature. [2017 S]

Solution:

Given that;

Applied forward bias voltage, $(V) = 0.15 \text{ V}$

Forward current, $(I) = 1 \text{ mA} = 0.001 \text{ A}$

Considering room temperature,

$T = 293 \text{ K}$

Volt equivalent at room temperature, $(V_T) = \frac{293}{11600} = 0.02526 \text{ V}$

Reverse saturation current, $(I_0) = ?$

We know that;

$$I = I_0 \left(e^{\frac{V}{V_T}} - 1 \right)$$

$$\text{or, } 0.001 = I_0 \left[e^{\left(\frac{0.15}{0.02526} \right)} - 1 \right]$$

$$\text{or, } I_0 = 2.6436 \times 10^{-6} \text{ A}$$

$$\therefore I_0 = 2.6436 \mu\text{A}$$

Hence, reverse saturation current is 2.643 μA .

27. A zener regulator has 9 volt zener voltage with variable load resistance as shown.

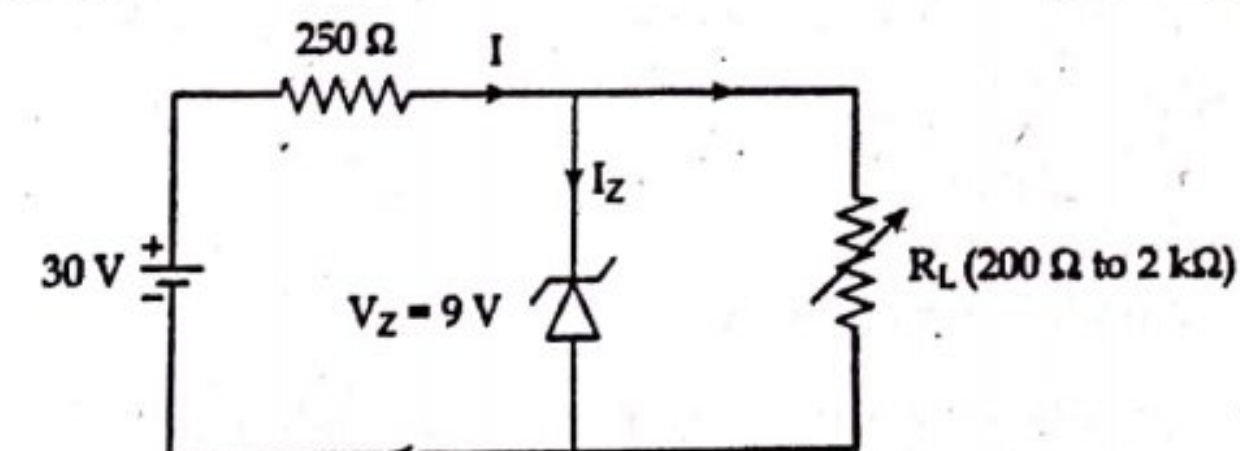
Calculate:

i) Current through series resistance

ii) Maximum and minimum load current

iii) Maximum and minimum zener current

iv) Maximum and minimum power dissipation in the zener diode. [2017 S, 2019 F]



Solution:

Given that;

$$V_Z = 9 \text{ V}$$

$$R_{L_{\min}} = 200 \Omega$$

$$R_{L_{\max}} = 2000 \Omega$$

$$V_{\text{in}} = 30 \text{ V}$$

$$R_S = 250 \Omega$$

$$\text{i) Current through series resistance, } (I_S) = \frac{V_{\text{in}} - V_Z}{R_S} = \frac{30 - 9}{250} = 0.0840 \text{ A}$$

$$\text{ii) Minimum load current, } (I_{L_{\min}}) = \frac{V_{\text{out}}}{R_{L_{\max}}} = \frac{V_Z}{R_{L_{\max}}} = \frac{9}{2000} = 0.0045 \text{ A}$$

$$\text{Maximum load current, } (I_{L_{\max}}) = \frac{V_{\text{out}}}{R_{L_{\min}}} = \frac{V_Z}{R_{L_{\min}}} = \frac{9}{200} = 0.045 \text{ A}$$

$$\text{iii) Minimum zener current, } (I_{Z_{\min}}) = I_S - I_{L_{\max}} = 0.0840 - 0.045 = 0.0393 \text{ A}$$

$$\text{Maximum zener current, } (I_{Z_{\max}}) = I_S - I_{L_{\min}} = 0.0840 - 0.0045 = 0.07950 \text{ A}$$

$$\text{iv) Minimum power dissipation in zener diode, } (P_{\min}) = V_Z - I_{Z_{\min}} = 9 \times 0.0390 = 0.351 \text{ W}$$

$$\text{Maximum power dissipation in zener diode, } (P_{\max}) = V_Z - I_{Z_{\max}} = 9 \times 0.07950 = 0.7955 \text{ W}$$

28. Find a piecewise linear model of a silicon diode with $I_S = 10^{-11} \text{ A}$ and $\eta = 1.6$ in vicinity of the operating point $I_D = 1 \text{ mA}$. [2018 S]

Solution:

Given that;

$$I_D = 1 \text{ mA} = 0.001 \text{ A}$$

$$I_S = 10^{-11} \text{ A}$$

$$\eta = 1.6$$

Considering room temperature;

$$\text{Volt equivalent at 293 K, } (V_T) = \frac{293}{11600} = 0.02526 \text{ V}$$

We know that;

$$I_D = I_S e^{\left(\frac{V_D}{\eta V_T}\right)}$$

Taking log on the both sides; we get,

$$V_D = \eta V_T \ln \left(\frac{I_D}{I_S} \right) = 1.6 \times 0.02526 \times \ln \left(\frac{0.001}{10^{-11}} \right)$$

$$= 1.6 \times 0.02526 \times 1.8420 \times 10^1$$

$$\therefore V_D = 0.745 \text{ V}$$

Also,

$$r_f = \frac{\eta V_T}{I_D}$$

$$= \frac{1.6 \times 0.745}{0.001}$$

$$\therefore r_f = 1191.18 \Omega$$

Then,

$$V_F = V_D - I_D r_f = 0.745 - 0.001 \times 1191.18$$

$$\therefore V_F = -0.446 \text{ V}$$

29. Write short notes on full wave centre tap rectifier. [2018 F]

Solution: See the definition part 1.8.2.1

30. Write short notes on full wave rectifier. [2018 S]

Solution: See the definition part 1.8.2

31. Semiconductors are neither good conductors nor good insulators but still they are used for conductivity. Justify this statement. [2018 S]

Solution:

The conducting properties of semiconductor may be altered in useful ways by deliberate, controlled introduction of impurities (doping) into the crystal structure. They can display a range of useful properties such as passing current more easily in one direction than the other, showing variable resistance and sensitivity to light or heat etc. Because electrical properties of a semiconductor material can be modified by doping or by the application of electrical fields or light, devices made from semiconductors can be used for amplification, switching and energy conversion.

Also, the semiconductors can act both as an insulator as well as a conductor based on the voltage input.

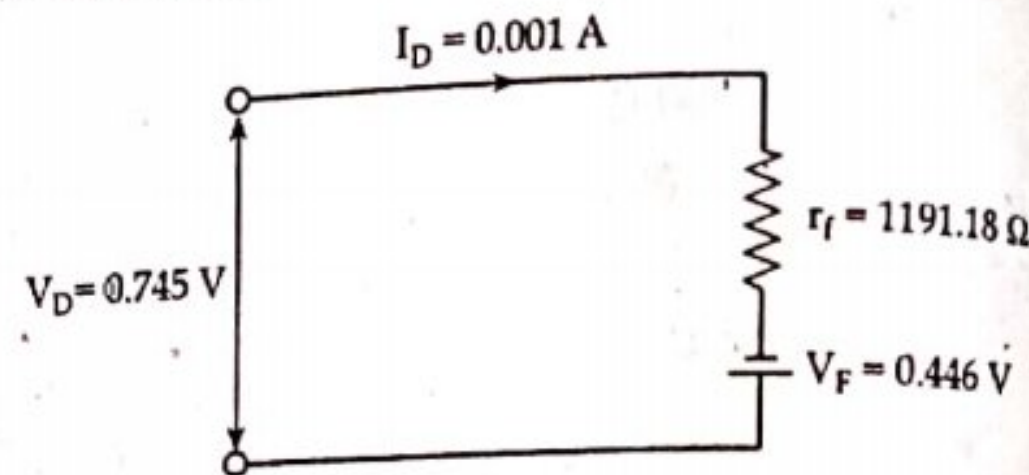
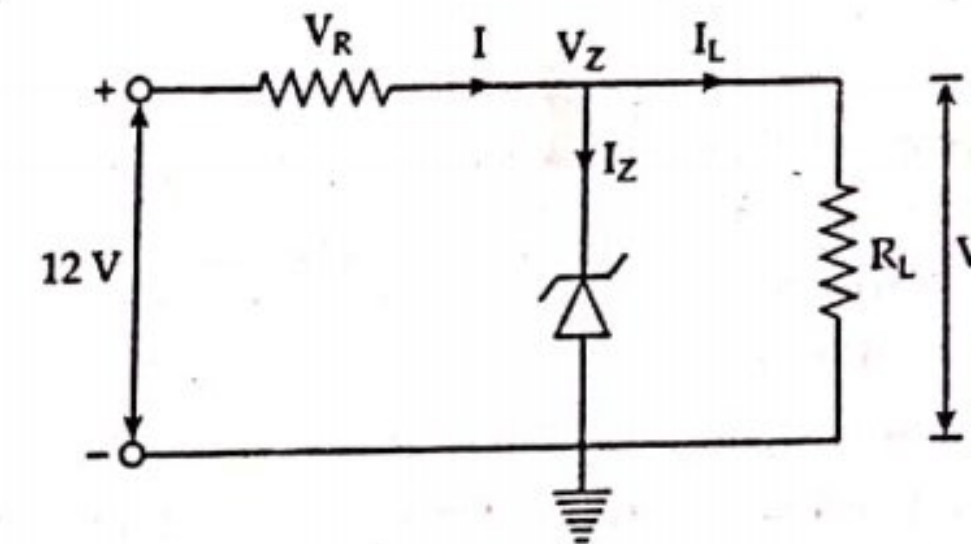


Figure: Circuit diagram of piecewise linear model

32. A 7.2 V zener diode is used in the circuit shown below and the load current is to vary from 12-100 mA. Find the value of series resistance R to maintain a voltage of 7.2 V across the load. The input voltage is constant at 12 V and the minimum zener current is 10 mA. [2018 S]



Solution:

Given that;

$$V_{in} = 12 \text{ V}$$

$$V_Z = 7.2 \text{ V}$$

$$I_{Z_{min}} = 10 \text{ mA} = 0.010 \text{ A}$$

$$R = \frac{V_{in} - V_0}{I_Z + I_L}$$

The voltage across R is to maintain constant at $12 - 7.2 = 4.8 \text{ V}$ as the load current changes from 12 to 100 mA. The minimum zener current will occur when load current is maximum.

$$\text{i.e., } R = \frac{V_{in} - V_0}{I_{Z_{min}} + I_{L_{min}}} = \frac{12 - 7.2}{0.010 + 0.10}$$

$$\therefore R = 43.62 \Omega$$

The $R = 43.62 \Omega$ is inserted in the circuit; the output voltage will remain constant over the regulating range.

33. Explain the formation of P-N junction and its operation under forward and reverse bias. [2018 F]

Solution: See the definition part 1.3.1, 1.4.1 and 1.4.2

34. Explain the operation of PN junction diode under:

- No bias
- Forward bias
- Reverse bias

[2019 F]

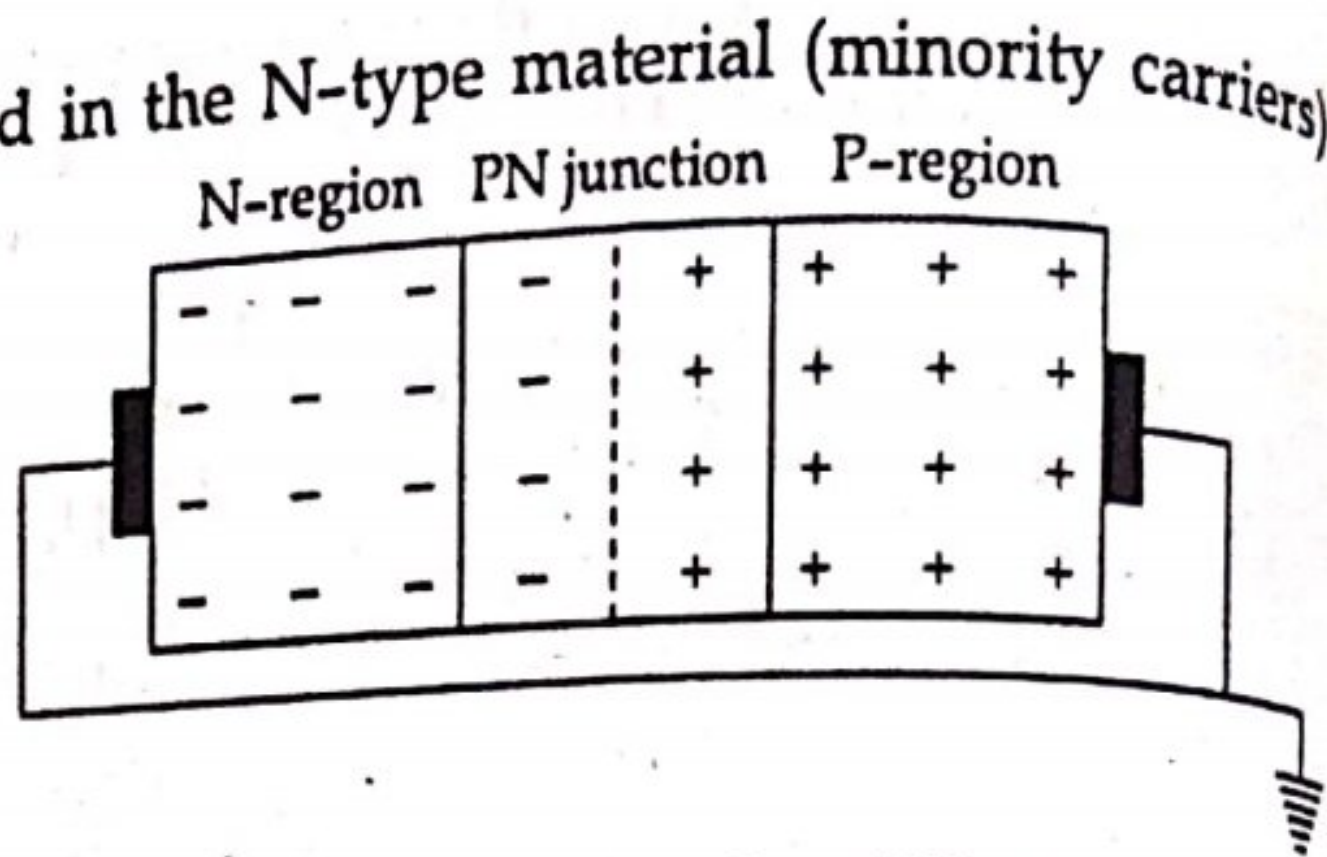
Solution:

i) No bias

When a diode is connected in a zero bias condition or no bias condition, no external potential energy is applied to the PN junction. However, if the diodes terminals are shorted together, a few holes (majority carriers) in the P-type material with enough energy to overcome the potential barrier

with move across the junction against this barrier potential. This is known as forward current.

Likewise, holes are generated in the N-type material (minority carriers) and they move across the junction in the opposite direction. This is known as the reverse current. This transfer of electrons and holes back and forth across the PN junction is shown.



The potential barrier that now exists discourages the diffusion of any more majority carriers across the junction. However, the potential barrier helps minority carriers (few free electrons in the P-region and few holes in the N-region) to drift across the junction. Then, an equilibrium or balance will be established when the majority carriers are equal and both moving in opposite direction, so that the net result is zero current flowing in the circuit. When this occurs, the junction is said to be in the state of dynamic equilibrium.

ii) Forward bias

See the definition part 1.4.1

iii) Reverse bias

See the definition part 1.4.2

35. What is a rectifier? Why are bridge rectifier circuits preferred over centre tapped rectifier? Explain the circuit diagram and waveforms for full wave bridge rectifier. [2019 F]

Solution:

Rectifier

See the definition part 1.8 (ii)

For the second part

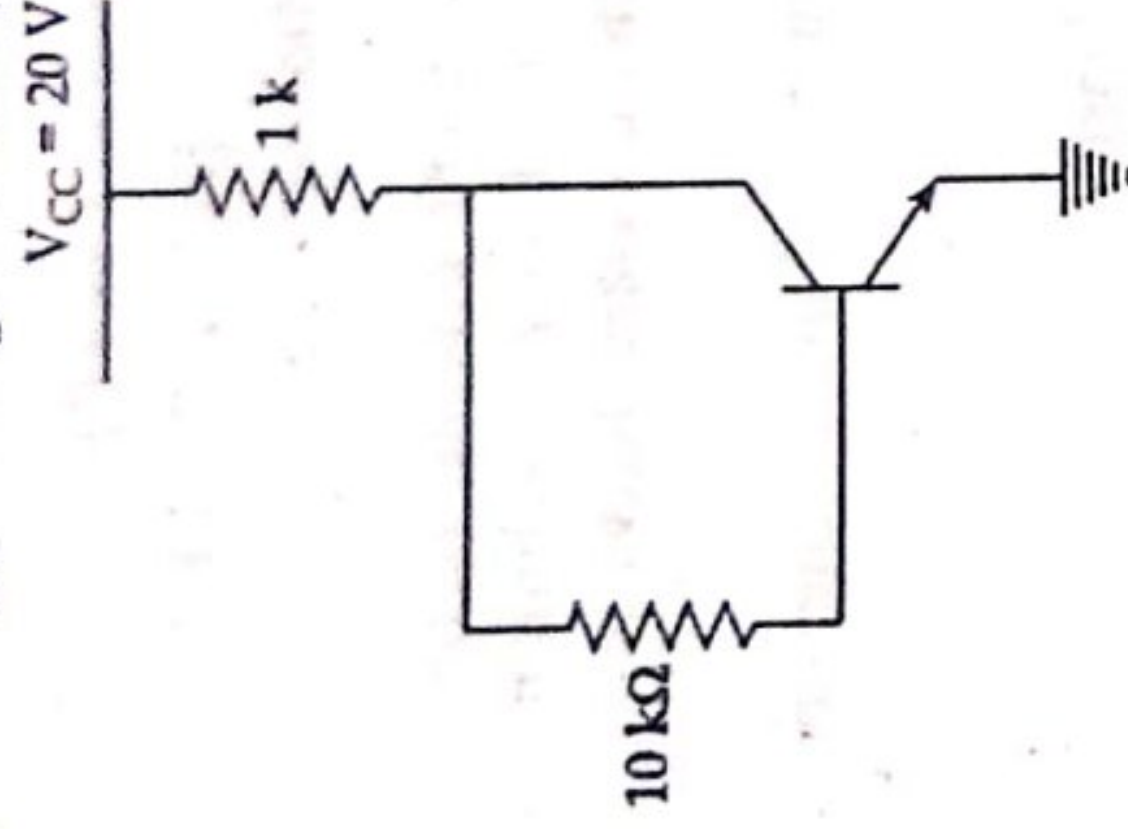
Bridge rectifier circuits are preferred over center tapped rectifier because the peak inverse voltage (PIV) ratings of the diodes in bridge rectifier is half than that of needed in a center tapped full wave rectifiers. The transformer utilization factor (TUF) is also more in bridge rectifier as compared to the centre tapped full wave rectifier. Also, the size of the transformer used in bridge rectifier is smaller than that in the center tapped rectifier.

No center tapping transformer are needed in bridge rectifier. Due to high PIV of bridge rectifier, they are suitable for high voltage applications.

For circuit diagram and waveforms for full wave bridge rectifier
See the definition part 1.8.2.2

2.7 SOLVED BOARD EXAMINATION PROBLEMS

1. For the silicon transistor biased by feedback resistor method, determine the operating points given that $\beta = 100$. [2013 F, 2013 S]



Solution:

Given that;

$$\beta = 100$$

$$V_{CC} = 20 \text{ V}$$

$$R_E = 10 \text{ k}\Omega = 10000 \Omega$$

$$R_C = 1 \text{ k}\Omega = 1000 \Omega$$

Applying K.V.L. in input side; we have,

$$V_{CC} = (I_C + I_B)R_C + I_B R_B - V_{BE}$$

$$\text{or, } V_{CC} = (1 + \beta)R_C I_B + I_B R_B + V_{BE}$$

$$\text{or, } I_B = \frac{V_{CC} - V_{BE}}{(1 + \beta)R_C + R_B}$$

$$\text{or, } I_B = \frac{20 - 0.7}{(1 + 100)1000 + 10000}$$

$$\therefore I_B = 0.00017387 \text{ A} = 0.17387 \text{ mA}$$

Now,

$$\text{Collector current, } I_C = \beta I_B$$

$$= 100 \times 0.00017387 = 0.017387 \text{ A}$$

Applying K.V.L. to output side; we have,

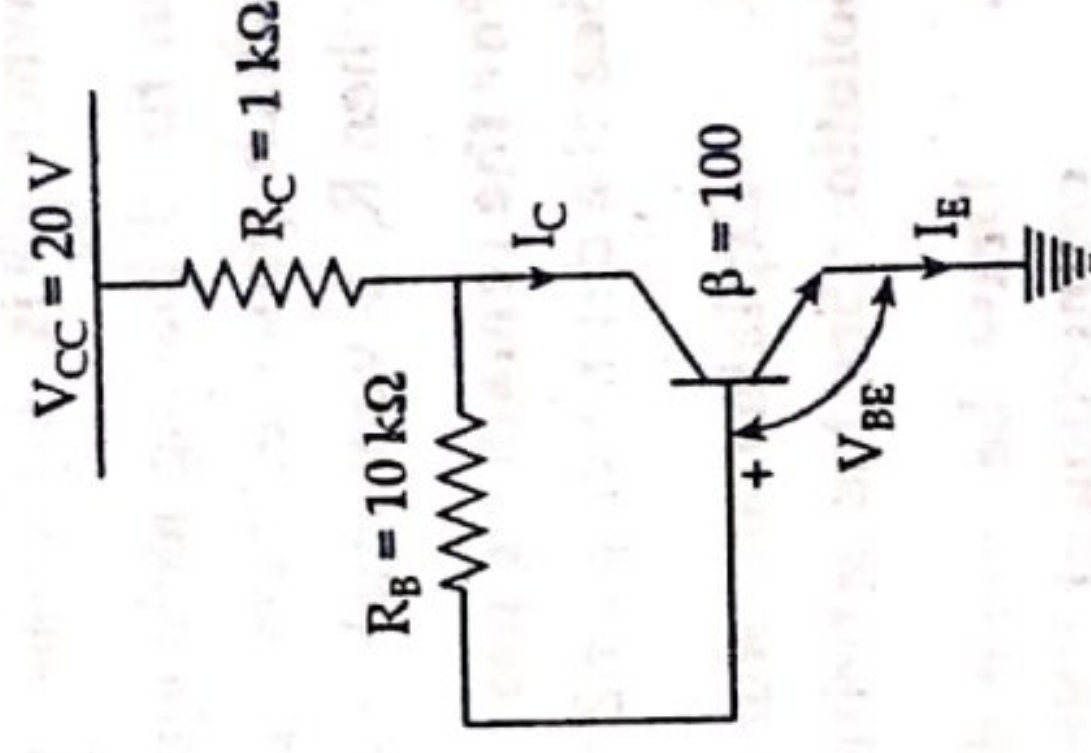
$$V_{CE} = V_{CC} - (I_B + I_C)R_C$$

$$\text{or, } V_{CE} = 20 - (0.00017387 + 0.017387) \times 1000$$

$$\text{or, } V_{CE} = 2.439 \text{ V}$$

Thus,

$$\begin{aligned} \text{Operating point } Q(V_{CE}, I_C) &= (2.439 \text{ V}, 0.017387 \text{ A}) \\ &= (2.439 \text{ V}, 17.387 \text{ mA}) \end{aligned}$$



2. Draw the circuit diagram of CE connection of a transistor. Also, draw its input and output characteristics.
[2013 F, 2015 S, 2016 F, 2016 S, 2017 S, 2018 F]

Solution: See the definition part 2.3.2, 2.3.2.1 and 2.3.2.2

3. Difference between BJT vs FET.
[2013 F, 2015 F, 2016 S, 2017 F, 2017 S, 2018 F]

Solution: See the definition part 2.5

4. What are the different biasing techniques used in amplifier? Among them which biasing technique is most preferred and why? Explain with necessary derivation. [2013 F, 2017 S]

Solution:

Different biasing techniques used in amplifier are;

- Fixed biasing
- Self biasing
- Collector-to-base biasing

Among them self-biasing technique is most preferred because in self-bias when $\frac{R_{th}}{R_E}$ is very small, stability, $S \approx 1$; which provides good stability. But in fixed biasing technique, β is normally a large quantity and $S = 1 + \beta$ which provides very poor stability. In collector to base-bias method, when R_C is very small, $S \approx 1 + \beta$ which is equal to that of fixed bias.

For the remaining part

See the definition part 2.6.3

5. Explain transistor as an amplifier. [2014 F, 2015 S, 2017 F, 2018 F]

Solution: See the definition part 2.4.1

6. Draw the circuit diagram, input and output characteristics of CE connection of PNP transistor. [2014 S]

Solution:

In common emitter configuration of PNP transistor, emitter is the terminal common to both the input side and output side. The signal to be amplified between base and emitter forming the input circuit while the amplified output voltage is developed across load impedance in the collector-to-emitter forming

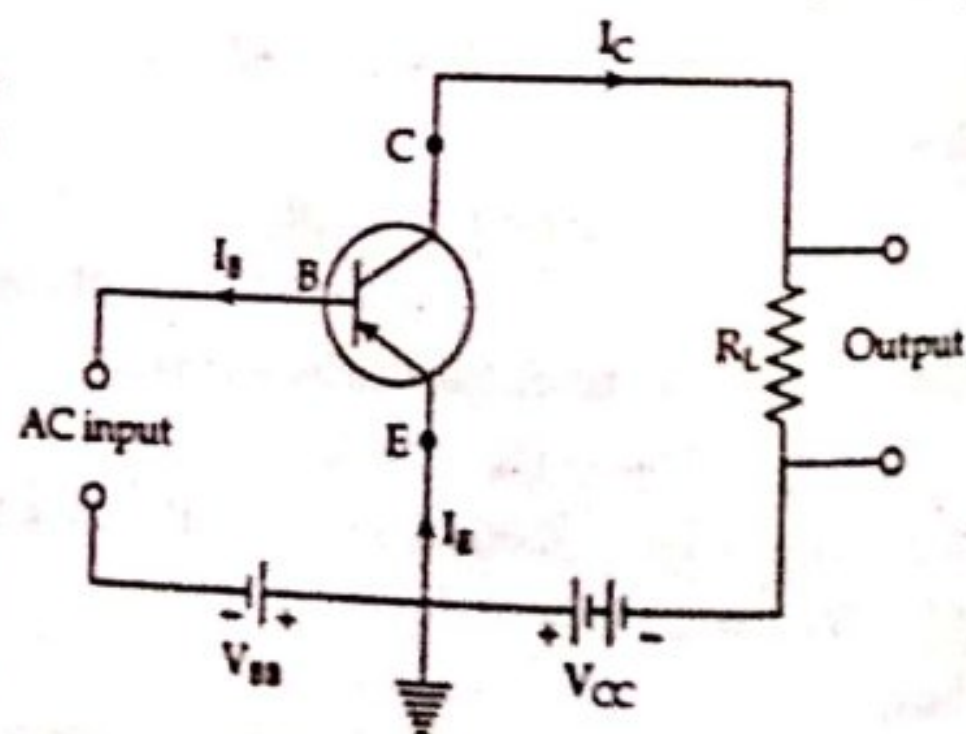


Figure: CE PNP transistor configuration

the output circuit.

Input and output characteristics of PNP transistor is same as that of NPN transistor.

7. Why transistor is called bipolar? [2014 S]

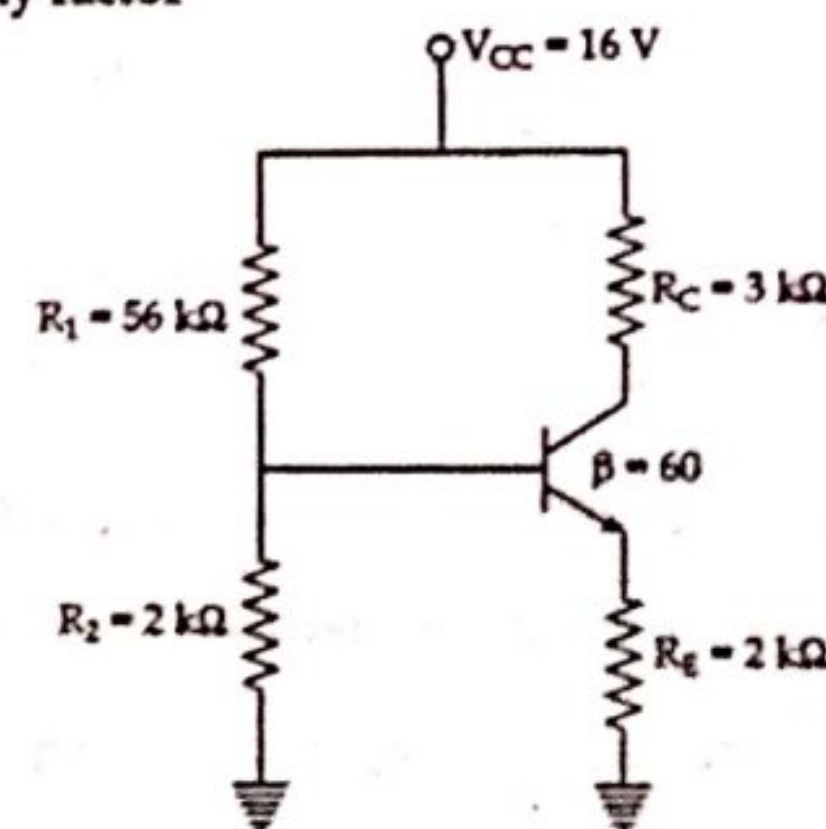
Solution:

Transistor is called bipolar because the transistor operation is carried out by two types of charge carriers (majority and minority carriers).

8. In common emitter (CE) germanium transistor amplifier self-bias is used. As given $V_{CC} = 16\text{ V}$, $R_C = 3\text{ k}\Omega$, $R_E = 2\text{ k}\Omega$, $R_1 = 56\text{ k}\Omega$, $R_2 = 20\text{ k}\Omega$ and $\beta = 60$. Calculate:

- operating point
- stability factor

[2014 S]



Solution:

Given that;

$$\begin{aligned} V_{CC} &= 16\text{ V} \\ R_C &= 3\text{ k}\Omega \\ R_E &= 2\text{ k}\Omega \\ R_1 &= 56\text{ k}\Omega \\ R_2 &= 20\text{ k}\Omega \\ \beta &= 60 \end{aligned}$$

Now,

$$R_{th} = \frac{R_1 \times R_2}{R_1 + R_2} = \frac{56 \times 20}{56 + 20} = 14.74\text{ k}\Omega$$

$$\begin{aligned} \text{i) Stability factor, } (S) &= (1 + \beta) \frac{1 + \frac{R_{th}}{R_E}}{(1 + \beta + \frac{R_{th}}{R_E})} \\ &= (1 + 60) \frac{1 + \frac{14.74}{2}}{(1 + 60 + \frac{14.74}{2})} = 7.47 \end{aligned}$$

ii) Here,

$$V_{th} = \frac{R_2}{R_1 + R_2} \times V_{CC} = \frac{20000}{56000 + 20000} \times 16 = 4.210 \text{ V}$$

Applying K.V.L. around the closed base circuit; we have,

$$V_{th} = I_B R_{th} + V_{BE} + I_E R_E$$

$$= I_B R_{th} + V_{BE} + (1 + \beta) I_B R_E \quad [\because I_E = (1 + \beta) I_B]$$

$$\text{or, } I_B = \frac{V_{th} - V_{BE}}{R_{th} + R_E(1 + \beta)} = \frac{4.210 - 0.3}{[14.74 + 2(1 + 60)] \times 1000}$$

$$\therefore I_B = 0.000029 \text{ A} = 0.02859 \text{ mA}$$

Thus,

$$\text{Collector current, } (I_C) = \beta I_B = 60 \times 0.000029 = 0.001740 \text{ A}$$

$$\text{Emitter current, } (I_E) = I_B + I_C = 0.000029 + 0.001740$$

$$= 0.001769 \text{ A} = 1.769 \text{ mA}$$

Applying K.V.L. to output loop, we have,

$$V_{CE} = V_{CC} - I_C R_C - I_E R_E$$

$$\text{or, } V_{CE} = 16 - (0.001740 \times 3000) - (0.001769 \times 2000)$$

$$\therefore V_{CE} = 7.242 \text{ V}$$

Thus,

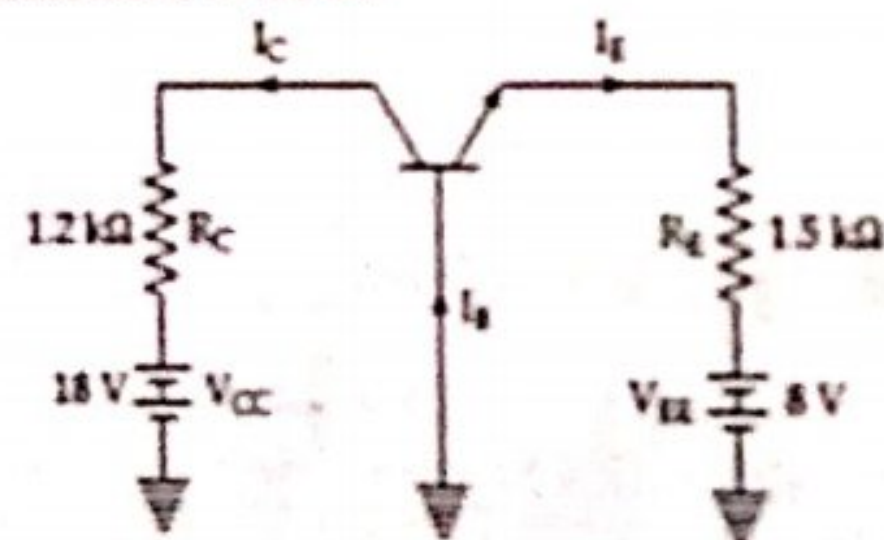
$$\text{Operating point } (V_{CEQ}, I_{CQ}) = (7.242 \text{ V}, 1.740 \text{ mA})$$

9. Why common emitter (CE) configuration is preferred over other configurations?
[2014 F, 2014 S, 2016 F, 2016 S, 2017 S, 2018 F, 2018 S]

Solution:

Common emitter configuration is preferred over other configuration as an amplifier because its current, voltage and power gains are quite high and the ratio of output impedance and input impedances is quite moderate.

10. For the common base circuit shown, determine I_C and V_{CE} . Assume transistor to be Si. [2014 F]



Solution:

For silicon transistor,

$$V_{BE} = 0.7 \text{ V}$$

Applying K.V.L. to the emitter-side loop, we have,

$$V_{EE} = I_E R_E + V_{BE}$$

$$\text{or, } I_E = \frac{V_{EE} - V_{BE}}{R_E}$$

$$\text{or, } I_E = \frac{8 - 0.7}{1.5} = 4.87 \text{ mA}$$

$$\therefore I_E \approx I_C = 4.87 \text{ mA}$$

Again, applying K.V.L. to the collector-side loop; we have,

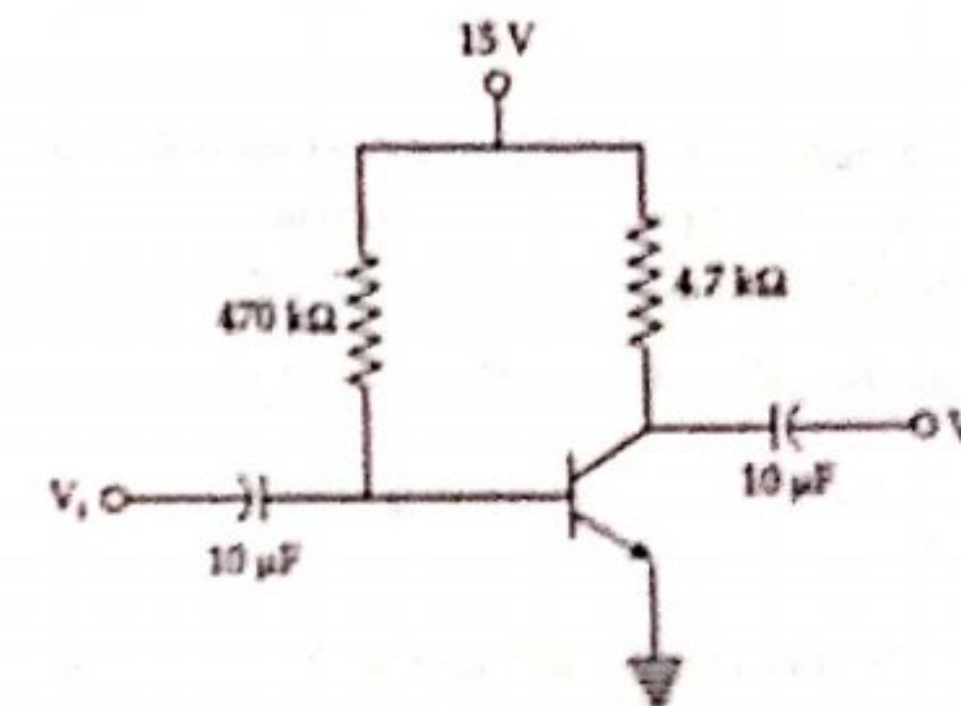
$$V_{CC} = I_C R_C + V_{CE}$$

$$\text{or, } V_{CE} = V_{CC} - I_C R_C$$

$$\text{or, } V_{CE} = 18 - 4.87 \times 1.2$$

$$\therefore V_{CE} = 12.16 \text{ V}$$

11. Determine the I_B , I_C , V_B , V_E , V_C , V_{CE} and V_{CB} in the given BJT circuit using DC analysis. [2015 F]



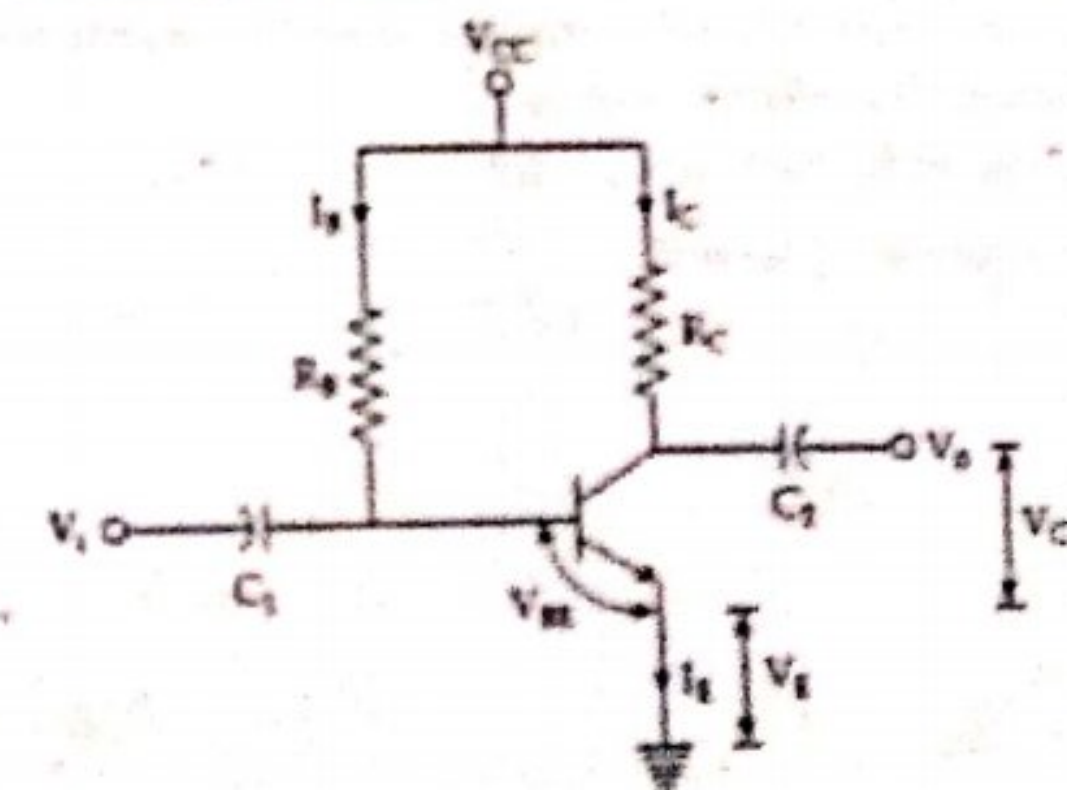
Solution:

Let, transistor used is silicon; then,

$$V_{BE} = 0.7 \text{ V}$$

$$\beta = 70$$

Redrawing the circuit; we have,



$$I_B = \frac{V_{CC} - V_{BE}}{R_B} = \frac{15 - 0.7}{470 \times 10^3} = 0.000030 \text{ A}$$

- i) $I_B = 0.000030 \text{ A}$
- ii) Collector current, $(I_C) = \beta I_B = 70 \times 0.000030 = 0.00210 \text{ A}$
- iii) Applying K.V.L. for output loop; we have,

$$V_{CC} = I_C R_C + V_{CE} + I_E R_E$$
or, $15 = 0.00210 \times 4.7 \times 10^3 + V_{CE} + 0$
 $\therefore V_{CE} = 5.130$
- iv) Collector voltage, $(V_C) = V_{CC} - I_C R_C$
 $= 15 - 0.00210 \times 4.7 \times 10^3$
 $= 5.130 \text{ V}$
- v) $V_B = V_{CC} - I_B R_B = 15 - 0.000030 \times 470 \times 10^3 = 0.90 \text{ V}$
- vi) $V_E = I_E R_E = 0 \text{ V}$
- vii) $V_{CB} = V_{CC} - I_C R_C = V_C = 5.130 \text{ V}$

12. State input/output resistance, ac/dc current gain of common emitter transistor amplifier configuration. [2015 S]

Solution: See the definition part 2.3.2.1 and 2.3.2.2

DC current gain is the ratio of the collector current to the base current. It is denoted by β .

$$\text{i.e., } \beta = \frac{I_C}{I_B}$$

AC current gain is defined as the ratio of the change in collector current to the change in base current at constant collector voltage. It is denoted by β_{ac} or β_0 .

$$\text{i.e., } \beta_0 = \left. \frac{\Delta I_C}{\Delta I_B} \right|_{V_C = \text{Constant}}$$

13. With the help of a characteristic curve, describe the different regions of operation of a bipolar junction transistor. Show the relationship between α and β . [2015 F]

Solution: See the definition part 2.3.1.2

Relationship between α and β

We have,

$$\beta = \frac{\Delta I_C}{\Delta I_B}$$

$$\text{and, } \alpha = \frac{\Delta I_C}{\Delta I_E}$$

$$\text{or, } \alpha = \frac{\Delta I_C}{\Delta I_C + \Delta I_B}$$

$$[\because \Delta I_E = \Delta I_C + \Delta I_B]$$

$$\text{or, } \frac{1}{\alpha} = \frac{\Delta I_C + \Delta I_B}{\Delta I_C}$$

$$\text{or, } \frac{1}{\alpha} = 1 + \frac{\Delta I_B}{\Delta I_C} = 1 + \frac{1}{\beta} = \frac{1 + \beta}{\beta}$$

$$\therefore \alpha = \frac{\beta}{1 + \beta}$$

$$\text{and, } \beta = \frac{\alpha}{1 - \alpha}$$

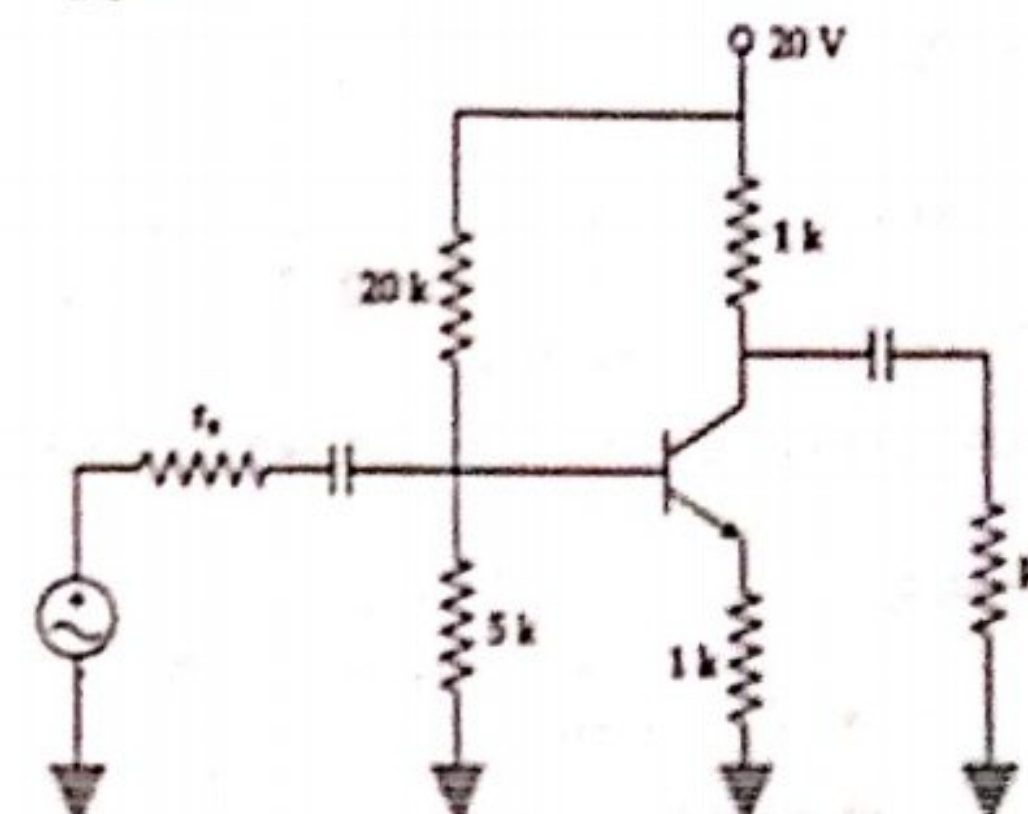
i.e., small variation in α corresponds to a large variation in β .

14. Briefly explain the working of transistor as a switch.

[2015 F, 2017 F, 2018 F, 2019 F]

Solution: See the definition part 2.4.2

15. In the given circuit, silicon transistor with $\beta = 100$ is biased by voltage divider method. Draw the dc load line and determine the operating point. [2016 F]



Solution:

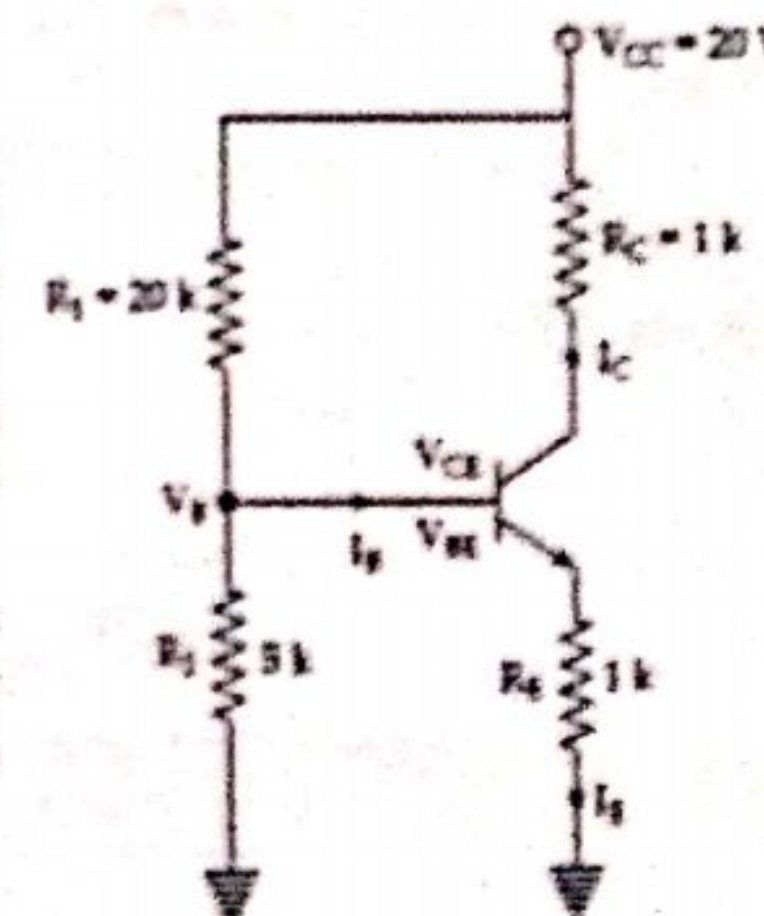
Given that;

$$\beta = 100$$

All capacitors are opened because capacitor behave as infinite resistance for dc and all ac voltage sources are grounded for finding operating point. The equivalent dc circuit is shown aside.

Now, applying Thevenin's theorem to the given potential divider network, we have Thevenin's resistance,

$$R_{th} = R_1 \parallel R_2$$



$$= \frac{R_1 \times R_2}{R_1 + R_2} = \frac{20 \times 5}{20 + 5}$$

$$= 4000 \Omega$$

$$\text{Thevenin's voltage, } (V_{th}) = \left(\frac{R_2}{R_1 + R_2} \right) \times V_{CC}$$

$$= \left(\frac{5}{20 + 5} \right) \times 20 = 4 \text{ V}$$

Applying K.V.L. to the input loop; we have,

$$V_{th} = I_B R_{th} + V_{BE} + I_E R_E$$

$$\text{or, } V_{th} = I_B R_{th} + V_{BE} + (1 + \beta) I_B R_E$$

$$\text{or, } I_B = \frac{V_{th} - V_{BE}}{R_{th} + (1 + \beta) R_E}$$

$$\text{or, } I_B = \frac{4 - 0.7}{4000 + (1 + 100) \times 1000}$$

$$\therefore I_B = 0.000032 \text{ A}$$

Then,

$$\text{Collector current, } (I_C) = \beta I_B$$

$$= 100 \times 0.000032$$

$$= 0.0032 \text{ A}$$

$$\text{Collector voltage, } (V_C) = V_{CC} - I_C R_C = 20 - 0.0032 \times 1000$$

$$= 16.86 \text{ V}$$

$$\text{Emitter voltage, } (V_E) = I_E R_E = (I_C + I_B) R_E$$

$$= (0.0032 + 0.000032) \times 1000$$

$$= 3.232 \text{ V}$$

Applying K.V.L. to output loop; we have,

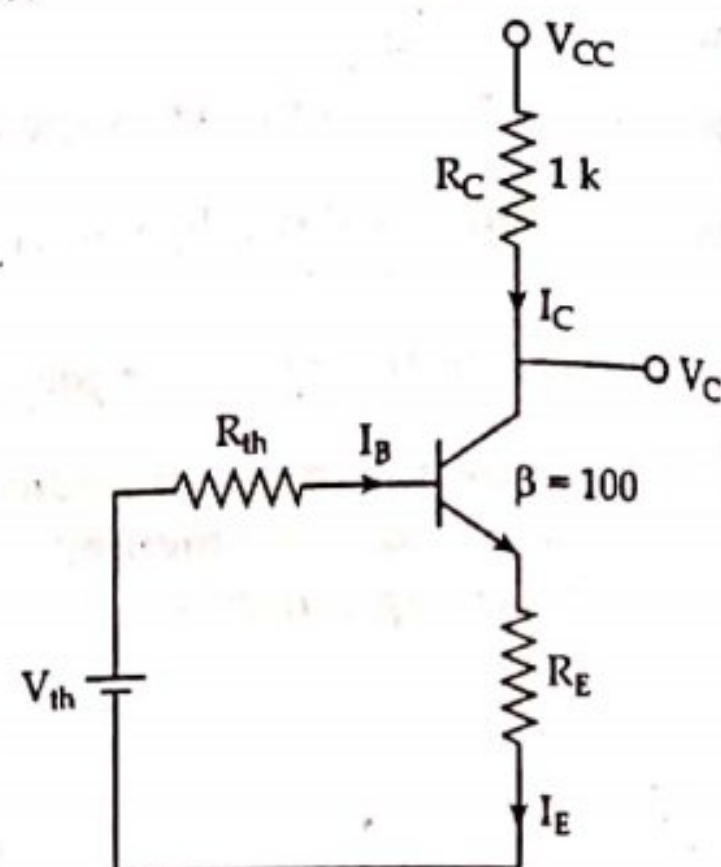
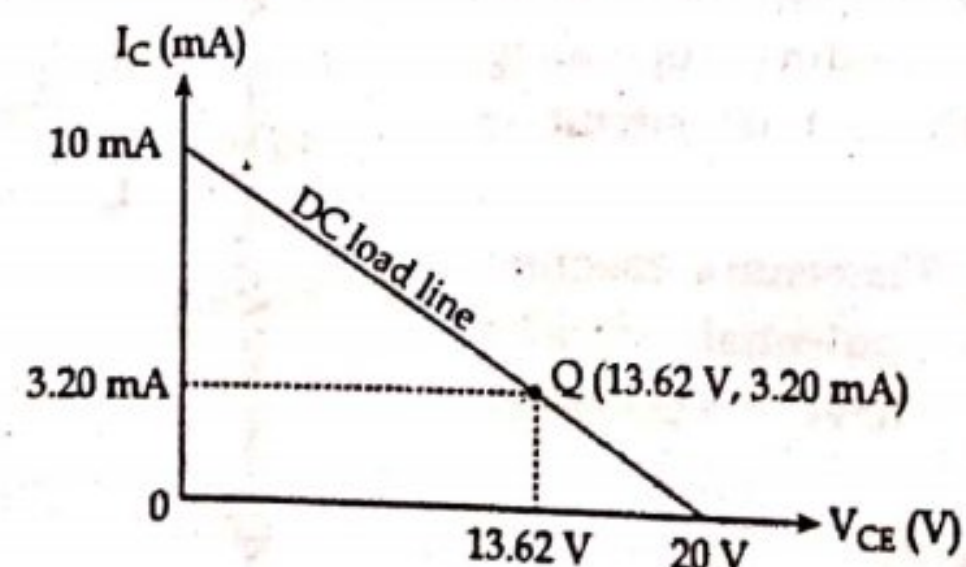
$$V_{CE} = V_C - V_E = 16.86 - 3.232 = 13.628 \text{ V}$$

Hence,

$$\text{Operating point } (V_{CEQ}, I_{CQ}) = (13.628 \text{ V}, 0.0032 \text{ A})$$

$$= (13.628 \text{ V}, 3.20 \text{ mA})$$

For $V_{CE} = 0 \text{ V}$;



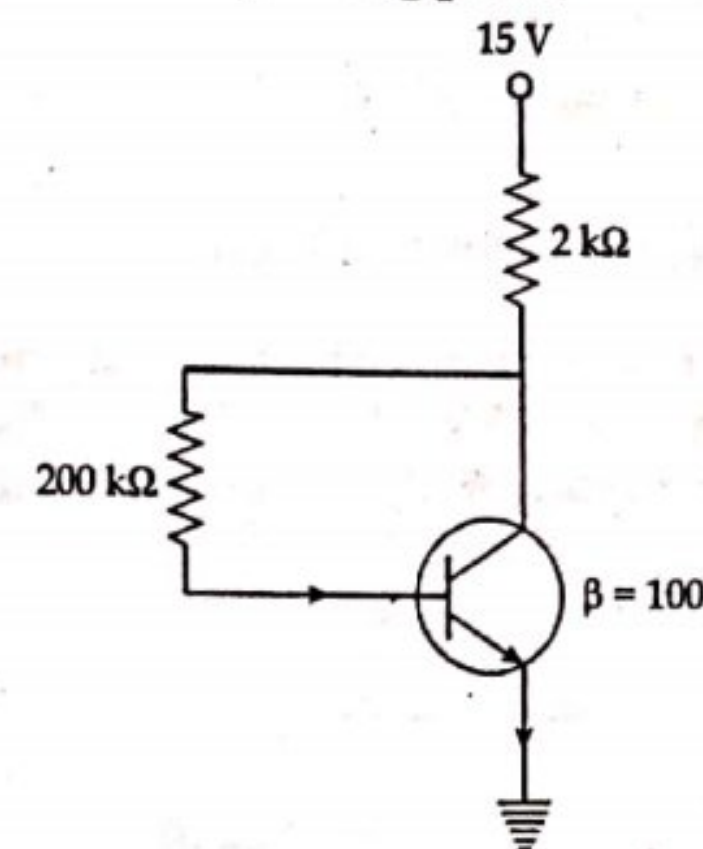
$$I_C = \frac{V_{CC}}{R_C + R_E}$$

$$\text{or, } I_C = \frac{20}{1000 + 1000} = 10 \text{ mA}$$

For $I_C = 0 \text{ A}$;

$$V_{CE} = V_{CC} = 20 \text{ V}$$

16. A silicon transistor biased by feedback resistor method is shown below. Determine the operating point. [2017 S, 2018 S]



Solution:

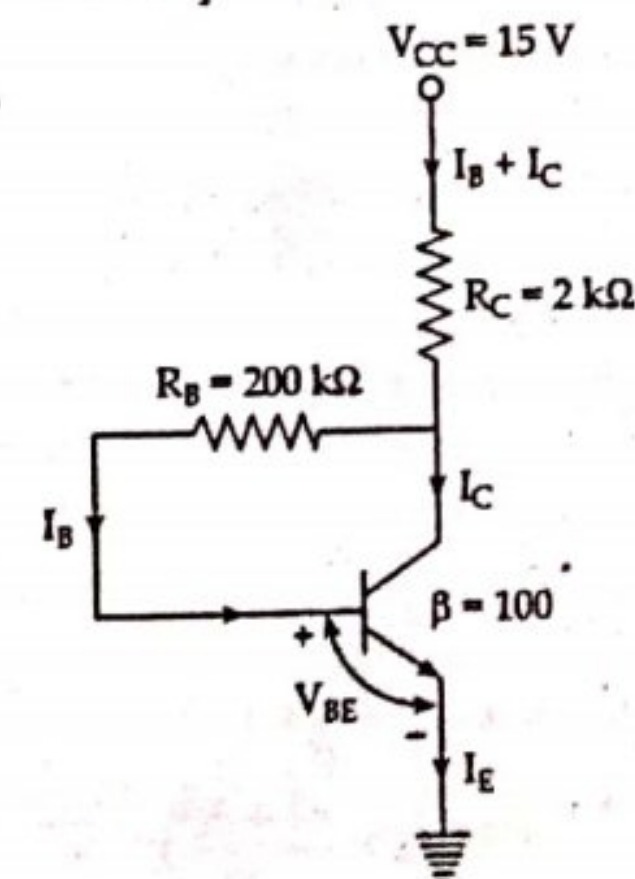
Applying K.V.L. in input side; we have,

$$V_{CC} = (I_C + I_B) R_C + I_B R_B + V_{BE}$$

$$\text{or, } V_{CC} = (1 + \beta) R_C I_B + I_B R_B + V_{BE}$$

$$\text{or, } I_B = \frac{V_{CC} - V_{BE}}{(1 + \beta) R_C + R_B}$$

$$\therefore I_B = \frac{15 - 0.7}{[(1 + 100) \times 2 + 200] \times 10^3}$$



$$= 0.000036 \text{ A}$$

$$= 0.0356 \text{ mA}$$

Now,

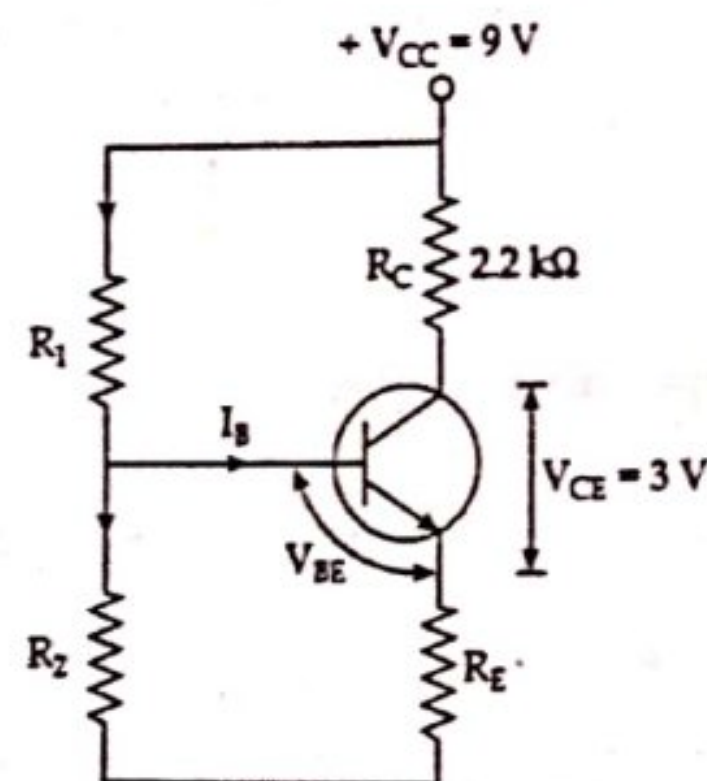
$$\begin{aligned} \text{Collector current, } (I_C) &= \beta I_B \\ &= 100 \times 0.000036 \\ &= 0.00356 \text{ A} \end{aligned}$$

Applying K.V.L. to output side; we have,

$$\begin{aligned} V_{CE} &= V_{CC} - (I_B + I_C)R_C \\ \text{or, } V_{CE} &= 15 - (0.000036 + 0.00356) \times 2 \times 10^3 \\ \therefore V_{CE} &= 7.808 \text{ V} \end{aligned}$$

Hence, operating point is Q(7.808 V, 0.00356 A).

17. In the circuit shown below, the operating point is chosen such that $I_C = 2 \text{ mA}$, $V_{CE} = 3 \text{ V}$, if $R_C = 2.2 \text{ k}\Omega$, $V_{CC} = 9 \text{ V}$ and $\beta = 50$; determine the values of R_1 , R_2 and R_E . Take $V_{BE} = 0.3 \text{ V}$ and $I_1 = 10I_B$. [2017 F]



Solution:

Given that;

$$I_C = 2 \text{ mA}$$

$$V_{CE} = 3 \text{ V}$$

$$R_C = 2.2 \text{ k}\Omega$$

$$V_{CC} = 9 \text{ V}$$

$$\beta = 50$$

$$I_1 = 10I_B$$

$$V_{BE} = 0.3 \text{ V}$$

We have,

$$\begin{aligned} \text{Emitter current, } (I_E) &= I_C \times \left(\frac{1 + \beta}{\beta} \right) \\ &= 0.002 \times \left(\frac{1 + 50}{50} \right) \end{aligned}$$

$$= 0.00204 \text{ A}$$

$$= 2.040 \text{ mA}$$

Applying K.V.L. to outer loop; we have,

$$\begin{aligned} V_{CC} &= I_C R_C + V_{CE} + I_E R_E \\ \text{or, } 9 &= 0.002 \times 2.2 \times 10^3 + 3 + 0.00204 \times R_E \\ \therefore R_E &= 785 \Omega \end{aligned}$$

Also,

$$\begin{aligned} R_C &= \frac{V_{CC} - V_{CE} - V_E}{I_E} \\ \text{or, } 2.2 \times 10^3 &= \frac{9 - 3 - V_E}{0.002} \end{aligned}$$

$$\begin{aligned} \text{or, } -V_E &= 0.002 \times 2.2 \times 10^3 - 9 + 3 \\ \therefore V_E &= 1.60 \text{ V} \end{aligned}$$

Thus,

$$I_B = \frac{I_C}{\beta} = \frac{0.002}{50} = 0.000040 \text{ A} = 0.040 \text{ mA}$$

$$I_1 = 10I_B = 10 \times 0.000040 = 0.0004 \text{ A}$$

Now,

$$I_1 = \frac{V_{CC}}{R_1 + R_2}$$

$$\therefore R_1 + R_2 = \frac{V_{CC}}{I_1} = \frac{9}{0.0004} = 22500 \Omega$$

$$\begin{aligned} \text{Voltage across } R_2, (V_2) &= V_{BE} + V_E \\ &= 0.3 + I_E R_E \\ &= 0.3 + 1.60 \\ &= 1.90 \text{ V} \end{aligned}$$

$$\therefore \text{Resistance, } (R_2) = \frac{V_2}{I_1} = \frac{1.90}{0.00040} = 4750 \Omega$$

$$\text{and, } R_1 = 22500 - 4750 = 17750 \Omega = 17.75 \text{ k}\Omega$$

18. Draw the NPN transistor and explain its working principle. Also, show that $I_C = \beta I_B + (1 + \beta) I_{CBO}$. [2018 S]

Or

Explain the current flow mechanism in NPN transistor. Derive the current equation in transistor. [2014 S, 2015 S, 2016 S]

Solution:

For NPN transistor diagram

See the definition part 2.2.1

The current equations are;

$$I_E = I_B + I_C$$

$$\text{and, } I_C = \alpha I_E + I_{CBO} = \alpha(I_B + I_C) + I_{CBO}$$

$$\text{or, } I_C(1 - \alpha) = \alpha I_B + I_{CBO}$$

$$\text{or, } I_C = \left(\frac{\alpha}{1 - \alpha}\right) I_B + \left(\frac{I_{CBO}}{1 - \alpha}\right) \quad (1)$$

$$\text{and, Total collector current, } (I_C) = \beta I_B + I_{CEO} \quad (2)$$

Comparing equation (1) and (2); we have,

$$\beta = \frac{\alpha}{1 - \alpha}$$

$$\text{and, } I_{CEO} = \left(\frac{1}{1 - \alpha}\right) I_{CBO}$$

Putting value of $I_{CEO} = \left(\frac{1}{1 - \alpha}\right) I_{CBO}$ in the equation (2); we have,

$$I_C = \beta I_B + \left(\frac{1}{1 - \alpha}\right) I_{CBO}$$

$$\therefore I_C = \beta I_B + (1 + \beta) I_{CBO} \quad \left[\because \frac{1}{1 - \alpha} = 1 + \beta\right]$$

19. Write short notes on transistor biasing.

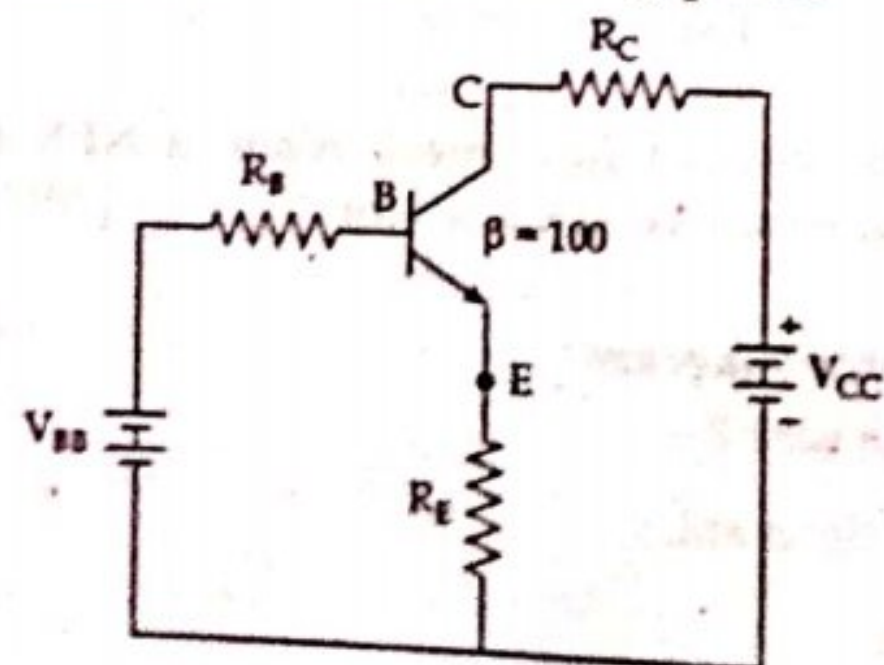
[2018 S]

Solution: See the definition part 2.6

The proper flow of zero signal collector current and the maintenance of proper collector-emitter voltage during the passage of signal is called transistor biasing. The circuit used for transistor biasing is called the biasing circuit. There is large number of circuits for biasing of a transistor. These circuits differ so as to their ability to keep the quiescent point fixed in spite of variations in transistors characteristics and also effects of temperature variations and ageing. There are three types of biasing circuit, namely;

- Fixed bias circuit
- Self bias circuit
- Collector to base bias circuit

20. For transistor circuit shown, $I_B = 50 \mu A$, $V_{BB} = 6 V$, $V_{BE} = 0.7 V$, $R_C = 2 k\Omega$, $R_E = 1 k\Omega$. Find I_E , R_B , V_{CE} if $V_{CC} = 20 V$ and $\beta = 100$. Also, draw the dc load line and locate Q-point. [2018 F, 2019 F]



Solution:

Given that;

$$I_B = 50 \mu A = 50 \times 10^{-6} A$$

$$V_{BB} = 6 V$$

$$V_{BE} = 0.7 V$$

$$R_C = 2 k\Omega = 2000 \Omega$$

$$R_E = 1 k\Omega = 1000 \Omega$$

$$V_{CC} = 20 V$$

$$\beta = 100$$

We have,

$$\begin{aligned} \text{Collector current, } (I_C) &= \beta I_B \\ &= 100 \times 50 \times 10^{-6} \\ &= 0.0050 A = 5 \text{ mA} \end{aligned}$$

$$\begin{aligned} \text{Collector voltage, } (V_C) &= V_{CC} - I_C R_C \\ &= 20 - 0.0050 \times 2000 \\ &= 10 V \end{aligned}$$

$$\begin{aligned} \text{Emitter current, } (I_E) &= I_B + I_C \\ &= 50 \times 10^{-6} + 0.0050 \\ &= 0.00505 A = 5.05 \text{ mA} \end{aligned}$$

$$\begin{aligned} \text{Emitter voltage, } (V_E) &= I_E R_E \\ &= 0.00505 \times 1000 \\ &= 5.050 V \end{aligned}$$

Now,

$$V_{CE} = V_C - V_E = 10 - 5.050 = 4.95 V$$

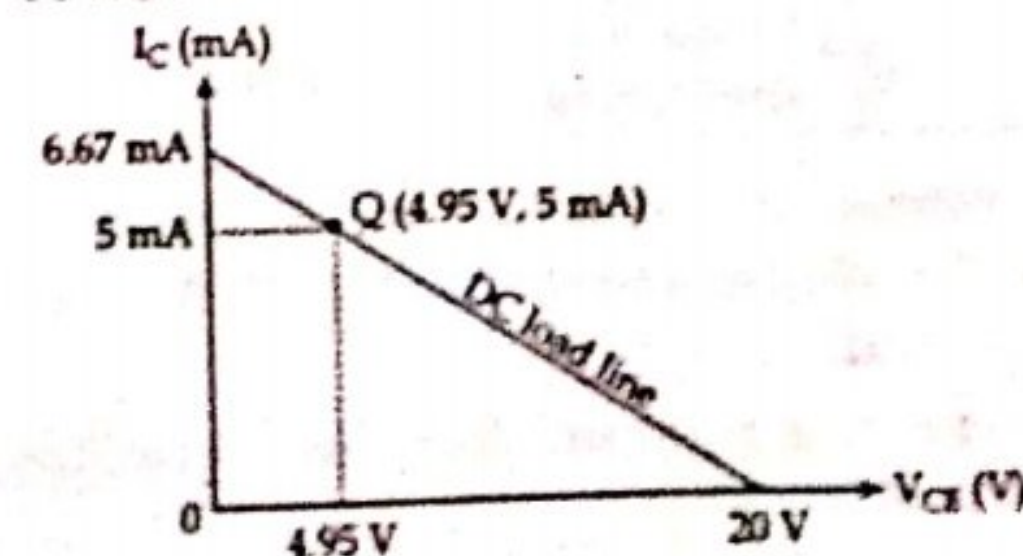
Applying K.V.L. to input loop; we have,

$$V_{th} = I_B R_{th} + V_{BE} + I_E R_E$$

$$\text{or, } V_{BB} = I_B R_B + V_{BE} + I_E R_E$$

$$\text{or, } 6 = 50 \times 10^{-6} \times R_B + 0.7 + 5.050$$

$$\text{or, } R_B = \frac{6 - 0.7 - 5.050}{50 \times 10^{-6}}$$



$$\text{or, } R_B = 5000 \Omega$$

$$\therefore R_B = 5 \text{ k}\Omega$$

Hence,

$$\text{i) } R_B = 5 \text{ k}\Omega$$

$$\text{ii) } I_E = 5.05 \text{ mA}$$

$$\text{iii) } V_{CE} = 4.95 \text{ V}$$

For $V_{CE} = 0 \text{ V}$;

$$I_C = \frac{V_{CC}}{R_C + R_E} = \frac{20}{1000 + 2000} = 0.0067 \text{ A} = 6.67 \text{ mA}$$

For $I_C = 0 \text{ A}$;

$$V_{CE} = V_{CC} = 20 \text{ V}$$

$\therefore$ Operating point or Q point (V_{CEQ}, I_{CQ}) = (4.95 V, 5 mA)

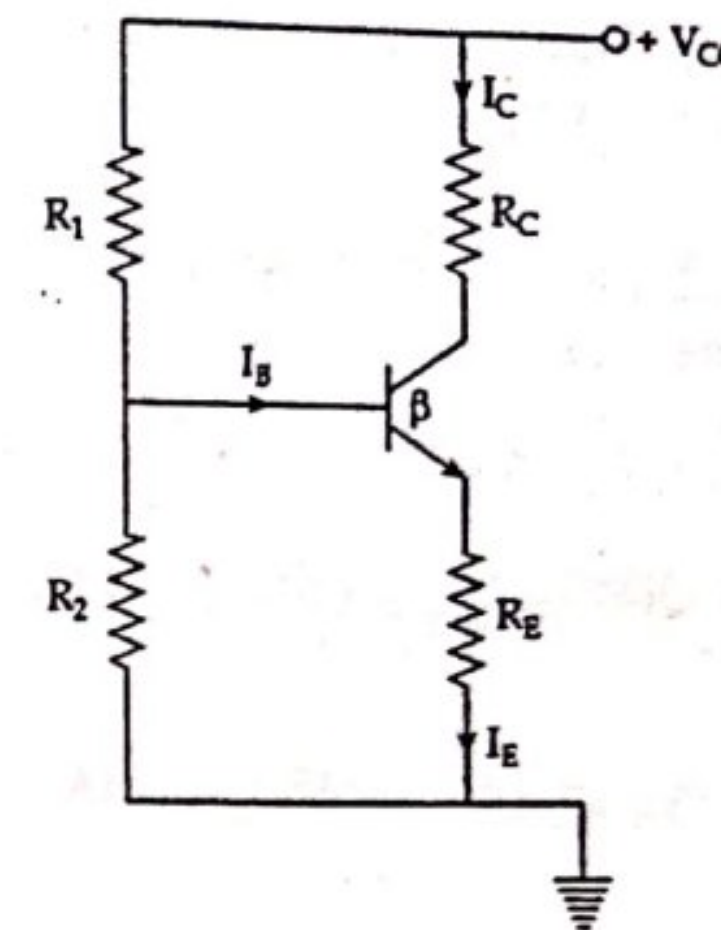
21. Compare common base, common emitter and common collector transistors in terms of input resistance, output resistance, voltage gain and applications. [2019 F]

Solution:

Configuration Characteristics	Common base	Common emitter	Common collector
Input resistance	Low (about 100 Ω)	Medium (about 800 Ω)	Very high (about 750 k Ω)
Output impedance	Very high (about 500 k Ω)	High (about 50 k Ω)	Low (about 500 Ω)
Voltage gain	About 150	About 500	Less than unity
Leak current	Very small	Very large	Very large
Current gain	Less than unit but usually more than 0.9	High (about 80)	High (about 10)
Output signal phase	In phase with input	Reverse	In phase with input
Applications	For high frequency applications	For AF applications	For impedance matching

22. For the voltage divider circuit, the following parameters are given as; $\beta = 100$, $R_1 = 60 \text{ k}\Omega$, $R_2 = 15 \text{ k}\Omega$, $R_C = 2 \text{ k}\Omega$, $R_E = 2 \text{ k}\Omega$ and $V_{CC} = 20 \text{ V}$

Find the operating point and draw the dc load line. Assume the transistor to be silicon.



Solution:

Given that;

$$R_1 = 60 \text{ k}\Omega$$

$$R_2 = 15 \text{ k}\Omega$$

$$R_C = 2 \text{ k}\Omega$$

$$R_E = 2 \text{ k}\Omega$$

$$V_{CC} = 20 \text{ V}$$

$$\beta = 100$$

Now,

$$R_{th} = \frac{R_1 \times R_2}{R_1 + R_2} = \frac{60 \times 15}{60 + 15} = 12 \text{ k}\Omega$$

$$V_{th} = \left(\frac{R_2}{R_1 + R_2} \right) \times V_{CC} = \left(\frac{15}{60 + 15} \right) \times 20 = 4 \text{ V}$$

Applying K.V.L. around the closed base circuit,

$$V_{th} = I_B R_{th} + V_{BE} + I_E R_E$$

$$\text{or, } V_{th} = I_B R_{th} + V_{BE} + (1 + \beta) I_B R_E$$

$$\text{or, } I_B = \frac{V_{th} - V_{BE}}{R_{th} + R_E(1 + \beta)}$$

$$\text{or, } I_B = \frac{4 - 0.7}{12000 + 2000(1 + 100)}$$

$$\therefore I_B = 1.542 \times 10^{-5} \text{ A} = 0.01542 \text{ mA}$$

Thus,

$$\text{Collector current, } I_C = \beta I_B = 100 \times 1.542 \times 10^{-5} = 1.542 \times 10^{-3} \text{ A}$$

$$\text{Emitter current, } I_E = I_B + I_C = 1.542 \times 10^{-5} + 1.542 \times 10^{-3} = 1.5574 \times 10^{-3} \text{ A}$$

Applying K.V.L. to output loop, we have,

$$\begin{aligned}
 V_{CE} &= V_{CC} - I_C R_C = I_E R_E \\
 &= V_C = V_E \\
 &= 20 - 1.542 \times 10^{-3} \times 2000 = 1.5574 \times 10^{-3} \times 2000 \\
 &= 20 - 3.084 = 3.148 \\
 &= 13.801 \text{ V}
 \end{aligned}$$

Thus,

$$\text{Operating point } Q(V_{CEQ}, I_{CQ}) = (13.801 \text{ V}, 1.542 \text{ mA})$$

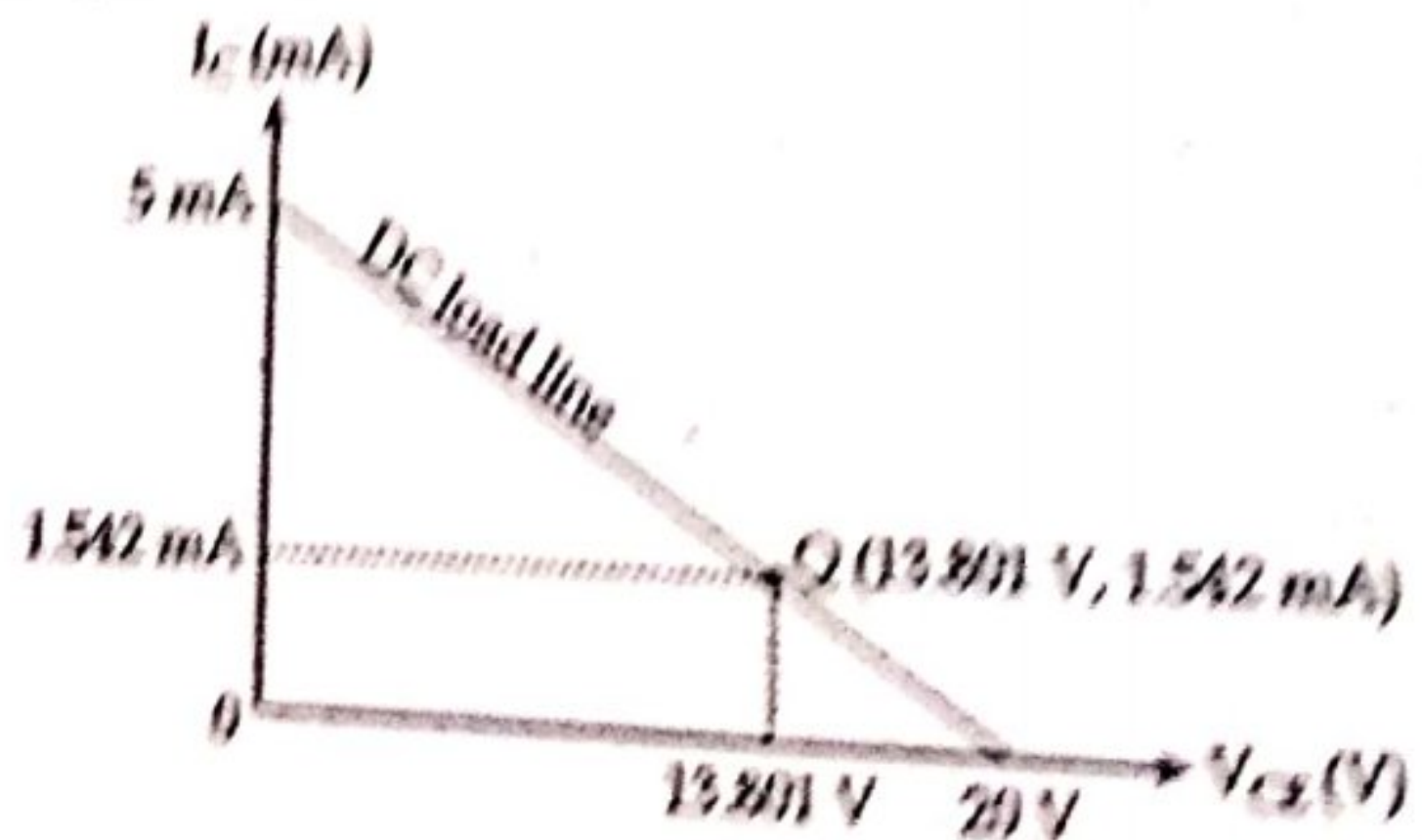
For $V_{CE} = 0 \text{ V}$;

$$I_C = \frac{V_{CC}}{R_C + R_E} = \frac{20}{2000 + 2000} = 0.005 \text{ A} = 5 \text{ mA}$$

For $I_C = 0 \text{ A}$;

$$V_{CE} = V_{CC} = 20 \text{ V}$$

DC load line diagram is;



3.7 SOLVED BOARD EXAMINATION PROBLEMS

1. Discuss the operation of summing operational amplifier. [2013 S]

Solution: See the definition part 3.5.1

2. Why is an oscillator used? What is the required condition for an oscillator? [2013 F, 2014 F]

Solution:

For the first part

Oscillators directly convert direct current (DC) from a power supply to an alternating current (AC) signal. Thus, oscillators are used to generate a repetitive waveform of fixed amplitude and frequency without any external input signal.

For the second part

See the definition part 3.6.2

3. Explain sinusoidal oscillator in brief. [2013 F, 2014 F]

Solution:

An electronic device that generates sinusoidal oscillations of desired frequency is known as a sinusoidal oscillator. Oscillator does not create energy but acts as an energy converter. It receives dc energy and changes it into ac energy of desired frequency. A sinusoidal oscillator can be realized by placing a frequency selective network in the feedback path of an amplifier (a transistor or an Op-amp). The circuit will oscillate at the frequency at which the total phase shift around the loop is zero, provided that the magnitude of the loop gain at this frequency is equal to or slightly greater than unity.

Types of sinusoidal oscillators

Sinusoidal electrical oscillations can be of two types viz; damped oscillations and undamped oscillations. The electrical oscillations whose amplitude goes on decreasing with time are called damped oscillations. The electrical system in which these oscillations are generated has losses and some energy is lost during each oscillations. Further, no means are provided to compensate for the losses and consequently the amplitude of the generated waves decreases gradually.

The electrical oscillations whose amplitude remains constant with time are called undamped oscillations. Although the electrical system in which

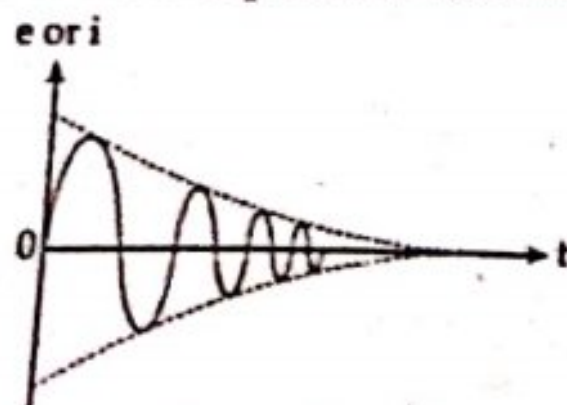


Figure: Damped oscillation

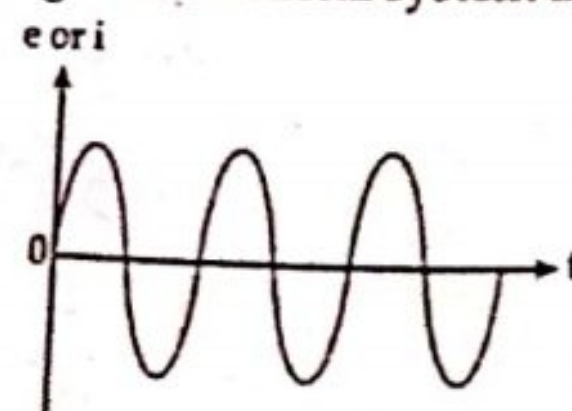


Figure: Undamped oscillation

these oscillations are being generated has also losses, but now right amount of energy is being supplied to overcome the losses. Consequently, the amplitude of the generated wave remains constant.

Types of sinusoidal or harmonic oscillators

i) Tuned circuit oscillators

These oscillators use a tuned circuit consisting of inductors and capacitors and are used to generate high frequency signals. Thus, they are also known as radio frequency oscillators. Such oscillators are Hartley and Colpitts oscillators.

ii) RC oscillators

These oscillators use resistors and capacitors and are used to generate low or audio frequency signals. Thus, they are also known as audio frequency oscillators. Such oscillators are phase shift and Wien bridge oscillators.

iii) Crystal oscillators

These oscillators use quartz crystal and are used to generate highly stabilized output signal with frequencies up to 10 MHz. The pierce oscillator is an example of a crystal oscillator.

iv) Negative-resistance oscillators

These oscillators use negative resistance characteristic of the devices such as tunnel diode. A tunnel diode oscillator is an example of negative resistance oscillator.

For having a sinusoidal oscillator, we require an amplifier with positive feedback. The idea is to use the feedback signal in place of an input signal. If the loop gain and phase are correct, there will be an output signal even though there is no external input signal. In other words, an oscillator is an amplifier that is modified by positive feedback to supply its own input signal.

4. Derive an expression for gain for non-inverting mode of an operational amplifier. [2013 S, 2016 F, 2018 S, 2019 F]

Solution: See the definition part 3.4.1

5. Derive an expression for gain for negative feedback amplifier. [2013 S, 2014 S, 2016 S, 2017 S, 2018 S]

Solution: See the definition part 3.2

6. Deduce the expression for the output voltage in inverting amplifier. [2013 F, 2014 F, 2017 F]

Solution: See the definition part 3.4.2

7. What are the main characteristics of an ideal operational amplifier? [2013 F, 2016 S, 2017 S, 2018 F]

Solution:

The main characteristics of op-amp are;

- i) Infinite open loop gain
- ii) Infinite input impedance
- iii) Zero output impedance
- iv) Perfect balance
- v) Infinite frequency bandwidth
- vi) Infinite slew rate
- vii) Infinite CMRR
- viii) Nil drift of characteristics with temperature

8. Write short notes on Barkhausen criteria for oscillation.

[2014 F, 2014 S, 2017 F, 2019 F]

Solution: See the definition part 3.6.2

9. What do you mean by ideal Op-amp?

[2014 F]

Solution:

An ideal op-amp is an amplifier with infinite input impedance, infinite loop gain, zero output impedance, infinite bandwidth and zero noise. It has positive and negative inputs, which allow circuits that use feedback to achieve a wide range of functions.

10. Define positive feedback in amplifier.

[2014 F, 2017 F]

Solution: See the definition part 3.6.1

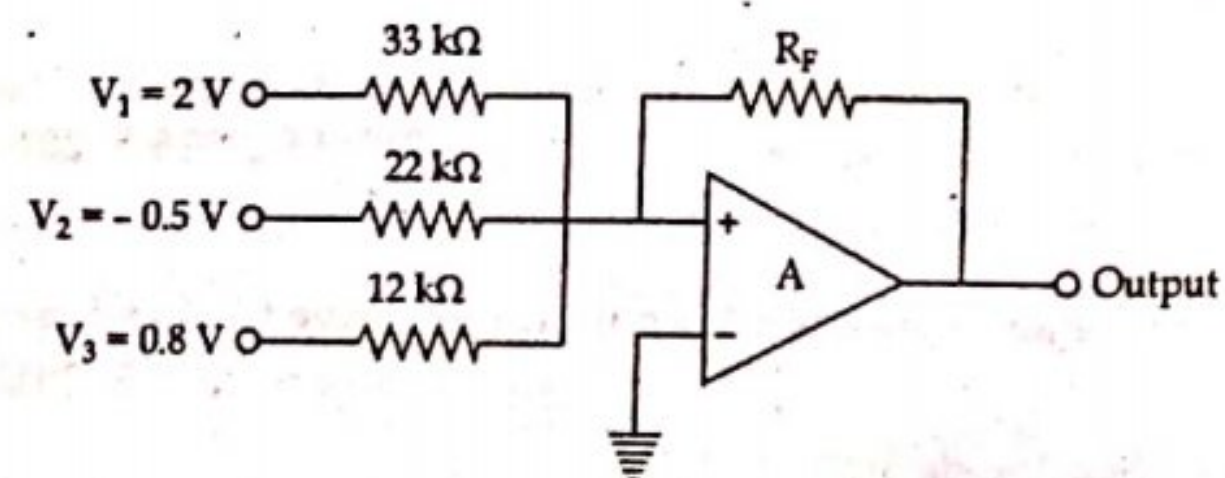
11. What is operational amplifier? Discuss the operation of differentiator in operational amplifier.

[2014 S]

Solution: See the definition part 3.3 and 3.5.4

12. Define virtual ground and CMRR of an op-amp. For the given operational amplifier circuit, calculate the output voltage for $R_f = 68 \text{ k}\Omega$.

[2015 F]



Solution:

Here,

$$V_1 = +2 \text{ V}$$

$$V_2 = -0.5 \text{ V}$$

$$V_3 = 0.8 \text{ V}$$

$$R_1 = 33 \text{ k}\Omega$$

$$R_2 = 22 \text{ k}\Omega$$

$$R_3 = 12 \text{ k}\Omega$$

$$R_f = 68 \text{ k}\Omega$$

We know that;

$$V_{out} = -R_f \left(\frac{V_1}{R_1} + \frac{V_2}{R_2} + \frac{V_3}{R_3} \right)$$

$$\text{or, } V_{out} = -68 \left(\frac{0.2}{33} - \frac{0.5}{22} + \frac{0.8}{12} \right)$$

$$\text{or, } V_{out} = -68(0.00606 - 0.02272 + 0.0666)$$

$$\therefore V_{out} = -3.40 \text{ V}$$

For CMRR and virtual ground

See the definition part 3.3.2 and 3.3.3

13. Explain an ideal characteristics of Op-amp. [2015 S, 2016 F, 2019 F]

Solution: See the definition part 3.3.1

14. Differentiate between amplifier and oscillator. "Negative feedback reduces the gain of an amplifier but enhances the circuit stability". Justify. [2015 F]

Solution:

	Amplifier	Oscillator
i)	An amplifier is a circuit, which amplifies the weak signal and raises the amplitude of the signal.	An oscillator is a circuit, which generates the ac waveforms of particular frequency for providing source to an electronic circuit.
ii)	It uses negative feedback.	It uses positive feedback.
iii)	The signal generated can be periodic or non-periodic.	The signal generated by it will be always periodic in nature.
iv)	The input is required for its operation, it cannot operate without input.	It do not require an input to perform operation.

For the second part

See the definition part 3.1.1 (i)

NOTE

Negative feedback reduces the gain of an amplifier by $\frac{1}{1 + \beta A}$. Since,

$$A_f = \frac{A}{1 + \beta A}$$

15. Design adder-subtractor circuit for the following equation
 $V_0 = -2V_1 - 4V_2 - 6V_3$ [2015 S, 2016 S]

Solution:

Given that;

$$V_0 = -(2V_1 + 4V_2 + 6V_3) \quad (1)$$

$$\text{and, } V_0 = -\left(\frac{R_F}{R_1}V_1 + \frac{R_F}{R_2}V_2 + \frac{R_F}{R_3}V_3\right) \quad (2)$$

Comparing equation (1) and (2); we have,

$$\frac{R_F}{R_1} = 2$$

$$\frac{R_F}{R_2} = 4$$

$$\frac{R_F}{R_3} = 6$$

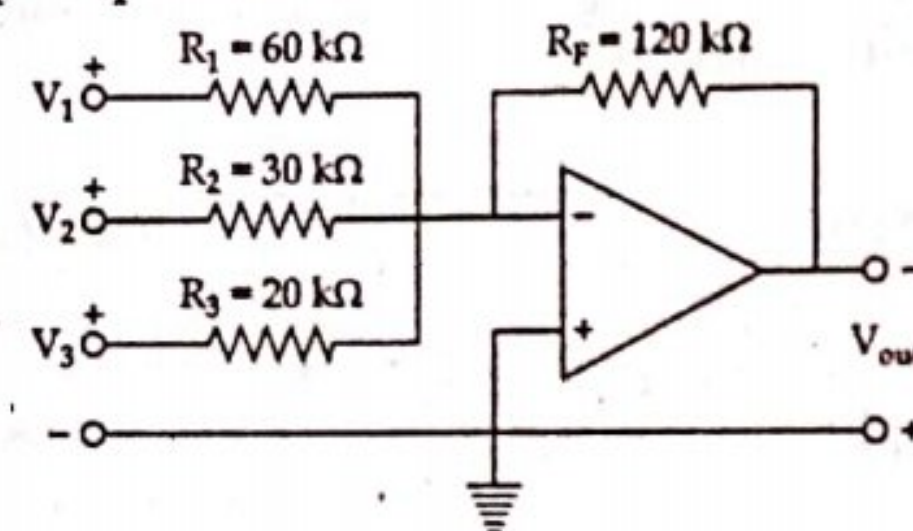
Assume, $R_F = 120 \text{ k}\Omega$

$$\therefore R_1 = 60 \text{ k}\Omega$$

$$R_2 = 30 \text{ k}\Omega$$

$$R_3 = 20 \text{ k}\Omega$$

The required op-amp circuit is;



16. Design adder-subtractor circuit for the following equation.

$$V_0 = -6V_1 - 24V_2 - 80V_3 \quad [2016 F]$$

Solution:

Given that;

$$V_0 = -(6V_1 + 24V_2 + 80V_3) \quad (1)$$

$$\text{and, } V_0 = -\left(\frac{R_F}{R_1}V_1 + \frac{R_F}{R_2}V_2 + \frac{R_F}{R_3}V_3\right) \quad (2)$$

Comparing equation (1) and (2); we get,

$$\frac{R_F}{R_1} = 6$$

$$\frac{R_F}{R_2} = 24$$

$$\frac{R_F}{R_3} = 80$$

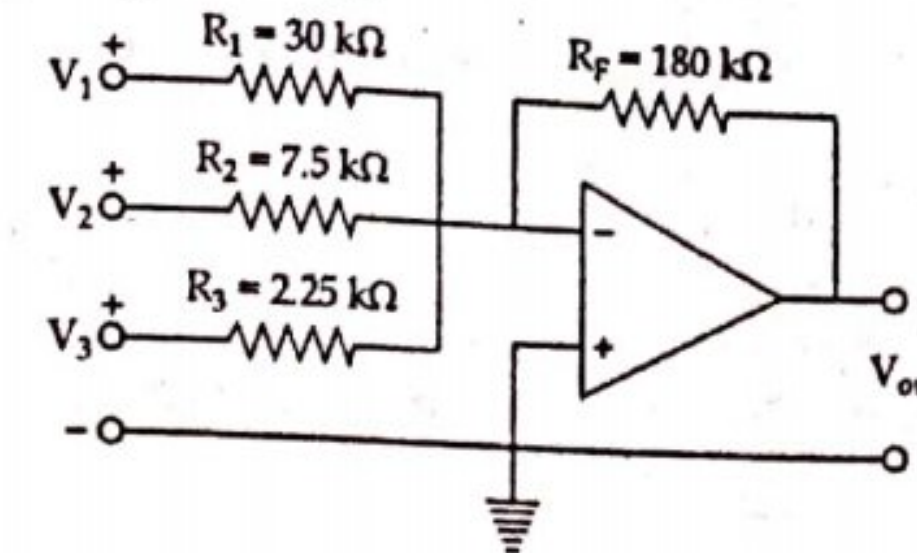
Assume $R_F = 180 \text{ k}\Omega$

$$\therefore R_1 = 30 \text{ k}\Omega$$

$$R_2 = 7.5 \text{ k}\Omega$$

$$R_3 = 2.25 \text{ k}\Omega$$

The required op-amp circuit is;



17. What is the importance of feedback in amplifier? An amplifier has a gain of 2000. If the feedback ratio is 0.04, then find the overall voltage gain of the amplifier with the negative feedback.

[2016 S]

Solution:

Feedback plays a major role in real-life circuits. Feedback allows circuit characteristics such as gain, input impedance, output impedance and bandwidth to be precisely controlled while making these parameters insensitive to variations in individual component parameters.

For the second part

Given that;

$$\text{Open-loop gain, } (A) = 2000$$

$$\text{Feedback ratio, } (\beta) = 0.04$$

$$\begin{aligned} \text{Overall voltage gain with negative feedback, } (A_F) &= \frac{A}{1 + \beta A} \\ &= \frac{2000}{1 + 0.04 \times 2000} \end{aligned}$$

$$\therefore A_F = 24.692$$

$$\begin{aligned} \text{and, Voltage gain with negative feedback in dB} &= 20 \log_{10} \left| \frac{A}{1 + \beta A} \right| \\ &= 20 \log_{10} \left| \frac{1}{1 + 0.04 \times 2000} \right| \end{aligned}$$

$$\therefore A_F \text{ in dB} = -38.169 \text{ dB}$$

18. Justify "Negative feedback is advantages in bandwidth boosting". [2017 F]

Solution:

The use of negative feedback increases the overall bandwidth of an amplifier. Bandwidth is defined by the difference between the higher (ω_H) and lower (ω_L) 3 dB cut off frequencies. The gain of an amplifier as a function of frequency is given by;

$$A(\omega) = -\frac{A_m}{1 + \left(\frac{j\omega}{\omega_H}\right)} \quad (1)$$

where, A_m is the mid-band gain.

ω_H is the upper 3 dB cut off frequency.

Assuming the feedback function β to be frequency independent the close-

loop gain $A_F = \frac{A}{1 - (A\beta)}$ becomes;

$$A_F(\omega) = \frac{A(\omega)}{1 - A(\omega)\beta}$$

Replacing the value of $A(\omega)$ from equation (1); we get,

$$A_F(\omega) = - \frac{\frac{A_m}{1 + \left(\frac{j\omega}{\omega_H}\right)}}{1 + \frac{A_m\beta}{1 + \left(\frac{j\omega}{\omega_H}\right)}} \quad (2)$$

$$\text{or, } A_F(\omega) = - \frac{A_m\omega_H}{j\omega + \omega_H(1 + A_m\beta)} \quad (2)$$

Comparing equation (1) and (2); we find that the new upper 3 dB frequency increases by a factor of $(1 + A_m\beta)$. Upper 3 dB frequency ω_{HF} is given by;

$$\omega_{HF} = \omega_H(1 + A_m\beta)$$

The upper 3 dB frequency increases by an amount equal to feedback factor.

Similarly, it can be shown that the feedback amplifier will have a lower 3 dB frequency ω_{LF} is given by;

$$\omega_{LF} = \frac{\omega_L}{1 + A_m\beta}$$

Lower 3 dB frequency is also reduced by an amount equal to feedback factor. The bandwidth of the amplifier with feedback is given by;

$$B\omega_F = \omega_H(1 + A_m\beta) - \frac{\omega_L}{1 + A_m\beta}$$

Thus, overall bandwidth increases by using negative feedback. The effect of negative feedback on the frequency response is as shown in the figure.

19. Draw and explain block diagram of an operational amplifier.

[2017 S]

Solution:

An operational amplifier being a multistage amplifier consists of some basic building blocks as shown in the block diagram.

The first stage is double ended high gain differential amplifier i.e., dual-input balanced output differential amplifier with a constant current

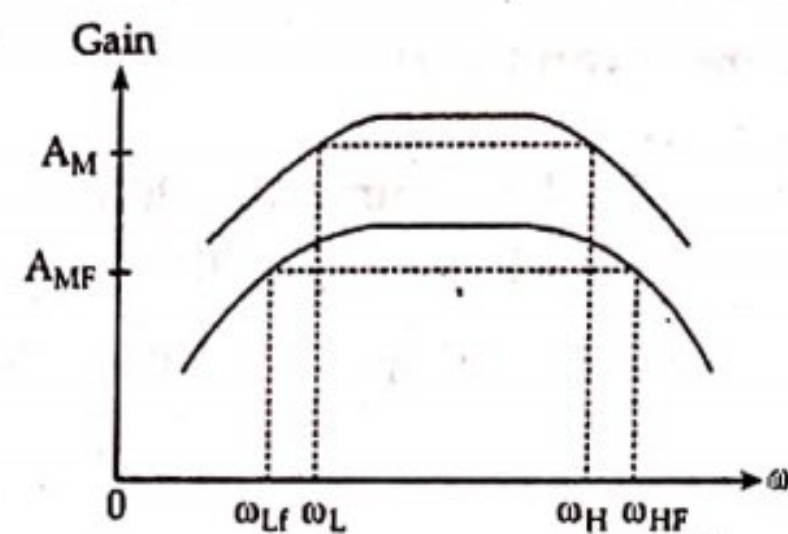


Figure: Effect of negative feedback on bandwidth

source (in order to increase CMRR). In this stage, high gain is desirable so that there would be a negligible effect on the output of any shortcoming in the following stages. This is the reason why this stage is generally responsible for most of the gain of op-amp. Also, this stage determines the input resistance of the op-amp.

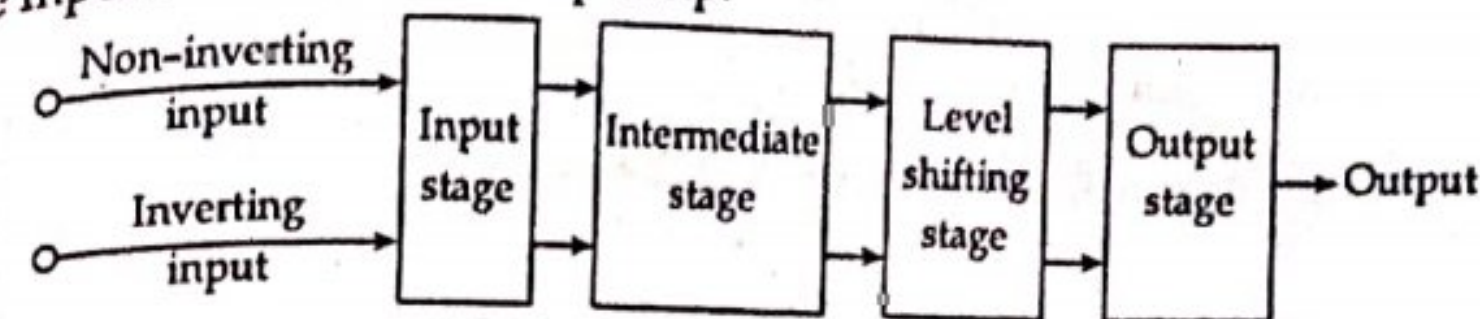


Figure: Block diagram of a typical Op-Amp

The second stage called the intermediate stage is usually another differential amplifier, which is driven by the output of first stage. In most amplifiers the intermediate stage is dual input, unbalanced output differential amplifier in order to increase the gain. The differential mode voltage gain of such an amplifier is half of the gain of the dual-input, balanced output differential amplifier. In this amplifier, output is measured at the collector of only one of the two transistors with respect to ground. In the quiescent condition some dc voltage exists at the output terminal and there is no other collector voltage at output to balance or nullify this output dc voltage. This is the reason why this type of amplifier is called unbalanced output type differential amplifier.

The third stage, known as level shifting stage, is usually an emitter follower circuit in order to shift the dc level at the output of the intermediate stage downward to zero volt with respect to ground. The error signal is developed in the intermediate stage due to direct coupling and gets amplified in the succeeding stages. This increase in dc level tends to shift the operating point of the succeeding stages, which also limits the output voltage swing or may distort the output signal.

The final stage called the output stage is usually a push-pull complementary amplifier. This stage increases the output voltage swing and the current supplying capability of the amplifier. A well designed output stage also provides low output resistance. The output of the op-amp is the voltage measured at output of this final stage push-pull amplifier with respect to ground.

20. Derive an expression for voltage gain of inverting configuration of Op-amp. [2017 F, 2018 F]

Solution:

For the first part

See the definition part 3.4.2

For the second part

Given that;

$$\text{Closed loop voltage gain, } (A_V) = \frac{V_{out}}{V_{in}} = -\frac{AR_F}{R_1 + R_F + AR_1}$$

$$\text{and, Output voltage, } (V_{out}) = A_V \times V_{in}$$

$$\therefore V_{out} = -\frac{AR_F}{R_1 + R_F + AR_1} \times V_{in}$$

But, open loop gain A is very large, $AR_1 \gg (R_1 + R_F)$;

$$\text{so, Closed loop voltage gain, } (A_V) = \frac{V_{out}}{V_{in}} = -\left(\frac{R_F}{R_1}\right)$$

$$\text{and, Output voltage, } (V_{out}) = -\frac{R_F}{R_1} \times V_{in}$$

21. What are the applications of an op-amp?

[2018 S]

Solution:

The applications of an op-amp are;

- | | |
|--|------------------------------|
| i) Addition amplifier | ii) Subtraction amplifier |
| iii) Zero cross detector | iv) Integrator |
| v) Differentiator | vi) Voltage follower |
| vii) Comparator | viii) Active filters |
| ix) Schmitt trigger | x) Instrumentation amplifier |
| xi) Voltage to current conversion and vice-versa | |

22. Derive the expression for voltage gain of negative feedback amplifier.

[2018 F, 2019 F]

Solution:

See the definition part 3.2

Hints

Replace X_{out} with V_{out} and X_{in} with V_{in}

Thus,

$$\text{Voltage gain, } (A_V) = \frac{V_{out}}{V_{in}}$$

$$\therefore A_V = \frac{1}{1 + \beta A}; \text{ for negative feedback}$$

$$\text{and, Output voltage, } (V_{out}) = V_{in} \times A_V = V_{in} \times \frac{1}{1 + \beta A}$$

23. Define the term feedback.

Solution:

[2018 S, 2018 F, 2019 F]

Feedback is an event that occurs when the output of a system is used as input back into the system as part of a chain of cause and effect. It is defined as the process in which a part of the output signal is returned back to the input.

4.8 SOLVED BOARD EXAMINATION PROBLEMS

1. Perform the following conversions

[2013 F]

- i) $(556.42)_{10} = (?)_2$
 ii) $(2AB.D8)_{16} = (?)_{10}$
 iii) $(1011011.0111)_2 = (?)_{16}$
 iv) $(24)_8 = (?)_{10}$

Solution:

i) Here,

Integer			Fraction
2	556	0	$0.42 \times 2 = 0.84$ with a carry of 0
2	278	0	$0.84 \times 2 = 0.68$ with a carry of 1
2	139	1	$0.68 \times 2 = 0.36$ with a carry of 1
2	69	1	$0.36 \times 2 = 0.72$ with a carry of 0
2	34	0	$0.72 \times 2 = 0.44$ with a carry of 1
2	17	1	$0.44 \times 2 = 0.88$ with a carry of 0
2	8	0	$0.88 \times 2 = 0.76$ with a carry of 1
2	4	0	$0.76 \times 2 = 0.52$ with a carry of 1
2	2	0	$0.52 \times 2 = 0.04$ with a carry of 1
	1		$0.04 \times 2 = 0.08$ with a carry of 0
			$0.08 \times 2 = 0.16$ with a carry of 0

Hence,

$$(556.42)_{10} = (100101100.01101011100)_2$$

ii) Here,

$$\begin{aligned} (2AB.D8)_{16} &= 2 \times 16^2 + 10 \times 16^1 + 11 \times 16^0 + 13 \times 16^{-1} \\ &\quad + 8 \times 16^{-2} \\ &= 512 + 160 + 11 + 0.8125 + 0.03125 \\ &= (683.8437)_{10} \end{aligned}$$

iii) Here,

$$\begin{array}{rcccc} (1011011.0111)_2 & = & 0001 & 0110 & \cdot & 0111 \\ & & \downarrow & \downarrow & & \downarrow \\ & & 1 & 6 & & 7 \end{array}$$

Hence,

$$(1011011.0111)_2 = (167)_{16}$$

iii) Here,

$$(24)_8 = 2 \times 16^2 + 4 \times 8^0 = (20)_{10}$$

2

State and prove De-Morgan's theorem for three variables.

[2013 F, 2013 S]

Solution:

De-Morgan's first theorem for three variable

The complement of the sum of three variables is equal to the product of the complements of the variables i.e., the complement of three variables ORed is the same as AND of the complements of each individual variable.

Proof

Let X, Y and Z are three variables. Then, the truth table is;

X	Y	Z	$\bar{X}$	$\bar{Y}$	$\bar{Z}$	$X+Y+Z$	$\overline{X+Y+Z}$	$\bar{X} \cdot \bar{Y} \cdot \bar{Z}$
0	0	0	1	1	1	0	1	1
0	0	1	1	1	0	1	0	0
0	1	0	1	0	1	1	0	0
0	1	1	1	0	0	1	0	0
1	0	0	0	1	1	1	0	0
1	0	1	0	1	0	1	0	0
1	1	0	0	0	1	1	0	0
1	1	1	0	0	0	1	0	0

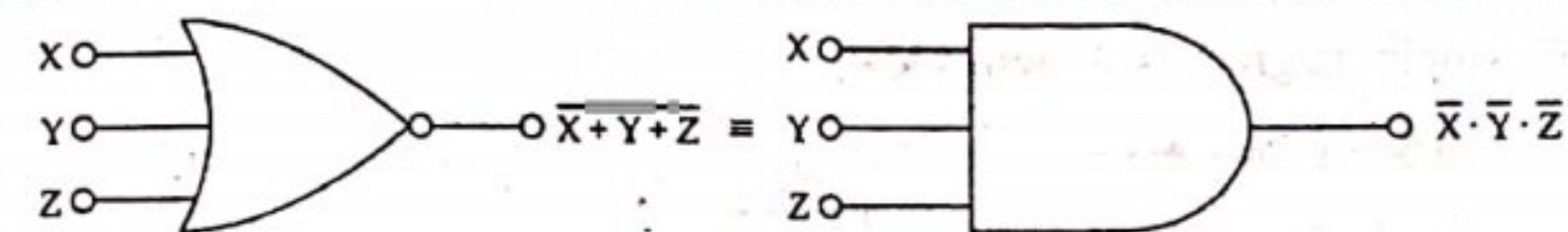


Figure: Logic diagram

De-Morgan's second theorem for three variable

The complement of the product of three variables is equal to the sum of the complements of the variables i.e., the complement of three variables ANDed is the same as OR of the complements of each individual variable.

Proof

X	Y	Z	$\bar{X}$	$\bar{Y}$	$\bar{Z}$	XYZ	$\overline{XYZ}$	$\bar{X} + \bar{Y} + \bar{Z}$
0	0	0	1	1	1	0	1	1
0	0	1	1	1	0	0	1	1
0	1	0	1	0	1	0	1	1
0	1	1	1	0	0	0	1	1
1	0	0	0	1	1	0	1	1
1	0	1	0	1	0	0	1	1
1	1	0	0	0	1	0	1	1
1	1	1	0	0	0	1	0	0

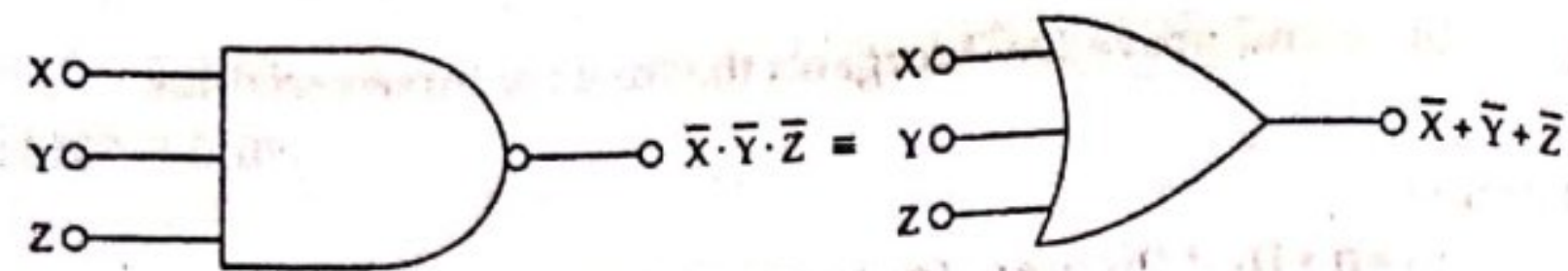


Figure: Logic diagram

3. Simplify the expression by using K-map. [2013 S]

$$Y = \bar{A}\bar{B}\bar{C} + \bar{A}B\bar{C} + \bar{A}\bar{B}C$$

Implement using basic gates.

Solution:

Here,

$$Y = \bar{A}\bar{B}\bar{C} + \bar{A}B\bar{C} + \bar{A}\bar{B}C$$

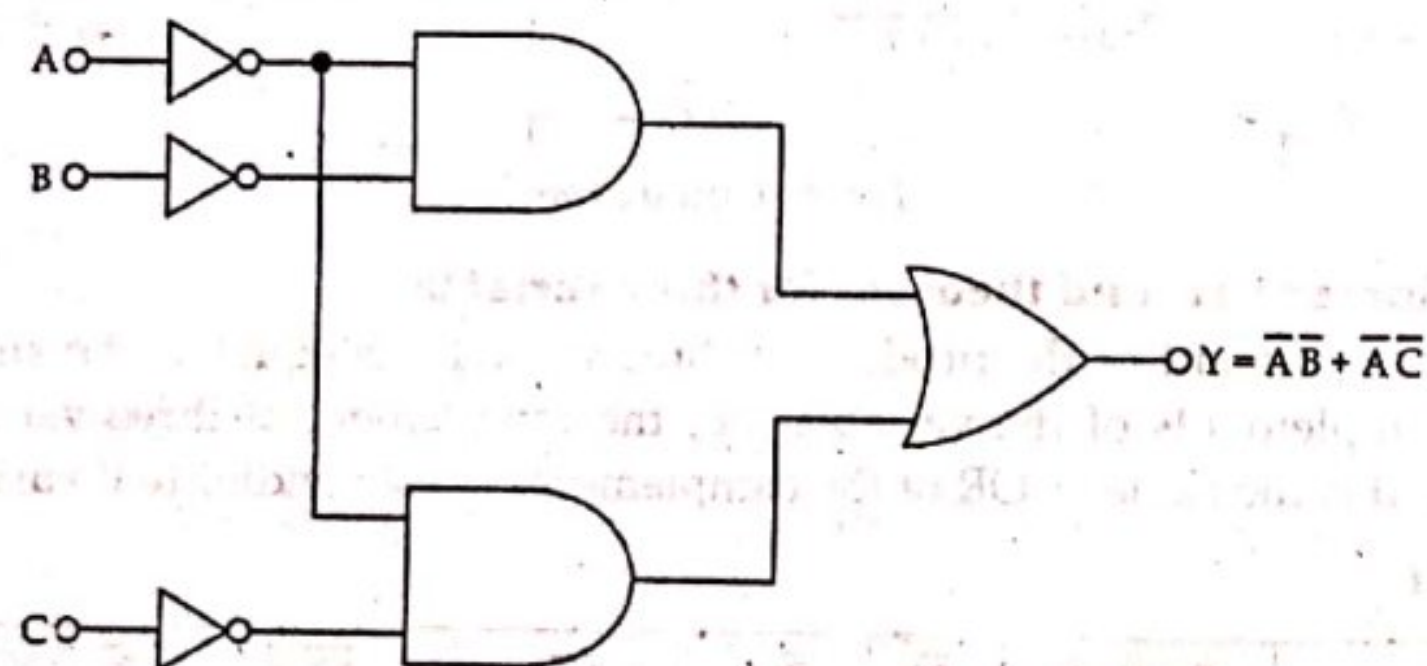
The K-map is;

BC \ A	00	01	11	10
0	1	1		1
1				

From K-map;

$$Y = \bar{A}\bar{B} + \bar{A}\bar{C}$$

The logic diagram is shown below;



4. Perform the following conversion.

i) $(24.6)_{10} = (\dots)_{\text{B}}$

ii) $(3FD)_H = (\dots)_2$

iii) $(756)_8 = (\dots)_{16}$

iv) $(725.62)_8 = (\dots)_6$

v) $(11011.111)_2 = (\dots)_{10}$

Solution:

i) Here,

Integer

Fraction

2	24	0	$0.42 \times 2 = 0.84$ with a carry of 0
2	12	0	$0.6 \times 2 = 0.2$ with a carry of 0
2	6	0	$0.2 \times 2 = 0.4$ with a carry of 0
2	3	1	$0.4 \times 2 = 0.8$ with a carry of 0
	1		$0.8 \times 2 = 0.6$ with a carry of 1
			$0.6 \times 2 = 0.2$ with a carry of 1

Hence,

$$(24.6)_{10} = (11000.00011)_B$$

ii) Here,

$$(3FD)_H = \begin{matrix} 3 & F & D \\ \downarrow & \downarrow & \downarrow \\ 0011 & 1111 & 1101 \end{matrix}$$

Hence,

$$(3FD)_H = (00111111101)_2$$

iii) Here,

$$(756)_8 = \begin{matrix} 7 & 5 & 6 \\ \downarrow & \downarrow & \downarrow \\ 111 & 101 & 110 \\ \downarrow & & \\ 0001 & 1110 & 1110 \\ \downarrow & \downarrow & \downarrow \\ 1 & 14 & 14 \\ \downarrow & \downarrow & \downarrow \\ 1 & E & E \end{matrix}$$

Hence,

$$(756)_8 = (1EE)_{16}$$

iv) Here,

$$\begin{aligned} (725.62)_8 &= 7 \times 8^2 + 2 \times 8^1 + 5 \times 8^0 + 6 \times 8^{-1} + 2 \times 8^{-2} \\ &= 448 + 16 + 5 + 0.75 + 0.03125 \\ &= (469.7813)_{10} \end{aligned}$$

Converting to base 6,

Integer

Fraction

6	469	1	$0.7813 \times 6 = 0.6878$ with a carry of 4
6	78	0	$0.6878 \times 6 = 0.1268$ with a carry of 4
6	13	1	$0.1268 \times 6 = 0.7608$ with a carry of 0
	2		$0.7608 \times 6 = 0.5648$ with a carry of 4
			$0.5648 \times 6 = 0.3888$ with a carry of 3

Hence,

$$(725.62)_8 = (2101.44043)_6$$

v) Here,

$$\begin{aligned}(11011.111)_2 &= 1 \times 2^4 + 1 \times 2^3 + 0 \times 2^2 + 1 \times 2^1 + 1 \times 2^0 \\ &\quad + 1 \times 2^{-1} + 1 \times 2^{-2} + 1 \times 2^{-3} \\ &= 16 + 8 + 0 + 2 + 1 + 0.5 + 0.25 + 0.125 \\ &= (27.875)_{10}\end{aligned}$$

5. Why NAND and NOR gates are called universal gates and show its implementation for basic gates. [2014 F]

Or

Write short notes on universality of NAND and NOR gate.

[2013 S, 2014 S, 2017 F, 2019 F]

Or

Write short notes on universal digital logic gates.

[2015 F]

Solution:

NAND or OR gates are said to be universal gates because any digital system can be implemented with these gates.

i) NAND gate as a universal gate

The table below shows how NAND gate can be connected to produce AND, OR and NOT gate.

Logic function	Symbol	Circuit using NAND gates only
NOT		
AND		
OR		

a) NOT gate from NAND gate

When two inputs of NAND gate are joined together so that it has one input, the resulting circuit is NOT gate. The truth table shows this fact.

A	B(= A)	$\bar{A}$
0	0	1
1	1	0

b) AND gate from NAND gate

For this purpose, two NAND gates are used. The output of first NAND gate is given to the second NAND gate acting as inverter. The resulting circuit is the AND gate. The output Y' of the first NAND gate is inverted output of AND gate. The second NAND gate acting as inverter further inverts it so that the final output Y is that of AND gate. The truth table shows this fact.

A	B	$\overline{AB}$	AB
0	0	1	0
0	1	1	0
1	0	1	0
1	1	0	1

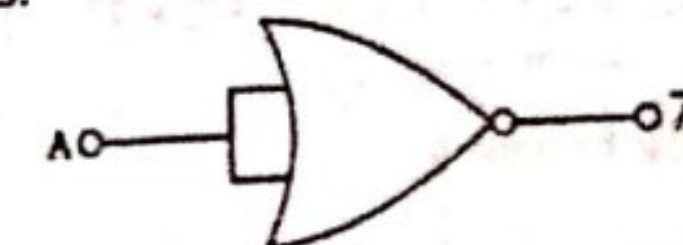
c) OR gate from NAND gate

For this purpose, three NAND gates are used. The first two NAND gates are operated as NOT gates and their outputs are fed to the third. The resulting circuit is OR gate. The truth table shows this fact.

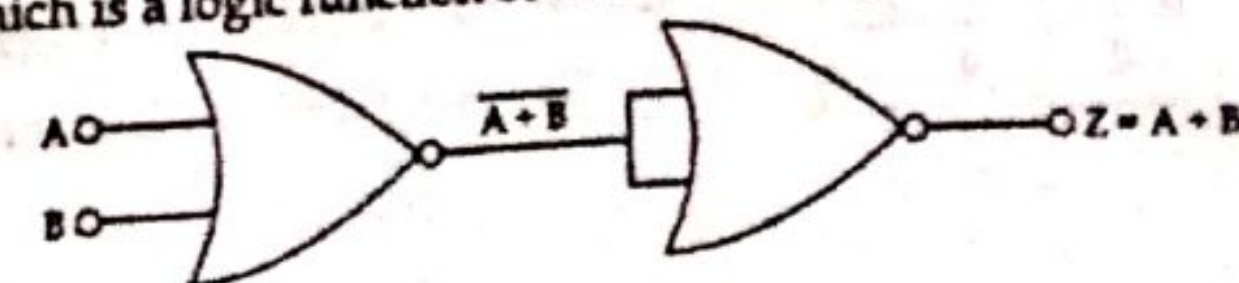
A	B	$\bar{A}$	$\bar{B}$	$A + B$
0	0	1	1	0
0	1	1	0	1
1	0	0	1	1
1	1	0	0	1

ii) NOR gate as a universal gate

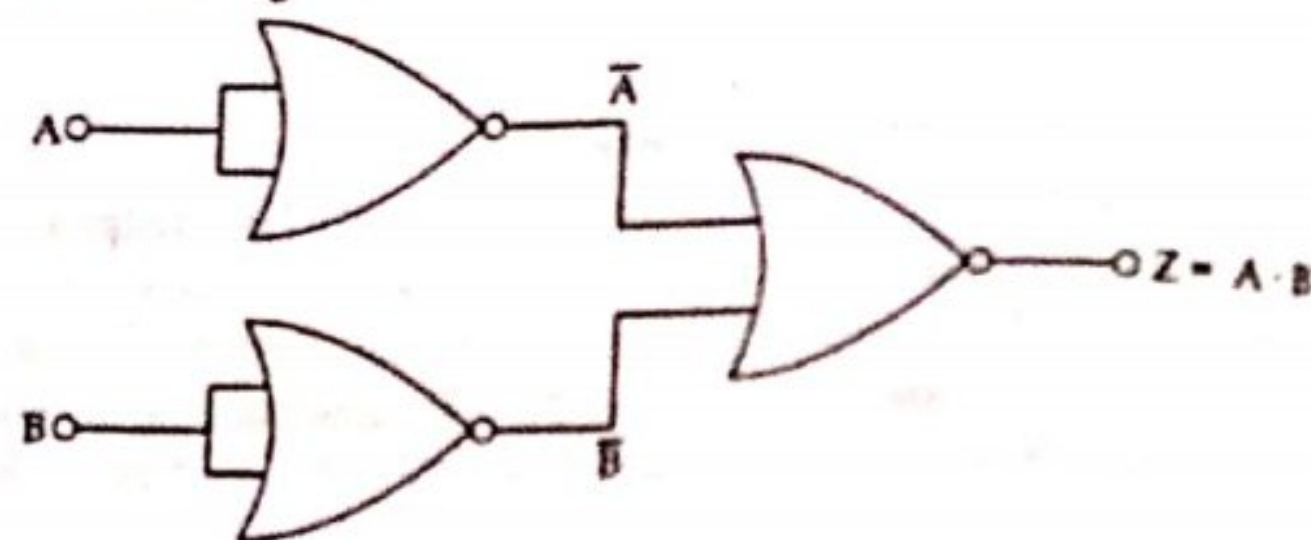
- a) NOT logic operation can be performed by tying together the two inputs.



- b) OR logic operation can be performed by using two NOR gate. The final output is inverted and is given as $Z = A + B$; which is a logic function of a normal OR gate.



- c) AND operation can be performed by using three NOR gates. The inputs have thus been inverted before they are applied to the NOR gate.



6. Perform the following conversion.

[2014 S]

- $(A1F.8E)_{16} = (\dots)_{10}$
- $(534.344)_8 = (\dots)_2$
- $(142.45)_7 = (\dots)_4$
- $(155.24)_{10} = (\dots)_5$
- $(54.35)_{10} = (\dots)_{16}$

Solution:

i) Here,

$$\begin{aligned} (A1F.8E)_{16} &= 10 \times 16^2 + 1 \times 16^1 + 15 \times 16^0 + 8 \times 16^{-1} + 14 \times 16^{-2} \\ &= (2591.554688)_2 \end{aligned}$$

ii) Here,

$$(534.344)_8 = \begin{array}{ccc|ccc} 5 & 3 & 4 & \cdot & 3 & 4 & 4 \\ \downarrow & \downarrow & \downarrow & & \downarrow & \downarrow & \downarrow \\ 101 & 011 & 100 & & 011 & 100 & 100 \end{array}$$

Hence,

$$(534.344)_8 = (101011100.011100100)_2$$

iii) Converting $(142.45)_7$ to decimal

$$\begin{aligned} (142.45)_7 &= 1 \times 7^2 + 4 \times 7^1 + 2 \times 7^0 + 4 \times 7^{-1} + 5 \times 7^{-2} \\ &= (79.67346)_{10} \end{aligned}$$

Now, converting to base 4;

Integer			Fraction
4	79	3	$0.6734 \times 4 = 0.6936$ with a carry of 2
4	19	3	$0.6936 \times 4 = 0.7744$ with a carry of 2
4	4	0	$0.7744 \times 4 = 0.0976$ with a carry of 3
	1		

Hence,

$$(142.45)_7 = (1033.223)_4$$

iv) Here,

Integer			Fraction
5	155	0	$0.24 \times 5 = 0.20$ with a carry of 1
5	31	1	$0.20 \times 5 = 1$ with a carry of 0
5	6	1	
	1		

Thus,

$$(155.24)_{10} = (1110.10)_{16}$$

v) Here,

Integer			Fraction
16	54	6	$0.35 \times 16 = 0.60$ with a carry of 5
	3		$0.60 \times 16 = 0.60$ with a carry of 9

Thus,

$$(54.35)_{10} = (36.59)_{16}$$

7. Perform the following conversions

[2014 F]

- $(2ABD8)_{16} = (?)_{10}$
- $(85.32)_{10} = (?)_2$
- $(10110110111)_2 = (?)_{16}$
- $(F5C.2B)_{16} = (?)_{10}$

Solution:

i) Here,

$$\begin{aligned} (2ABD8)_{16} &= 2 \times 16^4 + 10 \times 16^3 + 11 \times 16^2 + 13 \times 16^1 + 8 \times 16^0 \\ &= 2 \times 65536 + 10 \times 4096 + 11 \times 256 + 13 \times 16 + 8 \\ &= 8 + 208 + 2816 + 40960 + 131072 \\ &= (175064)_{10} \end{aligned}$$

ii) Here,

Integer			Fraction
2	85	1	$0.32 \times 2 = 0.64$ with a carry 0
2	42	0	$0.64 \times 2 = 0.28$ with a carry 1
2	21	1	$0.28 \times 2 = 0.56$ with a carry 0
2	10	0	$0.56 \times 2 = 0.12$ with a carry 1
2	5	1	$0.12 \times 2 = 0.24$ with a carry 0
2	2	0	$0.24 \times 2 = 0.48$ with a carry 0
	1		$0.48 \times 2 = 0.96$ with a carry 0

$$\therefore (85.32)_{10} = (1010101.0101000)_2$$

iii) Here,

$$(10110110111)_2 = (010110110111)_2$$

$$= \begin{array}{ccc} 0101 & 1011 & 0111 \\ \downarrow & \downarrow & \downarrow \\ 5 & 11 & 7 \end{array}$$

Thus,

$$(10110110111)_2 = (5117)_{16} = (5B7)_{16}$$

iv) Here,

$$(F5C.2B)_{16} = \begin{array}{ccccc} F & 5 & C & . & 2 & B \\ \downarrow & \downarrow & \downarrow & & \downarrow & \downarrow \\ (1111 & 0101 & 1100 & . & 0010 & 1011)_8 \end{array}$$

so, $\begin{array}{ccccccc} 111 & 101 & 011 & 100 & . & 001 & 010 & 110 \\ \downarrow & \downarrow & \downarrow & \downarrow & & \downarrow & \downarrow & \downarrow \\ 7 & 5 & 3 & 4 & . & 1 & 2 & 6 \end{array}$

Thus,

$$(F5C.2B)_{16} = (7534.126)_8$$

8. Simplify the Boolean function $F(A, B, C) = \sum(3, 5, 6, 7)$ using K-map. Implement the function using logic gates. [2014 S]

Solution:

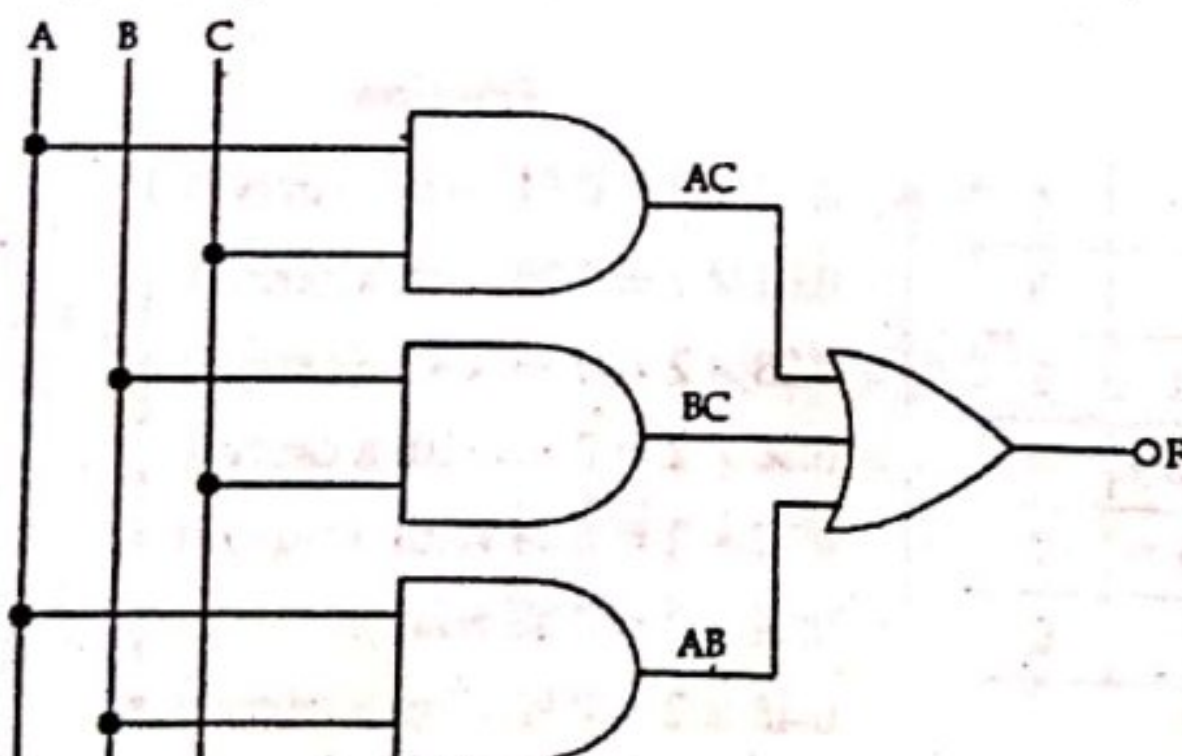
Here,

$$F(A, B, C) = \sum(3, 5, 6, 7)$$

BC \ A	00	01	11	10
0			1	
1		1	1	1

From K-map;

$$F(A, B, C) = AC + BC + AB$$



9. Perform the following conversions. [2015 F]

i) $(41.375)_{10} = (?)_2$

ii) $(561)_8 = (?)_5$

iii) $(513)_{10} = (?)_{BCD}$

iv) $(3DA2.FA)_{16} = (?)_2 = (?)_8$

Solution:

i) Here,

Integer

Fraction

2	41	1
2	20	0
2	10	0
2	5	1
2	2	0
	1	

$0.375 \times 2 = 0.75$ with a carry of 0
 $0.75 \times 2 = 0.50$ with a carry of 1
 $0.50 \times 2 = 1$ with a carry of 0

$$\therefore (41.375)_{10} = (101001.010)_2$$

ii) Converting into decimal system,

$$(561)_8 = 5 \times 8^2 + 6 \times 8^1 + 1 \times 8^0 = (369)_{10}$$

Now,

5	369	4
5	73	3
5	14	4
	2	

Thus,

$$(561)_8 = (2434)_5$$

iii) Here,

$$(513)_{10} = \begin{array}{ccc} 5 & 1 & 3 \\ \downarrow & \downarrow & \downarrow \\ 0101 & 0001 & 0011 \end{array}$$

Thus,

$$(513)_{10} = (010100010011)_{BCD}$$

iv) Here,

$$(3DA2.FA)_{16} = \begin{array}{ccccc} 3 & D & A & 2 & . & F & A \\ \downarrow & \downarrow & \downarrow & \downarrow & & \downarrow & \downarrow \\ (0011 & 1101 & 1010 & 0010 & . & 1111 & 1010)_2 \end{array}$$

On arranging;

011	110	110	100	010	111	110	100
↓	↓	↓	↓	↓	↓	↓	↓
3	6	6	4	2	7	6	4

Thus,

$$(3DA2FA)_{16} = (36642764)_8 = (0011110110100010.11111010)_2$$

10. Explain and prove the De-Morgan's theorem of Boolean algebra. [2015 S, 2016 S, 2017 S, 2017 F]

Solution: See the definition part 4.5

11. Using K-map, find the simpler expression for $F(A, B, C, D) = \sum(5, 7, 8, 10, 12, 13, 14, 15)$ and draw logic diagram using basic gates. [2015 F, 2017 S, 2018 F]

Solution:

Here,

$$F(A, B, C, D) = \sum(5, 7, 8, 10, 12, 13, 14, 15)$$

Using K-map;

CD \ AB	00	01	11	10
00				
01		1	1	
11	1	1	1	1
10	1			1

From K-map;

$$F(A, B, C, D) = A\bar{D} + BD$$

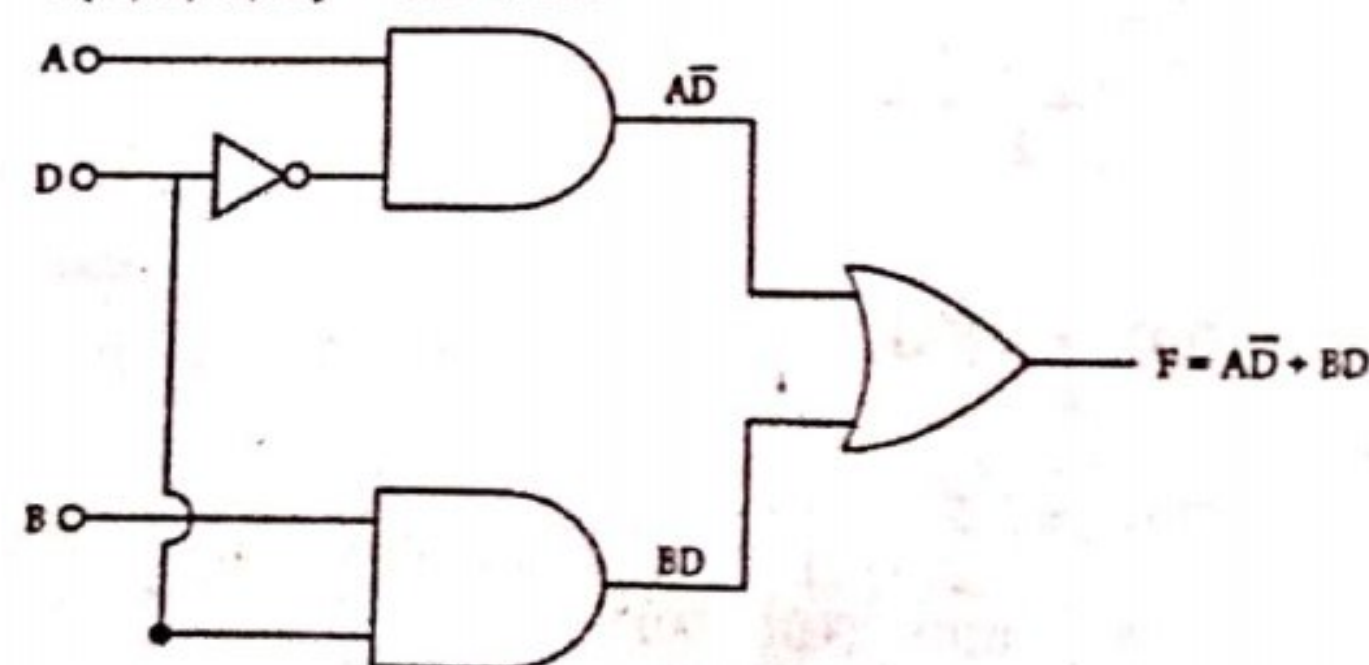


Figure: Logic diagram for $F = A\bar{D} + BD$

12. Simplify the following expression using K-map and implement it in circuit diagram using logic gates.

$$F = A \cdot \bar{B} \cdot \bar{C} + \bar{A} \cdot \bar{B} \cdot C + A \cdot \bar{B} \cdot C$$

[2015 S, 2017 S]

Solution:

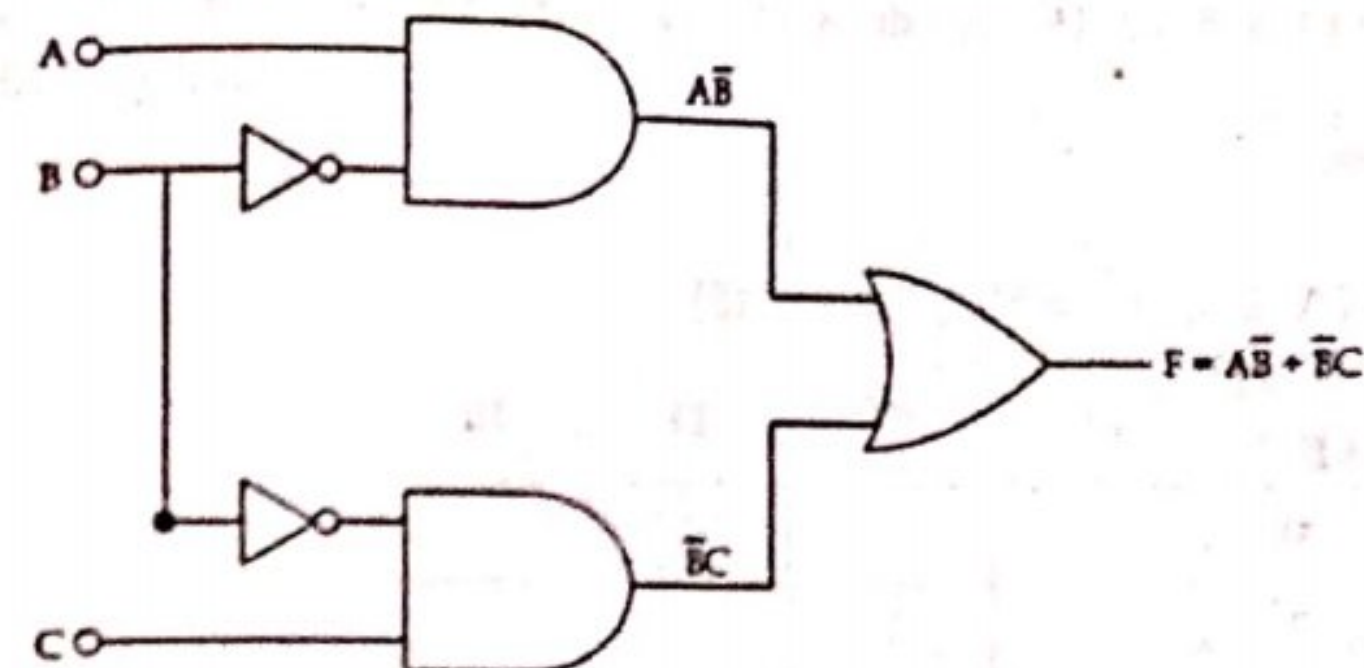
Here,

$$F = A \cdot \bar{B} \cdot \bar{C} + \bar{A} \cdot \bar{B} \cdot C + A \cdot \bar{B} \cdot C$$

A \ BC	00	01	11	10
0		1		
1	1	1		

From K-map;

$$F(A, B, C) = A\bar{B} + \bar{B}C$$



13. Perform the following conversion. [2016 S]

i) $(1011101.0111)_2 = (?)_{16}$

ii) $(72)_8 = (?)_{10}$

iii) $(345)_8 = (?)_{16}$

iv) $(2AED8)_{16} = (?)_{10}$

Solution:

i) Here,

$$(1011101.0111)_2 = 0101 \quad 1101 \quad \cdot \quad 0111$$

$$\downarrow \quad \downarrow \quad \cdot \quad \downarrow$$

$$5 \quad 13 \quad \cdot \quad 7$$

$$5 \quad D \quad \cdot \quad 7$$

Thus,

$$(1011101.0111)_2 = (5D7)_{16}$$

ii) Here,

$$(78)_8 = 2 \times 8^0 + 7 \times 8^1 = (58)_{10}$$

iii) Here,

$$(345)_8 = 3 \quad 4 \quad 5$$

$$\downarrow \quad \downarrow \quad \downarrow$$

$$011 \quad 100 \quad 101$$

$$\downarrow$$

$$0000 \quad 1110 \quad 0101$$

$$\downarrow \quad \downarrow \quad \downarrow$$

$$0 \quad 14 \quad 5$$

Hence,

$$(345)_8 = (0E5)_{16} = (E5)_{16}$$

iv) Here,

$$(2AE.D8)_{16} = 2 \times 16^2 + 10 \times 16^1 + 14 \times 16^0 + 13 \times 16^{-1} + 8 \times 16^{-2} \\ = (686.84375)_{10}$$

14. Solve the Boolean expression using K-map. $F(A, B, C, D) = \sum(3, 5, 8, 13, 15)$ and draw the logic diagram using basic gates. [2016 F, 2017 F]

Solution:

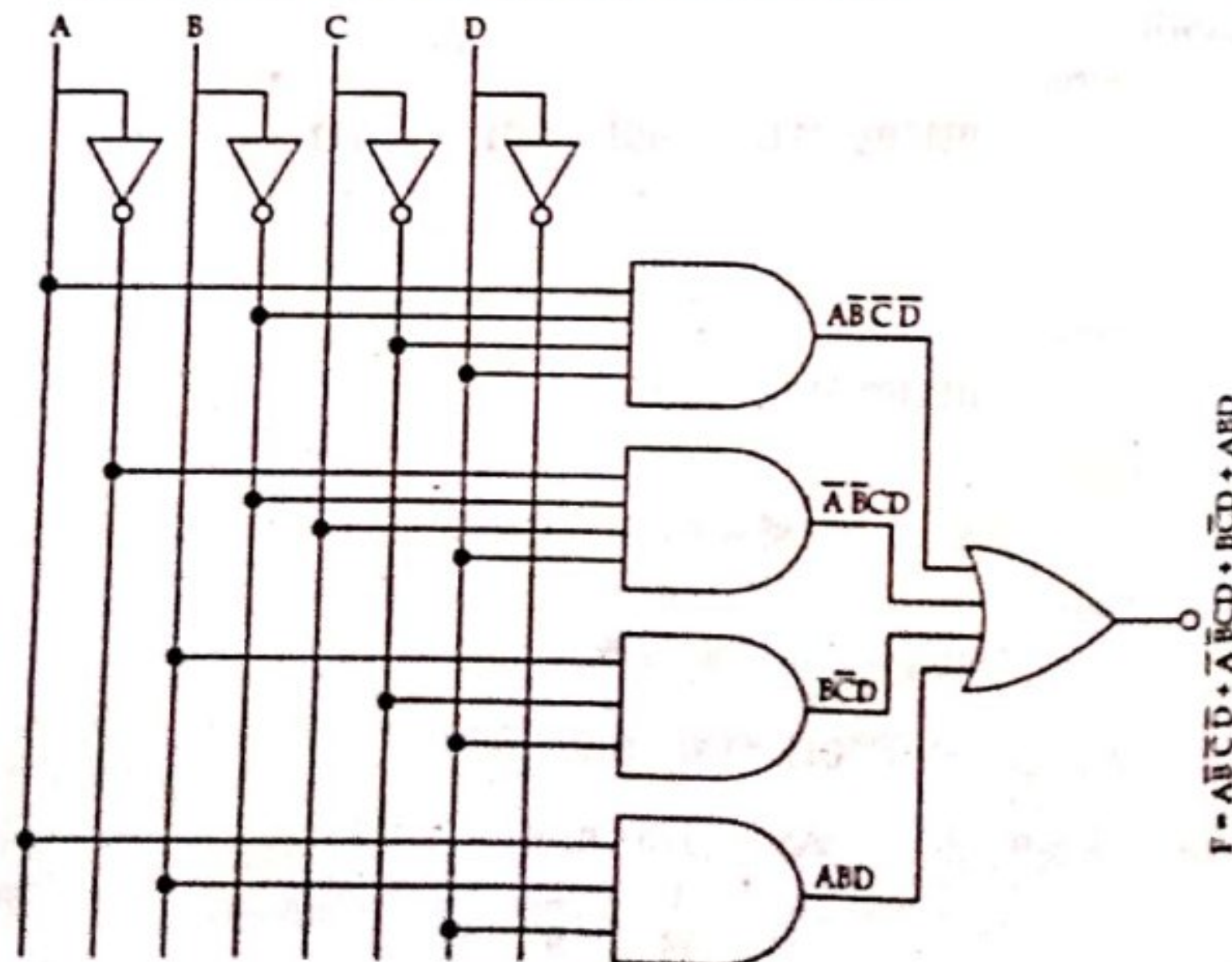
Here,

$$F(A, B, C, D) = \sum(3, 5, 8, 13, 15)$$

BC \ AB	00	01	11	10
0			1	
1		1		
11		1	1	
10	1			

From K-map;

$$F(A, B, C, D) = A\bar{B}\bar{C}\bar{D} + \bar{A}\bar{B}CD + B\bar{C}D + ABD$$



15. Convert the following numbers to indicated bases. [2016 F, 2017 F]
- $(B2C.B)_{16}$ to octal
 - $(1938.257)_{10}$ to hexadecimal
 - $(175.175)_{10}$ to binary
 - Hexadecimal 2AC5 to decimal

Solution:

i) Here,

$$(B2C.B)_{16} = \begin{array}{ccc} B & 2 & C \\ \downarrow & \downarrow & \downarrow \\ 1011 & 0010 & 1100 \end{array} \cdot \begin{array}{c} B \\ \downarrow \\ 1011 \end{array} \leftarrow \text{Binary}$$

so, arranging;

$$\begin{array}{ccc} 101 & 100 & 101 \\ \downarrow & \downarrow & \downarrow \\ 5 & 4 & 5 \end{array} \cdot \begin{array}{ccc} 101 & 100 \\ \downarrow & \downarrow \\ 5 & 4 \end{array}$$

Hence,

$$(B2C.B)_{16} = (5454.54)_8$$

ii) Here,

Integer

Fraction

16	1938	2	$\begin{array}{l} \uparrow 0.257 \times 16 = 0.112 \text{ with a carry of 4} \\ 0.112 \times 16 = 0.792 \text{ with a carry of 1} \\ 0.792 \times 16 = 0.672 \text{ with a carry of 12} \\ 0.672 \times 16 = 0.752 \text{ with a carry of 10} \\ 0.752 \times 16 = 0.032 \text{ with a carry of 12} \downarrow \end{array}$
16	121	9	
	7		

$$\therefore (1938.257)_{10} = (792.41CAC)_{16}$$

iii) Here,

Integer

Fraction

2	175	1	$\uparrow$ $0.175 \times 2 = 0.35$ with a carry of 0 $0.35 \times 2 = 0.70$ with a carry of 0 $0.70 \times 2 = 0.40$ with a carry of 1 $0.40 \times 2 = 0.80$ with a carry of 0 $0.80 \times 2 = 0.60$ with a carry of 1 $0.60 \times 2 = 0.20$ with a carry of 1 $0.20 \times 2 = 0.40$ with a carry of 0 $\downarrow$
2	87	1	
2	43	1	
2	21	1	
2	10	0	
2	5	1	
2	2	0	
	1		

$$\therefore (175.175)_{10} = (10101111.0010110)_2$$

iv) Here,

$$(2AC5)_{16} = 2 \times 16^3 + 10 \times 16^2 + 12 \times 16^1 + 5 \times 16^0 \\ = (10949)_{10}$$

16. Perform the following conversions.

[2017 S]

i) $(B2C.B)_{16}$ to octalii) Hexadecimal $(2AC5)$ to decimaliii) $(24)_8$ to decimaliv) $(535.344)_8$ to binary

Solution:

i) See the solution of Q. no. 6 (i)

ii) See the solution of Q. no. 6 (iv)

iii) Here,

$$(24)_8 = 2 \times 8^1 + 4 \times 8^0 \\ = (20)_{10}$$

iv) Here,

$$(535.344)_8 = \begin{array}{cccccc} 5 & 3 & 5 & . & 3 & 4 & 4 \\ \downarrow & \downarrow & \downarrow & & \downarrow & \downarrow & \downarrow \\ 101 & 011 & 101 & . & 011 & 100 & 100 \end{array}$$

Thus,

$$(535.344)_8 = (101011101.011100100)_2$$

17. Write short notes on digital to analog conversion and its significance. [2018 S]

Solution: See the definition part 4.1

18. Perform the following conversions.

[2018 F]

i) $(DAD.DAD)_{16} = (?)_{10}$ ii) $(88.966)_{10} = (?)_8$ iii) $(535.344)_8 = (?)_2$ iv) $(10001.001)_2 = (?)_{16}$

Solution:

i) Here,

$$(DAD.DAD)_{16} = 13 \times 16^2 + 10 \times 16^1 + 13 \times 16^0 + 13 \times 16^{-1} \\ + 10 \times 16^{-2} + 13 \times 16^{-3} \\ = (3501.8547)_{10}$$

ii) Here,

Integer

Fraction

8	88	0	↑	0.966 × 8 = 0.728 with a carry of 7
8	11	3		0.728 × 8 = 0.824 with a carry of 5
	1			0.824 × 8 = 0.592 with a carry of 6
				0.592 × 8 = 0.736 with a carry of 4
				0.736 × 8 = 0.888 with a carry of 5 ↓

Thus,

$$(88.966)_{10} = (130.75645)_8$$

iii) Here,

$$(535.344)_8 = \begin{array}{cccccc} 5 & 3 & 5 & . & 3 & 4 & 4 \\ \downarrow & \downarrow & \downarrow & & \downarrow & \downarrow & \downarrow \\ 101 & 011 & 101 & . & 011 & 100 & 100 \end{array}$$

Hence,

$$(535.344)_8 = (101011101.011100100)_2$$

iv) Here,

$$(10001.001)_2 = \begin{array}{ccc} 0001 & 0001 & . & 0010 \\ \downarrow & \downarrow & & \downarrow \\ 1 & 1 & . & 2 \end{array}$$

Hence,

$$(10001.001)_2 = (11.2)_{16}$$

19. Perform the following conversions.

[2018 S]

i) $(B2C.B)_{16} = (?)_8$ ii) $(74.562)_8 = (?)_2$ iii) $(11001.101)_2 = (?)_{10}$ iv) $(211)_x = (152)_8$. Find the value of X.

Solution:

i) See the solution of Q. no. 6 (i)

ii) Here,

$$(74.562)_8 = \begin{array}{ccccc} 7 & 4 & . & 5 & 6 & 2 \\ \downarrow & \downarrow & & \downarrow & \downarrow & \downarrow \\ 111 & 100 & . & 101 & 110 & 010 \end{array}$$

Hence,

$$(74.562)_8 = (111100.101110010)_2$$

iii) Here,

$$(11001.101)_2 = 1 \times 2^4 + 1 \times 2^3 + 0 \times 2^2 + 0 \times 2^1 + 1 \times 2^0 \\ + 1 \times 2^{-1} + 0 \times 2^{-2} + 1 \times 2^{-3}$$

$$= 16 + 8 + 0 + 0 + 1 + 0.50 + 0 + 0.125$$

$$= (25.625)_{10}$$

iv) Here,

$$(152)_8 = 1 \times 8^2 + 5 \times 8^1 + 2 \times 8^0$$

$$= (106)_{10}$$

Converting $(211)_x$ into decimal number;

$$2 \times x^2 + 1 \times x^1 + 1 \times x^0 = 106$$

$$\text{or, } 2x^2 + x + 1 = 106$$

$$\text{or, } 2x^2 + x - 105 = 0$$

$$\text{or, } x = 7 \text{ or } x = -7.5$$

Discarding negative value; hence,

$$x = 7$$

Thus,

$$(211)_7 = (152)_8 \text{ i.e., value of } x = 7$$

20. Simplify Boolean function $F(A, B, C, D) = \sum(0, 2, 5, 7, 8, 10, 13, 15)$ using K-map. Implement the function using logic gates.

[2018 S, 2019 F]

Solution:

Here,

$$F(A, B, C, D) = \sum(0, 2, 5, 7, 8, 10, 13, 15)$$

CD \ AB	00	01	11	10
00	1			1
01		1	1	
11		1	1	
10	1			1

From K-map;

$$F(A, B, C, D) = \overline{B} \overline{D} + BD$$

$$= \overline{B \oplus D}$$

$$= B \odot D$$

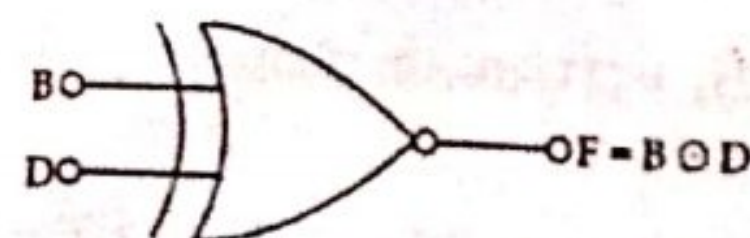


Figure: Logic gate implementation

21. Perform the following conversions

[2019 F]

i) $(1F.8)_{16} = (\dots\dots\dots)_{10}$

ii) $(1010.1011)_2 = (\dots\dots\dots)_{10}$

iii) $(1111001011001010)_2 = (\dots\dots\dots)_{16}$

iv) $(231.3)_4 = (\dots\dots\dots)_7$

Solution:

i) Here,

$$(1F.8)_{16} = 1 \times 16^1 + 15 \times 16^0 + 8 \times 16^{-1}$$

$$= (31.5)_{10}$$

ii) Here,

$$(1010.1011)_2 = 1 \times 2^3 + 0 \times 2^2 + 1 \times 2^1 + 0 \times 2^0 + 1 \times 2^{-1}$$

$$+ 0 \times 2^{-2} + 1 \times 2^{-3} + 1 \times 2^{-4}$$

$$= 8 + 0 + 2 + 0 + 0.5 + 0 + 0.125 + 0.0625$$

$$= (10.6875)_{10}$$

iii) Here,

$$(1111001011001010)_2 = \begin{matrix} 1111 & 0010 & 1100 & 1010 \\ \downarrow & \downarrow & \downarrow & \downarrow \\ F & 2 & C & A \end{matrix}$$

Hence,

$$(1111001011001010)_2 = (F2CA)_{16}$$

iv) Here,

$$(231.3)_4 = 2 \times 4^2 + 3 \times 4^1 + 1 \times 4^0 + 3 \times 4^0 + 3 \times 4^{-1}$$

$$= 32 + 12 + 1 + 0.25$$

$$= (45.25)_{10}$$

Converting to base 7,

Integer	Fraction
7 45 3	$0.25 \times 7 = 0.75$ with a carry of 1
6	$0.75 \times 7 = 0.25$ with a carry of 5
	$0.25 \times 7 = 0.75$ with a carry of 1

Thus,

$$(231.3)_4 = (63.151)_7$$

22. Write down the advantages and disadvantages of K-map.

Solution:

Advantages of K-map

- Data representation is simple.
- Easy and convenient to implement.
- Reduces the quantity and cost of logic gates.

- iv) Less numbers of steps needed compared to algebraic minimization technique.
- v) Does not need knowledge of Boolean algebraic theorems.
- vi) The K-map provides an easy, visual representation of a simple logic function to help with finding its primes, essential primes and minimize the logic.

Disadvantages of K-map

- i) Complexity of K-map process increases with the increase in the number of variables.
- ii) The minimum logical expression arrived at by using the K-map simplification procedure may or may not be unique depending on the choices made while forming the groups.

23. What is don't care condition?

Solution:

In certain digital systems, some input combinations never occur during the process of a normal operation because those input conditions are guaranteed never to occur. Such input combinations are called Don't-Care Combinations. The function output may be either 1 or 0 and these functions are called incompletely specified functions. These input combinations can be plotted on the Karnaugh map for further simplification of the function. The don't care combinations are represented by d or x or Φ .

24. Write down some advantages and disadvantages of BCD code.

Solution:

Some of the main advantages of using the BCD code are;

- i) It is easy to encode and decode decimals into BCD and vice versa.
- ii) It is also simple to implement a hardware algorithm for the BCD converter.
- iii) It is very useful in digital systems whenever decimal information is given either as inputs or displayed as outputs.
- iv) Digital voltmeters, frequency converters and digital clocks all use BCD as they display output information in decimal.

Disadvantages of BCD code are;

- i) BCD code for a given decimal number requires more bits than the straight binary code and hence there is difficulty to represent the BCD form in high speed digital computers in arithmetic operations, especially when the size and capacity of their internal registers are restricted or limited.
- ii) The arithmetic operations using BCD code require a complex design of Arithmetic and Logic Unit (ALU) than the straight binary number system.
- iii) The speed of the arithmetic operations that can be realized using BCD code is naturally slow due to the complex hardware circuitry involved.

5.5 SOLVED BOARD EXAMINATION PROBLEMS

1. Define modulation and demodulation. Discuss various types of modulation techniques. [2013 F, 2015 F]

Solution:

Modulation is the process of superimposing audio signal over the carrier (high frequency) wave. Demodulation or detection is a process of recovering the original modulating signal (intelligence) from the modulated carrier wave *i.e.*, the demodulation is a process of modulation.

	Amplitude modulation	Phase modulation	Phase modulation
i)	Amplitude of carrier waves varies as per amplitude or voltage of modulating signal input.	Phase of carrier wave varies as per voltage of modulating signal input.	Frequency of carrier wave varies as per voltage of modulating signal input.
ii)	It requires more bandwidth than carrier.	Bandwidth of phase modulation depends on modulating signal frequency.	Theoretically, it needs infinite bandwidth as there exists ∞ sidebands.
iii)	A.M. type includes DSB-SC, SSB, VSB, etc.	Digital P.M. types include BPSK, QPSK, QAM.	Digital F.M. types include FSK, GFSK, offset FSK.
iv)	Frequency and phase of carrier wave is kept constant.	Frequency and amplitude of carrier wave is kept constant.	Amplitude and phase of carrier wave is kept constant.

2. Differentiate between F.M. and A.M. [2013 S, 2014 S, 2016 S, 2017 S, 2018 F, 2019 F]

Solution:

The difference between F.M. and A.M. are as follows:

	Frequency modulation (F.M.)	Amplitude modulation (A.M.)
i)	Frequency of carrier wave varies as per voltage of input modulating signal.	Amplitude of carrier wave varies as per amplitude or voltage of modulating signal input.
ii)	Amplitude of carrier wave is constant.	Frequency of carrier wave is kept constant.
iii)	Bandwidth is large.	Bandwidth is much less than F.M.
iv)	F.M. receivers are immune to noise.	A.M. receivers are not immune to noise.
v)	Bandwidth = $2[\delta + f_m]$. The bandwidth depends on modulation index.	Bandwidth = $2f_m$. It is not dependent on the modulation index.

vi)	F.M. transmission and reception equipment are more complex.	A.M. equipments are less complex.
vii)	Space wave is used for propagation.	Ground wave and sky wave propagation is used. Therefore larger area is covered than F.M.
viii)	Infinite numbers of sidebands are present.	Only carrier and two sidebands are present.
ix)	F.M. broadcast operates in V.H.F. and V.H.F. range.	A.M. broadcast operates in M.F. and H.F. range.

3. Write short notes on optical fiber communication.

[2013 S, 2015 F, 2015 S, 2016 F, 2017 F, 2017 S, 2018 F]

Solution: See the definition part 5.4

4. Explain the need of modulation in the communication.

[2013 S, 2013 F, 2015 S, 2016 S, 2017 F, 2018 F, 2018 S, 2019 F]

Solution: See the definition part 5.1.2

5. Draw the block diagram of communication system and explain the function of each block. [2014 S, 2015 S, 2017 S, 2018 S, 2019 F]

Solution: See the definition part 5.1.1

6. Define amplitude modulation and draw the wave shapes of the message signal, carrier and modulated wave. [2014 F]

Solution:

Amplitude modulation is defined as a system in which the maximum amplitude of the carrier wave is made proportional to the instantaneous value (amplitude) of the modulating signal. It have the drawback of low efficiency (maximum efficiency of the A.M. is only 33.33%).

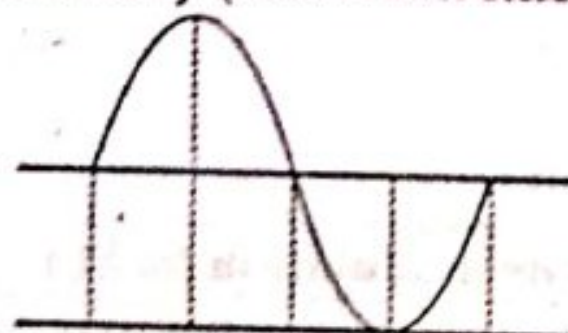


Figure: Modulating signal or message signal

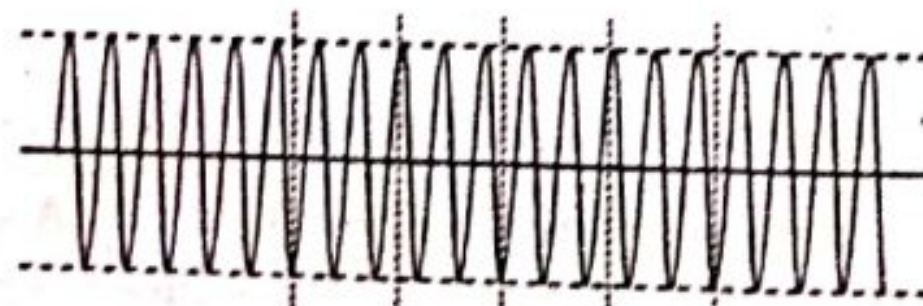


Figure: Carrier wave

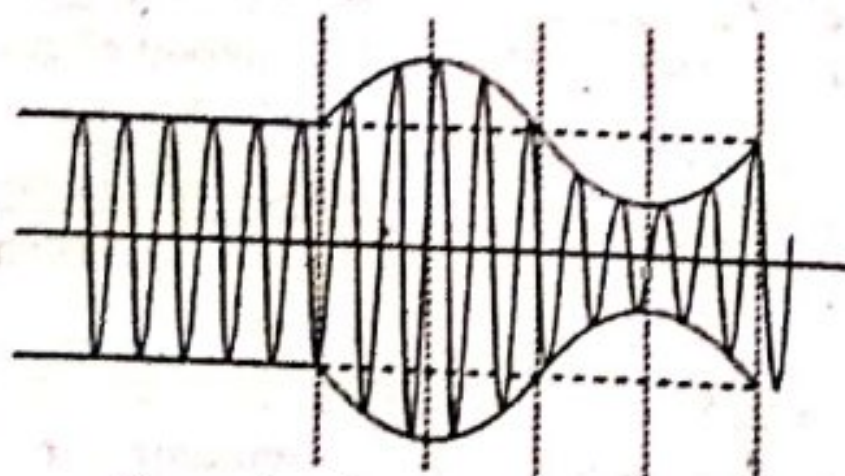


Figure: Amplitude modulated wave

Let the carrier and modulating voltage waves be represented as;

$$v_c = V_c \sin \omega_c t \quad (1)$$

$$\text{and, } v_m = V_m \sin \omega_m t \quad (2)$$

The amplitude of the carrier wave varies at a modulating signal frequency f_m . The amplitude of amplitude modulated wave is given as;

$$A = V_c + V_m$$

$$\text{or, } A = V_c + V_m \sin \omega_m t$$

$$\text{or, } A = V_c \left(1 + \frac{V_m}{V_c} \sin \omega_m t \right)$$

$$\text{or, } A = V_c (1 + m \sin \omega_m t)$$

where, m is the ratio of peak values of modulating signal and carrier wave and is known as modulation index. Its value is restricted between zero and unity.

The instantaneous value of amplitude-modulated wave is given by;

$$v = A \sin \omega_c t$$

$$\text{or, } v = V_c (1 + m \sin \omega_m t) \sin \omega_c t$$

$$\text{or, } v = V_c (1 + m \sin \omega_m t) \sin \omega_c t$$

$$\therefore v = V_c \sin \omega_c t + \frac{mV_c}{2} \cos(\omega_c - \omega_m)t - \frac{mV_c}{2} \cos(\omega_c + \omega_m)t$$

Above equation reveals that the amplitude-modulated wave is equivalent to the summation of three sinusoid; one having amplitude V_c and frequency $\frac{\omega_c}{2\pi}$, the second having amplitude $\frac{mV_c}{2}$ and frequency $\frac{\omega_c - \omega_m}{2\pi}$ and the third having amplitude $\frac{mV_c}{2}$ and frequency $\frac{\omega_c + \omega_m}{2\pi}$. The lower frequency component $\frac{\omega_c - \omega_m}{2\pi}$ is called the lower side frequency. The upper frequency component $\frac{\omega_c + \omega_m}{2\pi}$ is called the upper side frequency. The amplitude of each side frequency is $\frac{mV_c}{2}$. The amplitude of sideband frequencies can never exceed half the carrier amplitude because $m \leq 1$.

7. Explain the frequency modulation with the help of neat sketches. [2015 S]

Solution: See the definition part 5.2.1

8. Define modulation. Describe the principle of amplitude modulation in different modulation index. [2016 F]

Solution:

The process of superimposing audio signal over the carrier (high frequency) wave is called the modulation.

Principle of amplitude

See the solution of Q. no. 1

For the remaining part

The extent to which the amplitude of the carrier wave is varied by the modulating signal is called the degree of amplitude modulation or modulation index.

$$\text{i.e., } m = \frac{V_m}{V_c} = m_a$$

$$\text{or, } m = \frac{\text{Amplitude change of carrier wave}}{\text{Amplitude of normal or unmodulated carrier wave}}$$

Also,

$$m = \frac{V_{\max} - V_{\min}}{V_{\max} + V_{\min}}$$

where, $V_{\max}$ is the maximum value of amplitude of modulated carrier wave.

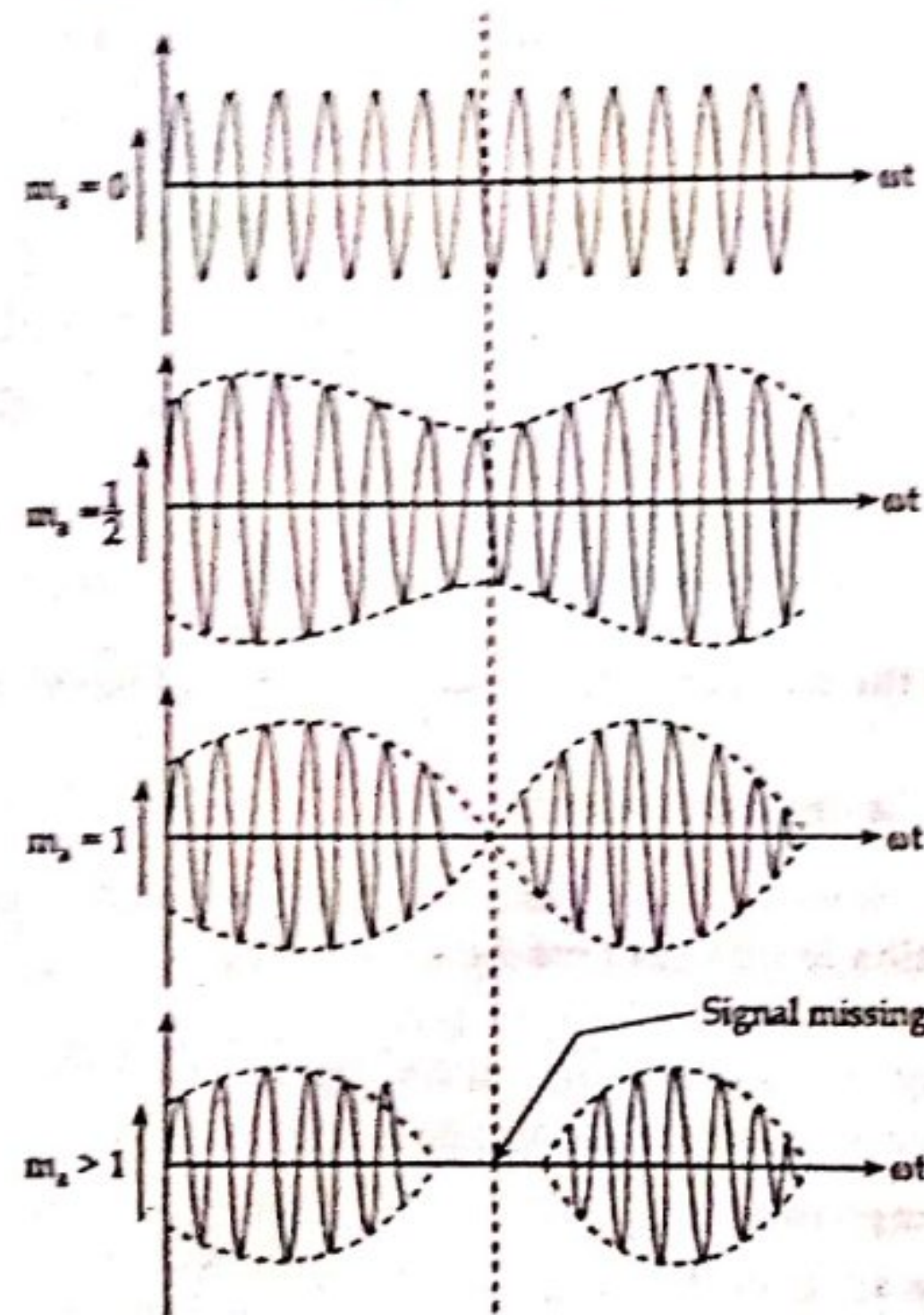
$V_{\min}$ is the minimum value of amplitude of modulated carrier wave.

$$\text{and, } V_c = \frac{V_{\max} + V_{\min}}{2}$$

$$\text{so, } V_{\max} - V_{\min} = 2V_c$$

Hence,

$$m_a = \frac{V_{\max} - V_{\min}}{2V_c} = \frac{V_{\max} - V_c}{V_c} = \frac{V_c - V_{\min}}{V_c}$$



Case I: $m_a < 1$

This case is referred to as the under-modulation and the A.M. wave is known as under-modulated A.M. wave. The message signal can be extracted from the under-modulated wave by using a simple circuit known as envelope detector.

Case II: $m_a = 1$

In this case, the envelope touches the zero-amplitude line for some time and this is referred to as 100% modulation. The A.M. wave is then called the 100% modulated A.M. wave. The message signal can be extracted from this wave by using envelope detector.

Case III: $m_a > 1$

In this case, some portion of the envelope crosses the zero-amplitude line. This condition is known as the over-modulation and the A.M. wave is called over-modulated A.M. wave. Modulation in excess of 100% is undesirable because it produces severe distortion and interference and spatter in the transmitter output.

9. Discuss in brief on the generation and detection of F.M. signal. [2018 S]

Solution: See the definition part 5.3.3 and 5.3.4

10. Classify types of modulation according to nature of carrier wave.

Solution:

Modulation can be classified into continuous wave modulation and pulse modulation according to the nature of carrier wave.

i) Continuous wave modulation

When the carrier wave is continuous in nature, the modulation process is called the continuous wave modulation or analog modulation. Amplitude modulation and angle modulation (frequency modulation and phase modulation) fall under this category.

ii) Pulse modulation

When the carrier wave is a pulse type waveform, the modulation process is referred to as pulse modulation. In pulse modulation, the carrier wave consists of a periodic sequence or rectangular pulse. Pulse modulation may be of analog or digital type.

11. What are the types of amplitude modulation?

Solution:

There are three types of amplitude modulation, namely;

- Double sideband suppressed carrier (DSBSC)
- Single sideband transmitted carrier (SSBTC)
- Single sideband suppressed carrier (SSBSC)

12. Write down the disadvantages of optical fiber.

Solution:

Disadvantages of optical fiber are as follows;

i) Limited application

Fiber optic cable can only be used on the ground, and it cannot leave the ground or work with mobile communication.

ii) Low power

Light emitting sources are limited to low power. Although high power emitters are available to improve power supply, it would add extra cost.

iii) Fragility

Optical fiber is rather fragile and more vulnerable to damage compared to copper wires.

iv) Distance

The distance between the transmitter and receiver should keep short or repeaters are needed to boost the signal.

v) Loss of light in fibre due to scattering (Attenuation).

vi) Expensive than other cables.

13. Write down the application of optical fiber communication.

Solution:

Fiber optic cables find many uses in a wide variety of industries and applications. Some uses of fiber optic cables include:

i) Medical

It is used as light guides, imaging tools and also as lasers for surgeries.

ii) Defense/government

It is used as hydrophones for seismic waves and SONAR, as wiring in aircraft, submarines and other vehicles and also for field networking.

iii) Data storage

It is used for data transmission.

iv) Telecommunications

Fiber is laid and used for transmitting and receiving purposes.

v) Networking

It is used to connect users and servers in a variety of network settings and help increase the speed and accuracy of data transmission.

vi) Industrial/commercial

It is used for imaging in hard to reach areas, as wiring where EMI is an issue, as sensory devices to make temperature, pressure and other measurements and as wiring in automobiles and in industrial settings.

vii) Broadcast/CATV

Broadcast/cable companies are using fiber optic cables for wiring CATV, HDTV, internet, video on-demand and other applications.

Fiber optic cables are used for lighting and imaging and as sensors to measure and monitor a vast array of variables. Fiber optic cables are also used in research and development and testing across all the above mentioned industries.

6.5 SOLVED BOARD EXAMINATION PROBLEMS

1. Discuss the application of electronics in civil engineering.
[2013 S, 2014 S, 2016 F, 2019 F]

Solution:

The main use of electronics in civil engineering is for structural health monitoring (SHM) of civil structures, particularly photonics (fiber optic Bragg grating). In SHM, civionics (civil engineering and electronic engineering) will provide engineers with feedback necessary to aid in optimizing design techniques and understanding infrastructure performance, behaviour and state of condition. The successful integration of intelligent sensing of innovative structures will allow civil structural engineers to expand the design envelope by taking risks to introduce new design concepts, materials and innovation in civil engineering. Some basic use of electronics in civil engineering are;

- i) Computer aided design of buildings
- ii) Usage of telemetry to perform measurement of plot
- iii) Structural health monitoring
- iv) Remote sensing to observe areas from a much better and bigger perspective
- v) Usage of sensors to measure stress on high profile buildings and suspensions bridge

2. Write short notes on EDM in surveying.
[2013 S, 2014 F, 2014 S, 2015 S, 2016 S, 2017 F, 2018 F, 2018 S]

Solution: See the definition part 6.2 and 6.2.3

3. What are the electronic instruments used in civil engineering?
[2015 F]

Solution:

The electronic instruments used in civil engineering are total station, electronic distance measurement (EDM) and strain gauge load cell.

4. Write short notes on strain gauge load cell. [2015 F, 2016 F]

Solution: See the definition part 6.3

5. Explain the operation of total station and its application in civil engineering. [2013 S, 2014 S, 2015 F, 2015 S, 2016 S, 2017 S, 2018 S]

Solution: See the definition part 6.1, 6.1.1 and 6.1.3

6. Define surveying. [2016 S]

Solution:

Surveying is defined as the profession or work of examining and recording the area and features of a piece of land so as to construct a map, plan or detailed description of it.

7. Write short notes on application of sensor in engineering. [2016 S]
Solution:

Sensors are sophisticated devices that are frequently used to detect and respond to electrical or optical signals. A sensor converts that physical parameter (*for example*; temperature, blood pressure, humidity, speed, etc.) into a signal which can be measured electrically. A sensor is always used with other electronics.

Sensors are an important part to any measurement and automation application. Sensors are pervasive. They are embedded in our bodies, automobiles, airplanes, cellular telephones, radios, chemical plants, industrial plants and countless other applications. Without use of sensors, there would be no automation. Some sensors and their application are;

i) Motion sensors

It monitors location of various parts in a system.

- absolute/relative position
- proximity
- acceleration
- angular/relative displacement

ii) Strain gauge

It is used to measure deflection, stress, pressure, etc.

iii) Temperature sensor

Thermostat (makes or breaks electrical connection with deflection)

iv) Accelerometer

They are used to measure acceleration along one or more axis and are relatively insensitive to orthogonal directions.

v) Light sensors

They are used in cameras, infrared detectors and ambient lighting applications.

vi) Magnetic field sensors

They are used for power steering, security and current measurements on transmission lines.

vii) Ultrasonic sensors

They are used for position measurements.

For example;

- Infrared sensors are frequently used to design TV remotes.
- Ultrasonic sensors are used for measuring distance of an object.

8. Explain how can we collect the information of remote object using remote sensing. [2016 F]

Solution: See the definition part 6.4

9. Explain the use of EDM in surveying. [2019 F]

Solution: See the definition part 6.2.3



SHORT ANSWER QUESTIONS

1. What is a P-N junction?

Ans: The contact surface between the layers of P-type and N-type semiconductor pieces placed together so as to form a P-N junction is called the P-N junction.

2. Why silicon is usually preferred over germanium for fabrication of semiconductor devices?

Ans: The silicon semiconductor devices have, in general, higher peak inverse voltage and current ratings and wider temperature range than germanium semiconductor devices and therefore, silicon is preferred over germanium in the manufacture of semiconductor devices.

3. How diode acts as a rectifier?

Ans: Rectifier is a device, which converts the sinusoidal ac voltage into unidirectional pulsating voltage. Diode is an electronic device consisting of two elements known as anode and cathode. Since, in a diode, electrons can flow in only one direction i.e., from cathode to anode, the diode provides the unilateral conduction necessary for rectification.

4. Avalanche breakdown occurs at large reverse voltage whereas zener breakdown occurs at low voltage. Give reasons.

Ans: The avalanche breakdown can occur at large reverse voltage whereas zener breakdown occurs at low voltage. The main reason for this is the difference in doping level of avalanche and zener diode. The doping level in zener diode is high as compared to the doping level of avalanche diode.

Because of higher doping in zener, the depletion width is smaller

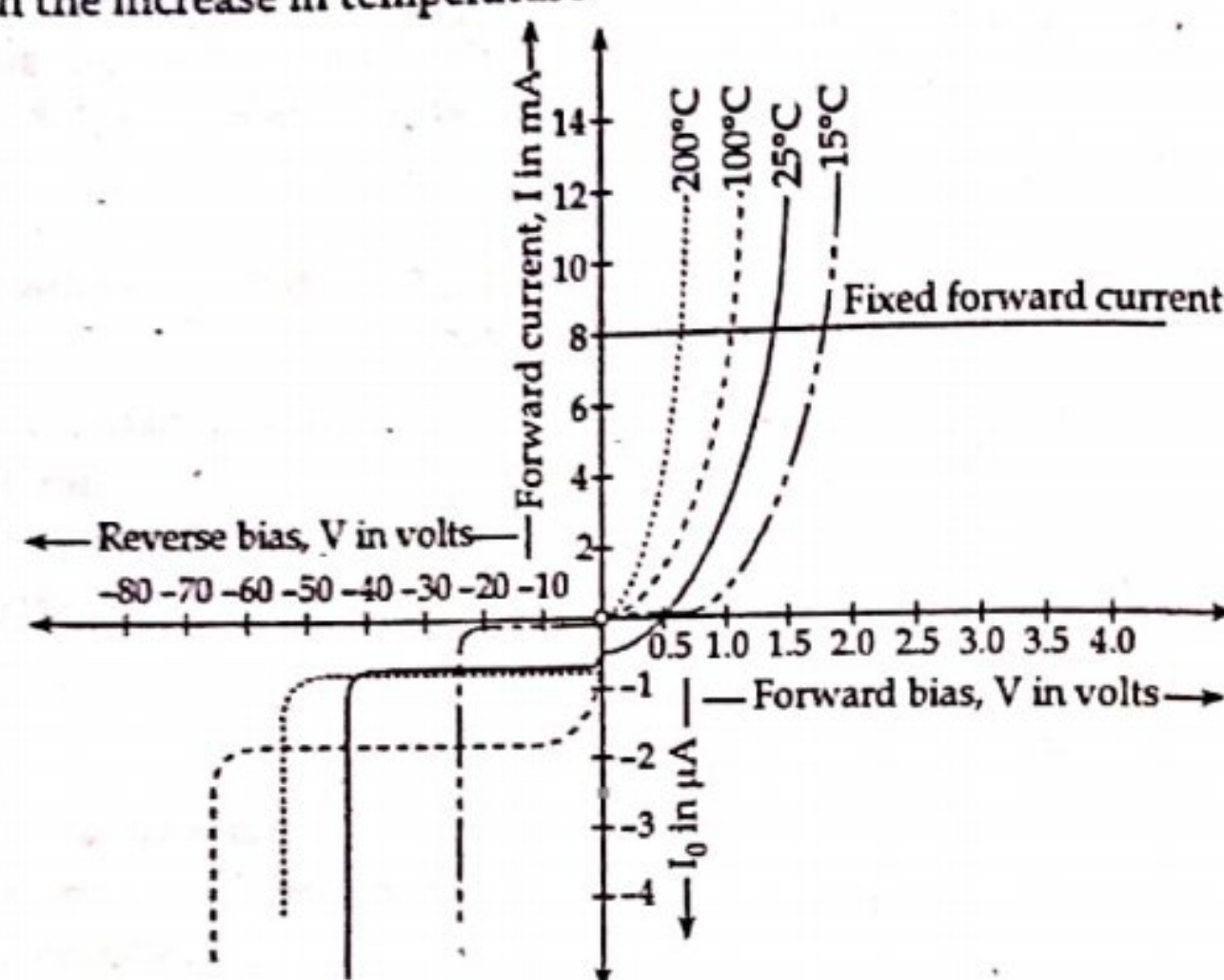
(with w being $\propto \frac{1}{\sqrt{\text{Doping}}}$) and due to this smaller width of the

breakdown voltage is smaller (breakdown voltage $V_{BR} \propto w \propto$

$$\frac{1}{\sqrt{\text{Doping}}})$$

5. Explain the effect of temperature on the V-I characteristics of the diode.

Ans: Reverse saturation current I_0 and volt equivalent of temperature V_T increase with the increase in temperature. I_0 doubles for every 10°C rise in temperature. This causes lowering of threshold of cut-in voltage. With the increase in temperature, forward characteristic becomes more ideal. In reverse bias, the breakdown voltage increases with the increase in temperature.



6. Write down the five applications of a diode.

Ans: Diode finds use in many applications as enumerated below;

- As rectifiers in dc power supplies
- In demodulation or detector circuits
- In clamping networks employed as dc restorers in TV receivers and voltage amplifiers
- As switches in digital logic circuits
- In clipping circuits used as wave shaping circuits in computers, radars, radio and TV receivers.

7. Why is zener diode used as a voltage regulator?

Ans: Zener diode has the property of behaving like a dc battery in 'ON' state (i.e., when the voltage across zener diode exceeds its zener voltage rating V_z). In 'ON' state, the voltage across zener diode remains constant until the voltage across it drops less than V_z . This property of zener diode makes its use as voltage regulator.

8. Explain how zener diode maintains constant voltage across the load?

Ans: Zener diode has the property of behaving like a dc battery in 'ON' state. If the zener diode is shunted across the load R_L and the voltage

across zener diode is more than zener voltage V_z , then zener diode is in 'ON' state and any variations in voltage across the zener diode due to variations either in supply voltage or in load resistance is not able to change the output voltage. Thus, zener diode maintains constant voltage across the load.

9. Why diodes are not operated in the breakdown region in rectifiers?

Ans: In breakdown region, a diode has a risk of getting damaged or burnt because the magnitude of current flowing through it increases in an uncontrollable manner. That is why diodes are not operated in the breakdown region in rectifiers.

10. Explain why a capacitor and an inductor can act as filtering components.

Ans: The filtering action of L and C depends on the facts that an inductor allows dc only and capacitor allow ac only to pass i.e., the filtering action of an inductor filter depends upon its property of opposing any change in current flowing through it whereas the action of capacitor filter depends upon its property of opposing any change in the voltage applied across its ends.

11. Why π -filters are not suitable for varying loads?

Ans: Voltage regulation in case of π -filters is very poor and therefore π filters are not suitable for varying loads.

12. Why is it necessary to include voltage stabilizer in a power supply?

Ans: The magnitude of output dc voltage may vary with the variation of either the input ac voltage or the magnitude of load current. So, at the output of a rectifier-filter combination (a voltage stabilizer) is required.

13. Why R-C filters are suitable for only light loads?

Ans: R-C filters have poor voltage regulation and need adequate ventilation to dissipate the heat developed in the resistor R. Thus, R-C filters are suitable only for light loads.

14. Explain why an ordinary transistor is called bipolar?

Ans: Because the transistor operation is carried out by two types of charge carriers (majority and minority carriers) an ordinary transistor is called bipolar.

15. Why transistor is called current controlled device?

Ans: The output voltage, current or power is controlled by the input current in a transistor, so it is called as the current controlled device.

16. What do you mean by operating point?

Ans: Quiescent point is a point on the dc load line which represents V_{CE} and I_C in the absence of ac signal and variations in V_{CE} and I_C take place around this point when ac signal is applied.

17. What is the requirement of good biasing circuit?

Ans: The requirements of good biasing circuit are;

- Establish the operating point in the middle of the active region of the characteristics, so that on applying the input signal the instantaneous operating point does not move either to the cutoff region or to the saturation region, even at the extreme values of the input signal.
- Stabilize the collector current I_C against temperature variations.
- Make the operating point independent of transistor parameters so that replacement of transistor by another transistor of the same type in the circuit does not shift the operating point.

18. Why collector to base bias circuit is seldom used?

Ans: The base resistor R_B not only provides a dc feedback for the stabilization of the operating point, but it also causes an ac feedback in a collector to base bias circuit. This is the reason that this circuit is not much used.

19. Why FET is called voltage controlled device?

Ans: In a FET, drain current is controlled by the effect of the extension of the field associated with the depletion region developed by the reverse bias on the gate, so it is called voltage controlled device.

20. What are the basic logic elements?

Ans: Basic logic elements are NOT gate, AND gate, OR gate and the flip-flop.

21. What is a truth table?

Ans: Truth table is a table that gives outputs for all possible combinations of inputs to a logic circuit.

22. What are the universal logic gates?

Ans: Universal logic gates are NAND and NOR gates.

23. Define positive and negative logic.

Ans: If the higher of the two voltages represents a 1 and the lower voltage represents a 0, the logic is called a positive logic. On the other hand, if the lower voltage represents a 1 and the higher voltage a 0, we have a negative logic.

24. State the limitation of Karnaugh's map.

Ans: The limitation of K-map is that beyond 6 variables it becomes very cumbersome and so it is generally used up to 6 variables.

25. What is the use of a Karnaugh map?

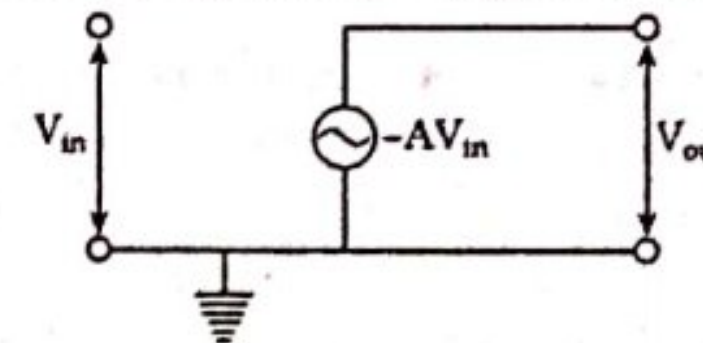
Ans: Karnaugh map is used for simplification and manipulation of Boolean expression.

26. What is the maximum value of output voltage obtainable from an Op-amp?

Ans: The output voltage magnitude is limited by supply voltage. The maximum output voltage will be about 2 to 3 volts less than the supply voltages V_{CC} .

27. Draw the equivalent circuit of an ideal Op-amp.

Ans: Equivalent circuit of an ideal Op-amp is shown below:



28. What are the applications of Op-amp in open loop configuration?

Ans: Applications of Op-amp in open loop configuration are;

- Differential amplifier
- Inverting amplifier
- Non-inverting amplifier

29. Name three important specifications of IC Op-amp.

Ans: The three important specifications of IC Op-amp are high voltage gain, low output impedance and high input impedance.

30. What is the voltage gain of a practical Op-amp?

Ans: Voltage gain of a practical Op-amp is of the order of 10^5 .

31. Give the examples of linear circuits.

Ans: Adder, subtractor, differentiator, integrator, voltage to current converter, current to voltage converter, instrumentation amplifier fall under the category of linear circuits.

32. Why integrators are preferred over differentiator?

Ans: Op-amp is mostly used as an integrator rather as a differentiator because in differentiator at high frequency, gain is high and so high frequency noise is also amplified which absolutely abstract the differential signal.

- 33. What is meant by radix or base of a number system?**
Ans: Radix or base of a number is the number of digits or distinct symbols it uses to represent various numbers.
- 34. What is meant by LSB and MSB?**
Ans: The right most bit of a binary number is called least significant bit (LSB) while the left most bit of a binary number is known as most significant bit (MSB).
- 35. Why hexadecimal code is widely used in digital systems?**
Ans: It is very convenient to enter binary data in a digital system using hex code.
- 36. What is meant by BCD code?**
Ans: BCD code is a decimal system with each digit encoded in its binary equivalent of 4 bits.
- 37. What is the difference between binary code and BCD?**
Ans: BCD is not a number system like binary. It is a decimal system with each decimal digit encoded in its binary equivalent. A straight binary code takes the complete decimal numbers and represents it in binary while the BCD code converts each decimal digit to binary individually.
- 38. What is modulation index in AM?**
Ans: The extent to which the amplitude of the carrier wave is varied by the modulating signal is called the modulation index and is defined as the ratio of change in amplitude of the carrier wave to the amplitude of normal carrier wave.
- 39. What is FM index?**
Ans: The ratio of frequency deviation to modulation frequency is called the FM index.
- 40. What is dark current of a photodiode?**
Ans: The dark current is the current that exists when no light is incident.
- 41. Give the application of optical fibers.**
Ans: Optical fiber finds wide use in communication, as sensors for measurement of strain, pressure, temperature and other physical parameters, as optical gyroscope, decoration purposes and as light guides in medical and other applications where bright light needs to be shined on a target without a clear line of sight path.
- 42. Give the advantages of optical fiber.**
Ans: The advantages of optical fibers are small size and weight, electrical isolation, flexibility, security and low transmission loss.

- 43. What is an optical fiber?**
Ans: Optical fiber is a glass or plastic fiber that is designed for guiding light along its length by total internal reflection.
- 44. In semiconductor diode, P-side is grounded and N-side is applied a potential of - 3 V. Will the diode conduct? Explain.**
Ans: The P-N diode connected with P-side grounded and with N-side to negative potential of 3 V, the diode will conduct because N-side is at negative potential with respect to positive diode.
- 45. What is the main difference between the characteristic of a simple switch and those of an ideal diode?**
Ans: The important difference between the characteristics of a simple switch and those of an ideal diode is that an ideal diode acts like an automatic switch—the switch is closed when the diode is forward biased and is opened when reverse biased.
- 46. Germanium is more temperature dependent than silicon. Why?**
Ans: Because the reverse saturation current in case of a germanium diode is approximately 1000 times larger.
- 47. Enumerate the properties that the junction diodes designed for use as rectifiers should possess.**
Ans: Junction diode designed for use in rectifiers should have the properties:
 i) V-I characteristics as close as possible to that of an ideal diode.
 ii) Negligible forward resistance
 iii) Large reverse breakdown voltage
 iv) Negligible reverse current i.e., large reverse resistance
 v) Small offset voltage in the forward direction
- 48. What is the importance of peak inverse voltage?**
Ans: If the applied voltage in reverse biased condition exceeds peak inverse voltage (PIV) rating of the diode, then the diode may get damaged.
- 49. Why diodes are not operated in the breakdown region in rectifiers?**
Ans: In breakdown region, a diode has a risk of getting damaged or burnt because the magnitude of current flowing through it increases in an uncontrollable manner. That is why diodes are not operated in the breakdown region.
- 50. Why half wave rectifiers are not generally used in dc power supply?**
Ans: The type of supply available from half wave rectifier is not satisfactory for general power supply that is why it is generally not used in dc power supply.

51. What are the regions of operation of a transistor?

Ans: The three regions of operations of a transistor are;

- i) cutoff region
- ii) active region
- iii) saturation region

52. Can a transistor be obtained by connecting two semiconductor diodes back to back?

Ans: No. Because in case of two discrete back to back connected diodes, there are four doped regions instead of three and there is nothing that resembles a thin base region between an emitter and a collector.

53. Which configuration of BJT is used for impedance matching and why?

Ans: Common collector configuration is used for impedance matching because it has relatively high input impedance and low output impedances.

54. Why emitter is always forward biased?

Ans: Emitter is always forward biased with respect to base so as to supply majority charge carriers to the base.

55. Why collector is always reverse biased with respect to base?

Ans: Collector is always reverse biased with respect to base so as to remove the charge carrier away from the base-collector junction.